M. Ihsan 489/00b

DEPARTMENT OF PHYSICS

Roll no. 226

Notes

Rs 340

M.Sc Part II

Nuclear Physics

By: Ali

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NUCLEAR PHYSICS

→ Course Outline:

- 73 • Basic Properties of nuclei 7th
  {Mass, Radius, Spin, Parity, Nuclear Binding energy, Magnetic dipole moment, Electric quadrupole moment, Statistics}
- • Detectors and Accelerators 7th
- 73–128 • Radioactive Decays {α, β, γ - decays}
- 273–305 • Nuclear Forces 7th
- 246–272 • Nuclear Models 7th, 8th
- 1??–159 • Nuclear Reactions 8th
- 16?–177 • Neutron Physics 8th
- 179–245 • Fission, Fusion 8th

→ Books:

→ Introductory Nuclear Physics
By Krane.

→ Nuclear Physics
By Ghoshal.

→ Nuclear Physics
By Verma, Gupta.

→ Nuclear Physics
By Kaplan.

→ Applications of Nuclear Physics: (Possible)

• Nuclear Power Generation

Nuclear Weapon Technology
Nuclear Medicine
Material Engineering
Radiocarbon Dating

Subject Introduction

→ Nuclear Physics (Definition).
It is the branch of physics which deals with the study of the components, structure, behaviour or properties of nuclei. It also deals with the interaction b/w nuclei, nuclear forces, radioactive decays etc.

Nucleus
→ Massive part of an atom
→ contains protons and neutrons (nucleons)

→ Mass no (A) : Total no. of nucleons (Protons + neutrons)

→ Charge No (Z), no. of protons in the nucleus

→ Neutron No (N) : A – Z

→ Symbol: $^{A}_{Z}X_{N}$ where X is chemical symbol for the element
e.g $^{1}_{1}H_{0}$, $^{238}_{92}U_{146}$

Isotopes:
A group of nuclei with same proton no. but different neutron no. are called isotopes. i.e Same Z, Diff A, N
example: $^{1}_{1}H_{0}$, $^{2}_{1}H_{1}$, $^{3}_{1}H_{2}$

Isotones:
A ...
neutron n...
i.e. Same ...
e.g $^{2}_{1}H$, ...
Isobars:
A ...
mass no. ...
i.e. S...
e.g $^{3}_{1}H$ ...
Mirror Nucl...
and proton...
Nuclear Iso...
of nucle...
equal m...
of excita...
→ Origin / H...
1896 → 1...
He fo...
radiation
they ...
in a ...
They tran...
spontaneo...
the...

Isotones:

A group of nuclei with same neutron no but different proton no.

i.e. Same : N Diff : Z, A

e.g. $^{2}_{1}H_1$, $^{3}_{2}He_1$, $^{13}_{6}C_7$, $^{14}_{7}N_7$

Isobars:

A group of nuclei with same mass no but different charge no.

i.e. Same : A Diff. Z

e.g. $^{3}_{1}H_2$, $^{3}_{2}He_1$, $^{17}_{7}N_{10}$, $^{17}_{8}O_9$

Mirror Nuclei:

Those nuclides whose neutron and proton no is exchanged.

e.g. $^{3}_{1}H_2$, $^{3}_{2}He_1$

→ **Nuclear Isomers:**

"Members of a set or group of nuclides with equal proton no and equal mass no but different states of excitation.

e.g. $^{99}_{43}Tc$, $^{99}_{43}Tc^*$ (Technetium)

→ **Origin/History of Nuclear Physics:**

→ 1896 → Henry Becquerel (discovery of radioactivity).

He found "Uranium compound" emitted radiation.

* they affected photographic plate wrapped in a black paper.

←(radioactive element)

They transform into other elements spontaneously, after emission of radiations in the process.

Few examples of radioactive elements: Polonium, Radium etc.

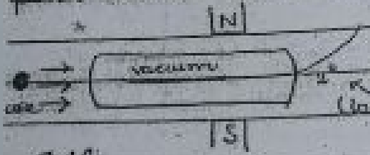
- Later work on radioactivity was performed by Curie and Rutherford.

- From their experiments it was found that there are 3 types of radiations. α, β, γ → difference in ionizing power, penetrating power, behaviour in B.

→ 1906 Rutherford was studying the properties of α-radiations.

He was working on deflection of α-particles in magnetic field

→ To minimize α-loss by scattering from air molecules



(He used vacuum apparatus)

(100 million volts per cm)

Self:

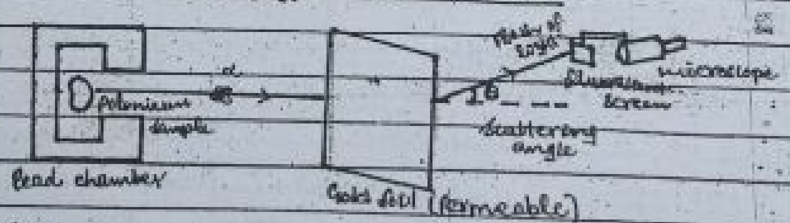
→ Rutherford announced the discovery of α-particles scattering by air in 1906 (Jan). He took a wire coated with radioactive material and passed the α-rays through a narrow slit. This resulted in a narrow, rectangular beam rather than a narrow, circular pencil shaped beam.

→ In vacuum the narrow slit showed perfectly sharp edges on a photographic plate the α-ray hits. However when the beam passed through air, the edges of the slit became diffused and widened.

→ In June 1906, Rutherford used his wire coated with α-emitting radioactive element, but he modified the slit the α-rays travelled through. Half the slit was left open and half was covered by a thin (0.003 cm) mica plate. The experiment was done in the vacuum; the mica taking the place of the air. The open side of the resulting photographic image was sharp and mica side was diffused.

He also subjected the α-beam to a magnetic field. Since e^-s bend the opposite way to α-particles any e^-s produced by the mica as the beam went through it be swept away. The image was the same as there was no magnetic field. The vacuum side was sharp and mica side was diffused.

→ Rutherford's Scattering Experiment :



Rutherford scattering is the elastic scattering of charged particles by the Coulomb interaction. It is a physical phenomenon explained by Ernest Rutherford in 1911. It is referred to as Coulomb's scattering because it relies only upon static electric forces, and the minimal distance b/w particles is set only by this potential.

α-particles from a radioactive source were allowed to strike a thin gold foil. Alpha particles produce a tiny but visible flash of light when they strike the fluorescent screen. Surprisingly α-particles were found at large deflection angles and some were even found

o be back scattered.
This experiment showed that the positive atter in atoms was concentrated in an acredibly small volume and give birth to e idea of nuclear atom.
If the gold foil were 1µm thick then using the diameter of gold atom from e periodic table suggests that the foil about 2800 atoms thick.

Rutherford and discovery of nucleus :

In 1906 Rutherford was working on deflection of α-rays in a magnetic field.
He noticed a large effect on the path of α-particles, if a small amount of air was present in the chamber.
At that time, people used to believe on Thomson's (Plum Pudding) atomic model. According to which atom has a uniform density.
But Rutherford thought that if atom has uniform density then it should not effect particle's path that ~~worked~~ much.
⇒ We need to investigate structure of atom in detail (as Probably it won't have uniform density).
For this he proposed an experiment in (1909).

" Scattering
Gold fo
therford's

$F = \frac{kq_1q_2}{r^2}$

Rutherford

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Discovery

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α-partic
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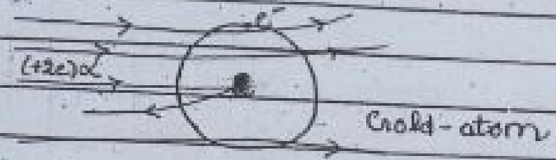
Interac
itrogen
Hydroge
itrogen
rther i
at hyd
presen
named

Atom

"Scattering of α-particles by Gold foil."

Rutherford's Scattering Experiment

$F = K\frac{q_1 q_2}{r^2}$



Rutherford Atomic Model:

Atom (neutral) → Electrons (-ve charge) + Nucleus (+ve charge)

Discovery of proton:

1919 – by Rutherford

Prout's hypothesis → also contributed for the discovery of +vely charged particle. So Rutherford credited him.

→ α-particle was bombarded on Nitrogen as a target

$${}^{4}_{2}\alpha + {}^{14}_{7}N \rightarrow {}^{17}_{8}O + {}^{1}_{1}H \rightarrow \text{(H-nucleus)}$$

Interaction of α-particles with Nitrogen knocked off hydrogen nuclei.

→ Hydrogen nucleus is present in nitrogen nucleus.

Further investigations / experiments showed that hydrogen nucleus (lightest nucleus) is present in all other nuclei.

named as proton (a particle which has +ve charge)

So

Atom → Electrons + Nucleus → consists of protons

Proton discovery $=\!=\!=$ Neutron discovery
1919 $\simeq 13$ years 1932 Chadwick

What happened in this gap?
were scientists satisfied with present structure of atom?

Ans No because of atomic mass problem. This is due to that theoretical mass was not equal to the experimental one.

Net nuclear charge $= +Ze$ (no. of protons or electrons) } overall atom was neutral

Net -ve charge $= -Ze$

As $\Rightarrow m_p \simeq 1836\, m_e$

$\rightarrow$ Mass of an atom $\simeq$ Mass of a nucleus

Consider an example of Helium atom

Experimentally mass of ${}^4_2He$ was 4amu
$m(He) \neq m(2p) + m(2e^-)$ Wrong!

$$\frac{m(He)}{\text{fundamental mass unit}} = \frac{4\,amu}{amu} = \underset{\text{mass no.}}{4(A)}$$

$\rightarrow$ A was found to be roughly double of Z i.e $A = 2Z$
but if we have Z protons in

the nucleus → mass wrong!

Assumptions:

① For correct mass, it was assumed that there are "A ÷ 2Z" no. of protons in the nucleus.

Problem: atom not neutral.

e.g Consider deutrium ^2_1H.

1 e^- around the nucleus = −1e charge.

2 protons inside nucleus − +2e charge = +1e net charge

② To avoid this difficulty, another assumption was made. According to this,

→ there are A no. of Protons in the nucleus.

→ there are Z no. of Electrons around the nucleus.

→ there are 'A−Z' no. of electrons inside the nucleus (nuclear electrons).

With this assumption.

^2_1H — neutral
- A=2, no. of protons inside nucleus
- Z=1 no. of electrons outside nucleus
- A−Z = 2−1 = 1, no. of e^-s inside nucleus.

→ Mass is now also correct.

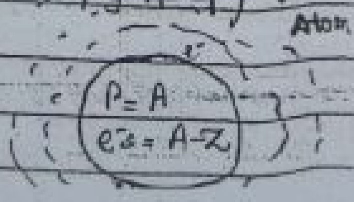
Q. Why presence of e^-s inside the nucleus is unsatisfactory?

Q. Why idea of nuclear e^-s were discarded?

For answer see Krane (Ch#1 ; Pg#4).

Imp.

Ans: ① Electron-Nuclear Interaction:



∴ There should be very strong force (attractive force) b/w nuclear e^-s and protons (stronger than Coulomb force). But there was no evidence of such strong force.

(ii) **Nuclear Spin:**

Calculation of nuclear spin for some nuclei where $A-Z =$ odd no. did not agree with the experimentally observed value

e.g Deutrium ^2_1H

$$A-Z = 2-1 = 1 \text{ (nuclear } e^-)$$

⇒ In Deutrium nucleus we have 2 protons and 1 nuclear e^-. Both e^- & P are spin-1/2 particles.

$$|s_1 - s_2| \leq s \leq (s_1 + s_2)$$

when $s' = 0$ $\quad |s' - s_3| \leq s \leq (s' + s_3)$

$$|0 - 1/2| \leq s \leq 1/2$$

$$1/2 \leq s \leq 1/2$$

when $s' = 1$ $\quad |1 - 1/2| \leq s \leq (1 + 1/2)$

$$1/2 \leq s \leq 3/2$$

$$s = 1/2,\ 3/2$$

Note: $0 \leq s \leq 1$ ∵ $s_1 = 1/2$, $s_2 = 1/2$, $s_3 = 1/2$

Rule for the addition of spin angular momentum

Calculated nuclear spin of Deutrium by Q.M $= 3/2$ or $1/2$ but experimental value of deutrium nuclear spin was found to be 1 → $\boxed{\text{idea of nuclear } e^-\text{s is wrong.}}$

Two types of Properties:
1) Static Properties
2) Dynamic Properties

1) Static Properties:
- Properties corresponding to ground state.
- When both charge and matter distribution inside the nucleus are assumed to be static.
eg electric charge, Mass, Radius ... etc.

2) Dynamic Properties:
When charge and matter distribution is assumed to be dynamic.
It include properties of nuclei exhibited in nuclear decay, nuclear reactions, nuclear excitations, fusion etc.

Note: Unit for Nuclear mass: u (unified atomic mass unit)

$1u = \frac{1}{12}$ (mass of ^{12}C-atom)

1 → gram $\quad 1u = 1.6605 \times 10^{-27}$ kg

or $\quad 1u = 1.6605 \times 10^{-24}$ g

Mole
Amount of substance of a system which contains as many entities (ions, atoms, molecules) as there are atoms in 12 g of ^{12}C (symbol mol)

No. of atoms in 12g of ^{12}C =

6.02×10^{23} atoms

(N_A - Avogadro no)

..its and Dimensions: SI units — not convenient in nuclear physics.

..ngth: In nuclear physics, we concern with lengths $\sim 10^{-15}$ m (1 fm)

..nergy:
Nuclear energies are conveniently measured in MeV
where $1\,MeV = 10^6\,eV$, $1\,eV = 1.6\times10^{-19}\,J$

..ass:
Nuclear mass; measured in terms of ..nified atomic mass unit (u)
..here $u = \frac{1}{12}$th of mass atom of ^{12}C taken to be exactly 12 units.
$1\,u = \frac{1}{12}$ (mass of ^{12}C-atom)

Since 12 nucleons in ^{12}C
⇒ 1 nucleon (Proton or neutron) mass $\simeq$ 1 u

Time:
For nuclear phenomenon — enormous range of time.
from 10^{-20} sec — 10^{32} years.

CHAPTER 1 BASIC PROPERTIES OF NUCLEI

Nucleus bound; Quantum system like an atom
It has different quantum states depending on energy, angular momen..
lowest energy state → Ground state
→ Nuclei normally exist in ground state

Using in $E = mc^2 \rightarrow 931.502\ MeV = (1u)c^2$
so $1u = 931.502\ \frac{MeV}{c^2}$

Also $\boxed{c^2 = 931.502\ \frac{MeV}{1u}}$

Basic Properties of Nuclei:

(1) Nuclear Mass: (Experimental methods to measure nuclear mass)

→ Easy way: $^{16}O_8$ $A = 16$
mass of $^{16}O_8 = (A)(1u)$
$\Rightarrow$ $= 16u$
∵ since mass of 1 nucleon = 1u

To measure nuclear mass precisely there is following procedure:

- First take out all e^-s (not possible or tedious)
- Measure atomic masses experimentally using mass spectroscopes
- Get nuclear mass 'M_{nuc}' from atomic mass $M(A,Z)$ by subtracting mass of Z-electrons

i.e. $M_{nuc} = M(A,Z) - Zm_e \rightarrow$ (1)

We also have to correct eq. (1) by taking into account binding energy of e^-s in the atom.

$$M_{nuc} = M(A,Z) - Zm_e + \sum_{i=1}^{Z} \frac{B_i}{c^2} \rightarrow (2)$$

→ There are 3 methods to measure nuclear masses.

(1) Mass spectrometry

Mass doublet method
Nuclear reaction method

Mass Spectrometry:
— First analytical technique where masses of nuclei were measured with high precision.
This technique involves the calculation of charge to mass ratio of charged particles
This method, first introduced by J.J Thomson.
To determine nuclear masses and relative abundances in a sample of ordinary matter, there must be a way to separate the isotopes first
because different isotopes carry different nuclear masses
→ For this an instrument called as "Mass Spectroscope" is used
First one was developed by Aston in 1919 This machine is used for;
① Separating isotopes
② Measuring masses of isotopes
③ Measuring the relative abundances of nuclei.

Mass Spectroscope

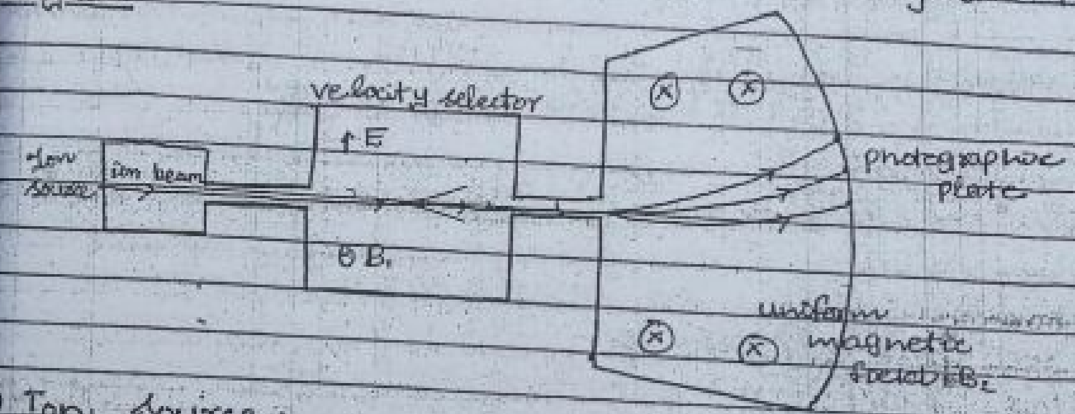
Mass Spectrograph	Mass Spectrometer
If separated isotopes/masses are focused to make an image on photographic plate.	If separated masses are passed through detecting slit and recorded electronically

→ A mass spectroscope has 3 main elements/components

① Ion source ② Velocity selector
③ Momentum selector.

Working principle: Deflection of ions under the action of applied magnetic field

Diagram



① **Ion source:**

First element of a mass spectroscope produces a beam of ionized atoms. (thermal distribution)

Ionization Methods: → vapour of material under study is bombarded with electrons.
spark discharge b/w the electrodes coated with material. (Add) Ions produced here have a broad range of velocities.

② **Velocity selector:**

- Used to select charge particles with a specific velocity.
- It consists of 1 electric and magnetic field.

- Upward force 'qE' by electric field $\vec{E}$.
- downward force "qvB" by magnetic field $\vec{B}$.

- Adjust $\vec{E}$ and $\vec{B}$ such that of their masses

This difference is measured to be (experimentally from instrument)

$$\Delta = 0.09390032 \pm$$

$$\Delta = m(C_9H_{20}) - m(C_{10}H_8)$$

$$= 9m(^{12}C) + 20m(^1H) - 10m(^{12}C) - 8m(^1H)$$

$$\Delta = 12m(^1H) - m(^{12}C)$$

$$m(^1H) = \frac{1}{12}\left[\Delta + m(^{12}C)\right]$$

Using values of 'Δ' and $m(^{12}C) = 12.000000\,u$

$$m(^1H) = 1.0078825 \pm$$

Calculation of mass of ^{14}N:

- Set the apparatus (spectroscope) for mass = 28 u
- Measure the difference b/w the molecular masses of C_2H_4 and N_2.

where $C_2H_4 = 2\times12 + 4\times1 = 28\,u$

$N_2 = 2\times14 = 28\,u$

Both the molecules follow the same trajectory but at the end strike at slightly different points with different radii.

→ Slight difference is due to the measurement of their masses.

Their difference is measured to be (experimentally from instrument).

$$\Delta = 0.025152196 \pm$$

$$\Delta = m(C_2H_4) - m(N_2)$$

So we have,

$$\Delta = 2m(^{12}C) + 4m(^{1}H) - 2m(^{14}N)$$

$$2m(^{14}N) = 2m(^{12}C) + 4m(^{1}H) - \Delta$$

$$m(^{14}N) = \frac{1}{2}\left[2m(^{12}C) + 4m(^{1}H) - \Delta\right]$$

Using values of 'Δ', $m(^{12}C)$ and $m(^{1}H)$, we get $m(^{14}N) = 14.00307396u$

→ **Nuclear Reaction Method:**

- Used to measure masses of unstable nuclides (having very short lifetime)

∵ ^{12}N → very unstable isotope of nitrogen. Similarly ^{3}H is very unstable.

- Perform a nuclear reaction (which completes in a very short time) 10^{-16}–10^{-22} sec

Consider the nuclear reaction

$$\underset{\text{projectile (unstable)}}{x} + \underset{\text{Target}}{X} \rightarrow \underset{\text{outgoing Particle}}{y} + \underset{\text{Recoiled nucleus (unknown)}}{Y}$$

X → Target at rest $T_X = 0$ (∵ K.E is zero)

x → K.E known (T_x) from accelerator

y → Its K.E can be measured from detector so T_y – known

Y → Its K.E can be calculated by using law of conservation of energy.

- Difference in K.E can be calculated, which is called Q-value.
- Use this Q-value to calculate the unknown mass.

Writing the eq. for law of conservation of energy for the nuclear reaction.

$$E_i = E_f$$

(Energy = Rest mass energy + K.E) (Rest mass of e^-s, p etc are shown in books.)

$$E_x + E_X = E_y + E_Y$$

are total energy of (x) = $E_x = m_x c^2 + T_x$

$m_x c^2 + T_x + m_X c^2 + T_X = m_y c^2 + T_y + m_Y c^2 + T_Y$

$m_x c^2 + m_X c^2 - m_y c^2 - m_Y c^2 = T_y + T_Y - T_x$

$(m_x + m_X - m_y - m_Y)\, c^2 = T_Y + T_y - T_x$ (Q-value)

Example

Calculate mass of ^{12}N using nuclear reaction method.

$$^{14}N + {}^{1}H \rightarrow {}^{12}N + {}^{3}H$$

From mass doublet experiment

$m(^{14}N) = 14.003074$, $m(^{1}H) = 1.007825u$

and $m(^{3}H) = 3.016049u$

The measured Q-value is,

Q-value = $-22.1355 \pm$ ... (MeV)

so $m(^{12}N) = m(^{1}H) + m(^{14}N) - m(^{3}H) - \frac{Q}{c^2}$

so $m(^{12}N) = 12.018613 \pm$ ...

- Main contribution to the uncertainity of the deduced mass comes from Q-value.
- Half life of $^{12}N = 0.01s$

Nuclear Binding Energy:

Before starting it, we define

Mass Defect:

As Nucleus = Protons + Neutrons (Nucleons)

It has been observed

∴ Mass of nucleus always < sum of the masses of all nucleons present in the nucleus. (experimental)

This difference is called "Mass Defect"
Let 'M_{nuc}' is the mass of a nucleus, consisting of Z protons and N neutrons then

$$\Delta m = (Zm_p + Nm_n) - M_{nuc} \rightarrow ①$$

where $N = A - Z$

Also $M_{nuc} = \underbrace{M(A,Z)}_{\text{atomic mass}} - Zm_e \rightarrow ②$

Using ② in ①

$$\Delta m = Zm_p + Nm_n - M(A,Z) + Zm_e$$
$$= Z(m_p + m_e) + Nm_n - M(A,Z)$$
$$\boxed{\Delta m = Z(m(^1H)) + Nm_n - M(A,Z)}$$

All masses $m(^1H)$, m_n, $M(A,Z)$ are used in U.

Example: Mass defect of deutrium

For deutrium $Z=1$ $N=1$
$m(^1H) = 1.00782U$ $m_n = 1.00866U$
and $M(^2_1H) = 2.014102U$ using
Δm for $^2_1H = 0.00288U$.

CHAPTER: 3 of book Q.7 (Do it yourself).

➤. The energy equivalent of the missing mass of nucleus is called Nuclear Binding Energy 'E_B'

$$E_B = \Delta m c^2$$
$$E_B = \{Zm(^1H) + Nm_n - M(A,Z)\}c^2$$

e.g $E_B(^2_1H) = (0.00288U)c^2$ using $c^2 = \frac{931.5 \text{ MeV}}{U}$

so $\boxed{E_B(^2_1H) = 2.22 \text{ MeV}}$

The minimum amount of energy which is
eded to breakup a nucleus of Z
tons and N neutrons completely, so
t they are all separated from one
other. This amount of energy is
lled Nuclear Binding energy.

Conversely,

If we start with Z protons and neutrons which are initially at rest
d completely separated from one another
in the amount of energy which is
leased when they are all brought
gether to constitute a nucleus is
called Nuclear Binding Energy.

eutron / Proton Separation Energies:

– Analogous
o the ionization energy in atomic physics.
· They tell about binding of the outermost/ valence nucleon.

- Separation energies show evidence for nuclear shell structure similar to the atomic shell structure.

Neutron Separation energy: (S_n)

– Amount of energy that is needed to remove a neutron ~~from a nucleus neutron~~ from a nucleus $^{A}_{Z}X_N$.

– It is equal to the difference in the binding energies b/w $^{A}_{Z}X_N$ and $^{A-1}_{Z}X_{N-1}$

i.e. $S_n = E_B(^{A}_{Z}X_N) - E_B(^{A-1}_{Z}X_{N-1})$

Using formula for E_B, we get

$$S_n = [Zm({}^1_1H) + Nm_n - M(A,Z) - Zm({}^1_1H) - (N-1)m_n + M(A-1,Z)]c^2$$

$$S_n = [(N-N+1)m_n - M(A,Z) + M(A-1,Z)]c^2$$

$$S_n = [M(A-1,Z) - M(A,Z) + m_n]c^2$$

or $\boxed{S_n = [m({}^{A-1}_{Z}X_{N-1}) - m({}^{A}_{Z}X_N) + m_n]c^2}$

Proton-Separation Energy (S_p)

– Amount of energy that is needed to remove a proton from a nucleus ${}^{A}_{Z}X_N$.

– It is equal to the difference in the binding energy b/w ${}^{A}_{Z}X_N$ and ${}^{A-1}_{Z-1}X_N$.

i.e $S_p = E_B({}^{A}_{Z}X_N) - E_B({}^{A-1}_{Z-1}X_N)$

Using formula for E_B, we get

$$S_p = [Zm({}^1_1H) + \cancel{Nm_n} - M(A,Z) - (Z-1)m({}^1_1H) - \cancel{Nm_n} + M(A-1, Z-1)]c^2$$

$$S_p = [m({}^1_1H) + M(A-1, Z-1) - M(A,Z)]c^2$$

or $\boxed{S_p = [m({}^{A-1}_{Z-1}X_N) - m({}^{A}_{Z}X_N) + m({}^1_1H)]c^2}$

Note: For a certain nucleus, S_p and S_n have different values. ${}^{17}_{8}O$; $S_p = 13.78$ MeV, $S_n = 4.14$ MeV

e.g in ${}^{16}_{8}O$; $S_p = 12.13$ MeV, $S_n = 15.67$ MeV

Since nuclei of the atoms are more strongly bound therefore energies of the order of the few MeV are required to break away a nucleon from the nucleus compared to only a few electron volt energy to separate or to detach an orbital electron from an atom to ionize it

<u>nding energy per nucleon: (f_B)</u> - The average energy needed to break off just one nucleon from the nucleus

$$\frac{B.E}{\text{nucleon}} = \frac{E_B}{\text{Total no. of nucleons}}$$

$$f_B = \frac{E_B}{A} = \frac{\{Zm(^1H) + Nm_n - M(A,Z)\}c^2}{A}$$

Example:

$E_B(^2_1H) = 2.224$ MeV $\quad f_B^{(^2_1H)} = 1.112$ MeV

$E_B(^4_2He) = 28.3$ MeV $\quad f_B^{(^4_2He)} = 7.705$ MeV

$E_B(^{16}_8O) = 127.62$ MeV $\quad f_B^{(^{16}_8O)} = 7.98$ MeV

i.q CH#3 "(Do it Yourself)

→ f_B of different nuclei represent the relative strengths of their binding.

× Q.7 From the known masses of ^{15}O and ^{15}N, compute the difference in the binding energy

Sol:

Binding energy of ^{15}O

$E_B = \{Zm_a(^1H) + Nm_n - M(A,Z)\}c^2$

$= \{8(1.007825) + 7(1.00866) - 15.003065\}c^2$

$= \{8(1.007825) + 7(1.00866) - 15.003065\} \times 931.5$ MeV

$= 111.924825$ MeV

Binding energy of ^{15}N.

$E_B = \{Zm(^1H) + Nm_n - M(A,Z)\}c^2$

$E_B = \{7(1.007825) + 8(1.00860) - 15.000109\} \times 931.5$ MeV

$= 0.123946 \times 931.5$ MeV

$= 115.455699$

Difference

Q.9) Comp...
the bi...
(a) ^7Li
N = ...
M...
Z = 3
$E_B = \{$...
$= \{$...
=
=
Binding

(b) ^{20}Ne
Z = ...

(c) ^{56}Fe

Difference :

$$\boxed{3.5313165 \text{ MeV}}$$

Q.9) Compute the total binding energy and the binding energy per nucleon for

(a) ^{7}Li

$N = 4$, $m_n = 1.00866u$

$M(A,Z) = 7.016003u$

$Z = 3$, $m(^1H) = 1.007825u$

$E_B = \{Zm(^1H) + Nm_n - M(A,Z)\}c^2$

$= \{3(1.007825) + 4(1.00866) - 7.016003\} \times 931.5 \text{ MeV}$

$= 0.42112 \times 931.5$

$= 39.227328 \text{ MeV}$

Binding energy per nucleon $= f_B = \frac{E_B}{A}$

$$f_B = \frac{39.227328}{7}$$

$$\boxed{f_B = 5.603904 \text{ MeV}}$$

(b) ^{20}Ne

$Z = 10$, $N = 10$, $m_n = 1.00866u$

$E_B = \{Zm(^1H) + Nm_n - M(A,Z)\}c^2$

$= \{10(1.007825) + 10(1.00866) - 19.992436\}c^2$

$E_B = 160.60 \text{ MeV}$

so,

$$f_B = \frac{E_B}{A} = \frac{160.60}{20}$$

$$\boxed{f_B = 8.03 \text{ MeV}}$$

(c) ^{56}Fe

$E_B = \{Zm(^1H) + Nm_n - M(A,Z)\}c^2$

$= \{26(1.007825) + (1.00866)(30) - 55.9349\}$

$\times 931.5 \ldots\ldots$

$E_B = 478.74 \text{ MeV}$

$f_B = \frac{E_B}{A} = \frac{478.74}{56}$

$\boxed{f_B = 8.54 \text{ MeV}}$

d) ^{235}U

$E_B = \{Zm(^1H) + Nm_n - M(A,Z)\}c^2$

$E_B = \{92(1.007825) + 143(1.00866) - 235.043924\} \times 931.5 \text{ MeV}$

$E_B = 1783.22 \text{ MeV}$

so,

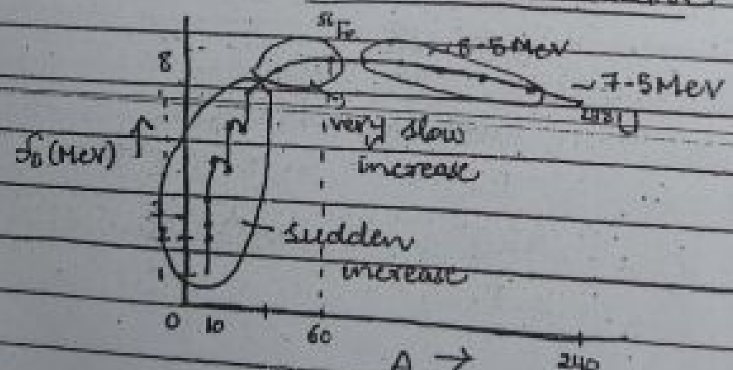
$f_B = \frac{E_B}{A}$

$f_B = \frac{1783.22}{235}$

$\boxed{f_B = 7.588 \text{ MeV}}$

Binding Energy per Nucleon (f_B):

'f_B' vs 'A' curve.



Understanding f_B vs A curve:

ⓐ → f_B for very light nuclei is very small

→ It rises rapidly with 'A' attaining a value of ~8 MeV/nucleon for A~20

→ It then rises slowly with 'A' and attains a max. value of 8.7 MeV/nucleon at $A \sim 56$ (Fe).

(a) → For $A > 56$, it decreases slowly.

(b) → For $20 < A < 180$, the variation of 'f_B' is very slight (it is within 10% of 8 MeV/nucleon).

→ Approximately constant in this region having a mean value of ~ 8.5 MeV/nucleon.

(c) → For heavy nuclei ($A > 180$), f_B decreases with 'A'. For the heaviest nuclei $f_B \sim 7.5$ MeV/nucleon.

(d) For very light nuclei, rapid fluctuations in the values of f_B (e.g. peaks observed at ${}^4_2He$, ${}^8_4Be$, ${}^{12}_6C$, ${}^{16}_8O$)

even-even nuclei A-Z

→ The appearance of peaks shows greater stability of corresponding nuclei in their immediate neighbourhood.

(e) → The curve suggests that we gain energy in two ways

(i) for $A < 60$ via fusion

(ii) for $A > 60$ via fission

Q. Why do we get this particular form of 'f_B' vs 'A curve'?
(What are the factors which affect the stability of a nucleus)?

1. It depends upon
- no. of nucleons present in the nucleus (either even or odd etc)
- interactions b/w nucleons
  ↳ There are two types of interactions b/w nucleons.
i) Coulomb Repulsion force b/w protons
ii) Strong Nuclear force b/w protons and Neutrons.

→ To understand f_e VS A curve, we use 'Liquid Drop Model'

- It was first proposed by N. Bohr in 1937.
- Applied by C.F von Weizsacker and H.A. Bethe to develop a 'Semi Empirical Mass formula' for the B.E of the nuclei

Semi- Empirical Mass Formula: (V.V.Imp)
- It was developed by using Liquid Drop Model.

→ Liquid Drop Model:

Similarities b/w nucleus and drop of a liquid.

{Add points from P#47 of given copy}

BASIC Assumptions of Liquid Drop Model

According to liquid drop model a nucleus is like a drop of liquid which is "incompressible" and "spherical"

Incompressible: It means density remains constant i.e (no. of nucleus in a unit volume remains constant)

$$\frac{A}{V} = \text{constant} \quad (\Rightarrow V \propto A)$$

$$A = V \cdot \text{constant}$$

$$A = \left(\frac{4}{3}\pi R^3\right) \text{constant}$$

$$A = R^3 \left(\frac{4}{3}\pi \times \text{constant}\right)$$

$$A = R^3 (\text{constant}).$$

$$A^{1/3} = R(\text{constant})^{1/3}$$

$$R = \frac{1}{\text{constant}} A^{1/3}$$

$$\boxed{R = R_0 A^{1/3}}$$

where $R_0 = 1.2 \times 10^{-15}$ m $R_0 = 1.2$ fm

Semi-Empirical Mass formula

- Also known as (Bethe-Weizsacker) formula.
- Based on Liquid Drop Model (LDM)
- To calculate nuclear masses (or nuclear B.E)

As we have

$$E_B = [Zm(^1H) + Nm_n - M(A,Z)]c^2$$

where $M(A,Z) = M_{nuc} + Zm_e$

or $E_B = [Zm_p + Nm_n - M_{nuc}]c^2 \rightarrow (2)$

or $M_{nuc} = Zm_p + Nm_n - \frac{E_B}{c^2} \rightarrow (3)$

where E_B can be given as a sum of five terms.

$$E_B = \underset{\text{Volume term}}{E_V} + \underset{\text{surface term}}{E_S} + \overset{\text{coulomb term}}{E_C} + \overset{\text{Asymmetric term}}{E_{Asym}} + \underset{\text{pairing term}}{E_P}$$

where these terms depend on no. of nucleons, interaction b/w nucleons

erivation of Semi Empirical Mass Formula (SEMF):

- Based on LDM

$$E_B = \underbrace{E_v + E_s + E_c}_{\downarrow} + E_{asy} + E_p$$

their origin due to interactions (strong nuclear, coulomb) among nucleons

$$E_B = a_v A - a_s A^{2/3} - a_c Z(Z-1) A^{-1/3} + \underbrace{E_{asy} + E_p}$$

Asymmetry Energy Term (E_{asym})

→ In the light nuclei, there is tendancy of $N \approx Z$ (symmetry)

In such nuclei (light), if $N = Z$, they found to be more stable than their neighbours where $N \neq Z$ (Asymmetry)

Due to this asymmetry, there will be a reduction in the binding energy by an amount 'E_a' proportional to $(N-Z)^2$

→ Please note that, in case of heavier nuclei, usually there is neutron excess.

Note:
In the heavier nuclei, increase in no. of protons tend to weaken the B.E due to Coulomb Repulsive force b/w them (long-range force). So some extra neutrons must be present to provided additional n-n bonds to compensate for this)

→ So the effect of assymetry term should decrease for heavier nuclei

$$E_{asym} \propto \frac{1}{A}$$

so overall $E_{asym} \propto -\frac{(N-Z)^2}{A}$

$$E_{asym} \propto -\frac{(A-2Z)^2}{A} \quad \text{using } N = A - Z$$

$$E_{asym} = -\frac{a_{sym}(A-2Z)^2}{A}$$

Pairing Energy Term: (E_p)

Note

Even A nuclei	(Z-even, N-even) e-e nuclei
	(Z-odd, N-odd) o-o nuclei
Odd A nuclei	(Z-even, N-odd) e-o nuclei
	(Z-odd, N-even) o-e nuclei

- A close study of B.E of different nuclei shows that e-e nuclei are more strongly bound (more stable - more abundant).
- e-o, o-e nuclei have B.E. less than e-e nuclei (less stable - less abundant than e-e nuclei).
- o-o nuclei less strongly bound (least stable - least abundant)

Why this happens?

- This effect arises due to the pairing of the nucleons of the same type with opposite spin.
- Such pairing tends to increase the strength of binding.

Maximum for e-e nuclei
(where both types of nucleons (p,n) are paired)

In e-o or o-e nuclei
* one unpaired (p or n)

In o-o nuclei
two unpaired nucleons (1p & 1n)

The pairing effect is incorporated by pairing energy term and it is taken to be

$E_P = +a_P A^{-3/4}$ for e-e nuclei

& $E_P = 0$ for e-o, o-e nuclei

$E_P = -a_P A^{-3/4}$ for o-o nuclei

So total SEMF

$$E_B = a_v A - a_s A^{2/3} - a_c Z(Z-1) A^{-1/3} - a_{asym}\frac{(A-2Z)^2}{A}$$

$$\pm\, a_P A^{-3/4} \begin{cases} + & \text{for e-e} \\ 0 & \text{for e-o, o-e} \\ - & \text{for o-o} \end{cases}$$

Using this 'E_B' the mass can be calculated from this formula

$$M(A,Z) = Zm(^1H) + Nm_n - \frac{E_B}{c^2}$$

→ The Constants

$a_v, a_s, a_c, a_{asym}, a_p$ were adjusted to give the best agreement with the experimental results.

$a_v = 15.5\,MeV$

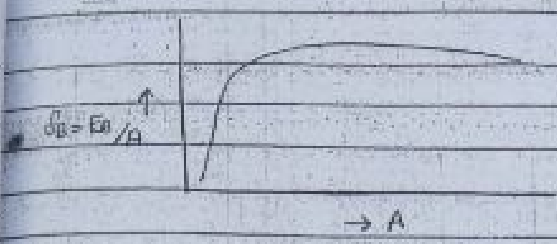
$a_s = 16.8\,MeV$

$a_c = 0.72\,MeV$

$a_{asy} = 23\,MeV$

$a_p = 34\,MeV$

$\boxed{\text{Q 8 → Q 13, Ex. Ch \# 3}}$



theoretical results for f_B vs A

Limitations of SEMF

The comparison of f_B vs curve (the experimental one and the one calculated by SEMF) shows that the extra stability of certain nuclei where the no. of protons or neutrons are 2, 8, 20, 28, 50, 82, or 126 not explained by this formula based on LDM.

Applications of Semi Empirical Mass Formula (SEMF) : (V. Imp).

→ **Alpha decay :** SEMF can be used to check which nuclei are stable against α-decay and which one can disintegrate via α-decay.

→ **Stability of nuclei against β-decay :** SEMF can be used to predict the isobar which does not undergo β-decay (most stable isobar).

① **Alpha decay (α-disintegration energy) :**

Consider a nucleus ${}^{A}_{Z}X_{N}$, which undergoes α-decay into the nucleus ${}^{A-4}_{Z-2}Y_{N-2}$, then

$${}^{A}_{Z}X_{N} \rightarrow {}^{A-4}_{Z-2}Y_{N-2} + {}^{4}_{2}He_{2}\,(\alpha) \rightarrow ①$$

The α-disintegration energy (Q_α) can be calculated using

Q_α = difference in the rest-mass energies of the reactants and products

$$= m({}^{A}_{Z}X_{N})c^2 - \{m({}^{A-4}_{Z-2}Y_{N-2}) + m({}^{4}_{2}He_{2})\}c^2$$

$$Q_\alpha = \{m({}^{A}_{Z}X_{N}) - m({}^{A-4}_{Z-2}Y_{N-2}) - m({}^{4}_{2}He_{2})\}c^2 \rightarrow ②$$

$$m({}^{A}_{Z}X_{N}) = Zm({}^{1}H) + Nm_n - \frac{E_B(A,Z)}{c^2}$$

$$m({}^{A-4}_{Z-2}Y_{N-2}) = (Z-2)m({}^{1}H) + (N-2)m_n - \frac{E_B(A-4,Z-2)}{c^2}$$

$$m({}^{4}_{2}He_{2}) = 2m({}^{1}H) + 2m_n - \frac{E_B(4,4)}{c^2}$$

where E_B's can be calculated by using SEMF.

Putting these in eq. ② we get

$$Q_\alpha = \{Zm({}^{1}H) + Nm_n - E_B(A,Z)/c^2 - Zm({}^{1}H) + 2m({}^{1}H) - Nm_n + 2m_n + E_B(A-4,Z-2)/c^2 - 2m({}^{1}H) - 2m_n + E_B(4,2)\} \times c^2$$

$$Q_\alpha = E_B(4,2) + E_B(A-4,Z-2) - E_B(A,Z)$$

B.E for ${}^{4}_{2}He$ is, $E_B({}^{4}_{2}He) = 28.3$ MeV

so $Q_\alpha = 28.3\text{ MeV} + E_B(A-4,Z-2) - E_B(A,Z) \rightarrow ③$

Now using semi empirical mass formula (neglecting pairing energy term E_P)

$$E_B(A,Z) = a_v A - a_s A^{2/3} - a_c \frac{Z^2}{A^{1/3}} - a_{sym}\frac{(A-2Z)^2}{A}$$

$$E_B(A-4,Z-2) = a_v(A-4) - a_s(A-4)^{2/3} - a_c\frac{(Z-2)^2}{(A-4)^{1/3}}$$

$- a_{asym} \dfrac{[(A-4)-2(Z-2)]^2}{A-4}$

...tting these values of E_B in eq. (B)

$= 28.3\,MeV + a_v(A-4) - a_s(A-4)^{2/3} - a_c\dfrac{(Z-2)^2}{(A-4)^{1/3}}$

$- a_{asym}\dfrac{[(A-4)-2(Z-2)]^2}{A-4} - a_vA + a_sA^{2/3} + a_c\dfrac{Z^2}{A^{1/3}}$

$+ a_{asym}\dfrac{(A-2Z)^2}{A}$

$= 28.3\,MeV + \cancel{a_vA} - 4a_v - a_c\dfrac{(Z-2)^2}{(A-4)^{1/3}} - a_{asy}\dfrac{[(A-4)-2(Z-2)]^2}{A-4}$

$- \cancel{a_vA} + a_sA^{2/3} - a_s(A-4)^{2/3} + a_c\dfrac{Z^2}{A^{1/3}} + a_{asym}\dfrac{(A-2Z)^2}{A}$

$= 28.3\,MeV - 4a_v + a_s\left[A^{2/3} - (A-4)^{2/3}\right]$

$+ a_c\left[\dfrac{Z^2}{A^{1/3}} - \dfrac{(Z-2)^2}{(A-4)^{1/3}}\right] + a_{asy}\left[\dfrac{(A-2Z)^2}{A} - \dfrac{[(A-4)-2(Z-2)]^2}{A-4}\right]$

$\rightarrow$ (A)

...onsidering $a_s\left[A^{2/3} - (A-4)^{2/3}\right]$

$= a_sA^{2/3}\left[1 - \left(1 - \dfrac{4}{A}\right)^{2/3}\right]$

$= a_sA^{2/3}\left[1 - \left\{1 + \left(\dfrac{2}{3}\right)\left(\dfrac{-4}{A}\right) + \dots\right\}\right]$ ∵ Binomial Expansion

$= a_sA^{2/3}\left[\cancel{1} - \cancel{1} + \dfrac{8}{3A}\right]$ At large A $4/A^2$ is even smaller so neglecting higher terms.

$= a_sA^{2/3}\left[\dfrac{8}{3A}\right]$

$a_s\left[A^{2/3} - (A-4)^{2/3}\right] = a_sA^{-1/3}\dfrac{8}{3} = \dfrac{8}{3}a_sA^{-1/3} \rightarrow$ (4)

...nsidering, $a_c\left[\dfrac{Z^2}{A^{1/3}} - \dfrac{(Z-2)^2}{(A-4)^{1/3}}\right]$

$= a_c\left[Z^2A^{-1/3} - (Z-2)^2(A-4)^{-1/3}\right]$

$= a_cA^{-1/3}\left[Z^2 - (Z-2)^2\left(1 - \dfrac{4}{A}\right)^{-1/3}\right]$

$$\left[\frac{Z^2}{A^{1/3}} - \frac{(Z-2)^2}{(A-4)^{1/3}}\right] = a_c A^{-1/3}\left[Z^2 - (Z^2-4Z+4)\left(1+\frac{4}{3A}+\dots\right)\right]$$

∵ Neglecting higher terms. Also neglect $1/Z^2$ terms, so

$$= a_c A^{-1/3} Z^2\left[1-\left(1-\frac{4}{Z}+\underbrace{\cancel{\frac{4}{Z^2}}}_{\text{neglect}}\right)\left(1+\frac{4}{3A}\right)\right]$$

$$= a_c A^{-1/3} Z^2\left[1-\left(1-\frac{4}{Z}\right)\left(1+\frac{4}{3A}\right)\right]$$

$$= a_c A^{-1/3} Z^2\left[1-\left(1+\frac{4}{3A}-\frac{4}{Z}-\frac{16}{3AZ}\right)\right]$$

$$= a_c A^{-1/3} Z^2\left[\cancel{1}-\cancel{1}-\frac{4}{3A}+\frac{4}{Z}+\frac{16}{3AZ}\right]$$

$$= a_c A^{-1/3} Z^2\left[\frac{4}{Z}-\frac{4}{3A}\right] \qquad \frac{16Z}{3A} \rightarrow \text{smallest term ignore it.}$$

$$= 4a_c A^{-1/3}\left[\frac{Z^2}{Z}-\frac{Z^2}{3A}\right]$$

$$a_c\left[\frac{Z^2}{A^{1/3}} - \frac{(Z-2)^2}{(A-4)^{1/3}}\right] = 4a_c A^{-1/3} Z\left[1-\frac{Z}{3A}\right] \rightarrow (5)$$

Considering $a_{asy}\left[\frac{(A-2Z)^2}{A} - \frac{[(A-4)-2(Z-2)]^2}{A-4}\right]$

$$= a_{asy}\left[\frac{(A-2Z)^2}{A} - \frac{(A-\cancel{4}-2Z+\cancel{4})^2}{A-4}\right]$$

$$= a_{asym}\left[\frac{(A-2Z)^2}{A} - \frac{(A-2Z)^2}{A-4}\right]$$

$$= a_{asym}(A-2Z)^2\left[\frac{A-4-A}{A(A-4)}\right]$$

$$= -4a_{asym}\frac{(A-2Z)^2}{A(A-4)} \rightarrow (6)$$

Using eq. (4), (5) and (6) in eq. (A)

$$Q_\alpha = 28.3\,\text{MeV} - 4a_v + \frac{8}{3}a_s A^{-1/3} + 4a_c Z\left(1-\frac{Z}{3A}\right)A^{-1/3} - \frac{4a_{asym}(A-2Z)^2}{A(A-4)} \rightarrow (B)$$

process (α-disintegration) occurs if $Q_\alpha > 0$

We can find A for which α-disintegration is observed spontaneously.

Using values of a_v, a_s, a_c, a_{asym} it gives $Q_\alpha > 0$ for $A > 160 \rightarrow \alpha$-disintegration.

Missing Lecture:

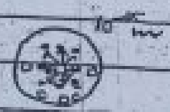
① Volume Energy Term (E_v).

The basis of the volume term is "strong nuclear force" (short range force $\sim 10^{-15}$ m)

Attractive force – Contribution in the binding of nucleons inside the nucleus.)

This term is

$$E_v \propto A$$



$$E_v = a_v A$$

where a_v – constant of proportionality

A = mass of nucleus

⇒ A given nucleon may only interact strongly with its closest neighbour.

→ As nuclear density is roughly constant.

→ Each nucleon has about the same no of nearest neighbours

and hence each nucleon contributes roughly the same amount of B.E.

So, $E_B = a_v A + E_s + E_c + E_{asy} + E_p$

Surface Energy Term (E_s):

→ The surface energy term is a correction to volume term.

Q. Why needed:

→ The nucleons at the surface

nucleus are surrounded by fewer
-hbours (as compared to those nucleons
-ich are at the centre.)
These nucleons do not contribute to
- quite as much as those in
- centre so they are less tightly bound
B.E is reduced by an amount
-oportional to the surface area of
-cleus of radius R.

As surface area $= 4\pi R^2$

so, $E_s \propto -4\pi R^2$

$$E_s \propto -4\pi (R_0 A^{1/3})^2$$

$$E_s \propto -4\pi R_0^2 A^{2/3}$$

$$E_s = a_s A^{2/3}$$

where $a_s \rightarrow$ Constant of proportionality

Now,

$$E_B = a_v A - a_s A^{2/3} + E_c + E_{asy} + E_P$$

③ Coulomb Energy Term (E_c)

→ This term consider the contribution from the coulomb repulsion b/w protons

→ Negative contribution in B.E (as the repulsion b/w protons make the nucleus less-tightly bound.)

→ Since Coulomb force - A long range force thats why acts over the entire volume of the nucleus.

→ Each proton repels all of the other protons, so this term should be proportional to $\frac{Z(Z-1)}{2} \sim \frac{Z^2}{2}$

→ As the work done in assembling a nucleus of Z protons is

$$W = \frac{3/5\ Z^2 e^2}{4\pi\epsilon_0 R}$$

Since P.E is the -ve of total work done, the Coulomb energy for the charged sphere is

$$E_c \propto -\frac{3/5\ Z^2 e^2}{4\pi\epsilon_0 R}$$

$$E_c = -a_c \frac{Z^2}{A^{1/3}} \quad \text{or} \quad E_c = -\frac{a_c Z(Z-1)}{A^{1/3}}$$

So,

$$E_B = a_v A - a_s A^{2/3} - a_c \frac{Z^2}{A^{1/3}} + E_{asy} + E_p$$

Applications of SEMF

→ Stability of nuclei against β-decay.
(Predict the most stable isobar)

- In nuclear $\bar{\beta}$-decay

$${}^{A}_{Z}X_{N} \xrightarrow{\beta^-} {}^{A}_{Z+1}X_{N-1} + e^- + \bar{\nu}_e$$

(fundament β-decay
$n \to p + e^- + \bar{\nu}_e$)

- In nuclear β^+-decay

$${}^{A}_{Z}X_{N} \xrightarrow{\beta^+} {}^{A}_{Z-1}X_{N+1} + e^+ + \nu_e \quad (p \to n + e^+ + \nu_e)$$

∵ This process is not possible for isolated proton.

∴ Mass no. of parent and daughter nucleus same (Isobars).

Since (No. of protons and neutrons different in parent & daughter nuclei) ⇒ (B.E of them is different)

Isobaric family — A family of different nuclides for which 'A' is same is called as isobaric family.

Let say X, Y, Z, L, M, N, ... etc are nuclei with same 'A', but different 'Z', they are said to make an isobaric family.

$$X \xrightarrow{\beta} Y + e^- + \bar{\nu}_e$$
$$\quad \lfloor \xrightarrow{\beta} Z + e^- + \bar{\nu}_e$$
$$- - - - - -$$

We have to predict that isobar from the family, which is most stable (i.e. which will not undergo β-decay).

Consider SEMF

$$M(A,Z) = Zm({}^1H) + Nm_n - \frac{E_B}{c^2} \rightarrow ①$$

where $E_B = a_v A - a_s A^{2/3} - a_c \frac{Z^2}{A^{1/3}} - a_{asym}\frac{(A-2Z)^2}{A}$

$$\pm a_p A^{-3/4} \rightarrow ②$$

Initially, we proceed our calculations without including the effect of pairing term (for simplicity).

Using (2) in (1)

$$M(A,Z) = Zm({}^1H) + Nm_n - \left[a_v A - a_s A^{2/3} - a_c \frac{Z^2}{A^{1/3}} - a_{asym}\frac{(A-2Z)^2}{A}\right]\frac{1}{c^2}$$

Using $N = A - Z$

$$M(A,Z) = Zm({}^1H) + Am_n - Zm_n - \frac{1}{c^2}\left[a_v A - a_s A^{2/3} - a_c \frac{Z^2}{A^{1/3}} - a_{asym}\frac{(A^2 - 4AZ + 4Z^2)}{A}\right]$$

$$M(A,Z) = \left\{ A\left(m_n - \frac{a_V}{c^2} + \frac{1}{c^2} a_{ASYM} \frac{A^2}{A} \right) + \frac{a_S}{c^2} A^{2/3} \right\}$$
$$- (m_n - m(^1H))Z - \frac{4a_{ASYM} AZ}{AC^2} + \frac{1}{A}\left(\frac{a_C}{c^2} A^{2/3} + \frac{4a_{ASYM}}{c^2} \right) Z^2$$

$$M(A,Z) = \left\{ A\left(m_n - \frac{a_V}{c^2} + \frac{1}{c^2} a_{ASYM} \right) + \frac{a_S}{c^2} A^{2/3} \right\}$$
$$- (m_n - m(^1H))Z - \frac{4a_{ASYM} Z}{c^2} + \frac{1}{A}\left(\frac{a_C}{c^2} A^{2/3} + \frac{4a_{ASYM}}{c^2} \right) Z^2$$

$$\therefore M(A,Z) = f_A + pZ + qZ^2 \rightarrow ③$$

where $f_A = \left\{ A\left(m_n - \frac{a_V}{c^2} + \frac{1}{c^2} a_{ASYM} \right) + \frac{a_S}{c^2} A^{2/3} \right\}$

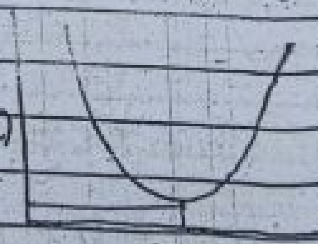
$$p = -(m_n - m(^1H)) - \frac{4a_{ASYM}}{c^2}$$

& $q = \frac{1}{A}\left(\frac{a_C}{c^2} A^{2/3} + \frac{4a_{ASYM}}{c^2} \right)$

Eq. ③ quadratic in 'Z'

↓

Parabola for given 'A'



For a given value of A (i.e. for a particular isobaric family) → There will be some particular value of 'Z' for which M(A,Z) is minimum ⇒ E_B is maximum for that particular nucleus → Most stable isobar of that family

To determine 'Z' for which M(A, Z) is minimum:

For a particular isobaric family (i.e. fixed A)

* Differentiate eq. ③ w.r.t 'Z'

$$\left(\frac{\partial M}{\partial Z}\right)_A = p + 2qZ \rightarrow ⑤$$

putting $\left(\frac{\partial M}{\partial Z}\right)_A = 0$, one can obtain the lowest point of parabola (where M(A, Z) is minimum).

Let say we take $Z = Z_A$ for that nucleus corresponding to min M(A, Z)

$$p + 2q Z_A = 0$$

$$\boxed{Z_A = \frac{-p}{2q}}$$

Using values of all constants, we have

$$\boxed{Z_A = \frac{A}{1.98 + 0.015 A^{2/3}}}$$

Since $M(A, Z) = f_A + pZ + qZ^2$

to calculate mass of the most stable isobar, use $Z = Z_A$

$$M(A, Z_A) = f_A + pZ_A + qZ_A^2$$

where $Z_A = \frac{-p}{2q}$

$$M(A, Z_A) = f_A - \frac{p^2}{2q} + \frac{qp^2}{4q^2}$$

$$\Rightarrow \boxed{M(A, Z_A) = f_A - \frac{p^2}{4q}}$$

Effect of evenness or oddness (or effect of pairing term) on the Prediction of most stable isobar:

When pairing energy term is taken into account, then the mass parabolas for the different isobars, defined by eq. ③

$$M(A,Z) = f_A + PZ + QZ^2$$

fall into two categories depending on whether

A = odd or if A = even

(o-e, e-o nuclei) (o-o, e-e nuclei)

For odd 'A'

Pairing energy term $E_p = 0$

we get a single parabola.

For even 'A'

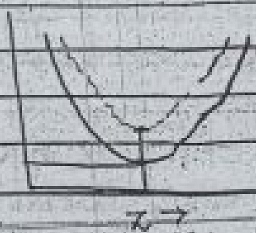
we get 2 parabolas for the same 'A' (for a given isobaric family).

1- for (e-e) nuclei - even-Z

(since E_p added)

2- for (o-o) nuclei - odd-Z

(Since E_p subtracted).



For same 'A' the parabola for o-o nuclei lies above that for e-e nuclei by an amount $2E_p$

Examples

For odd A = 135 Fig 9.2 Ghoshal

For even A = 102

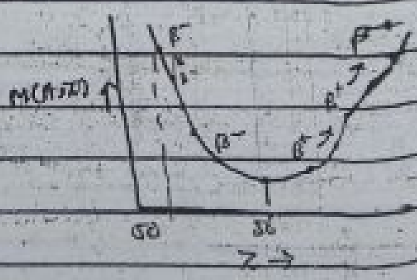
Example:

Let A = 135

Z_A, using formula

$Z_A = 56.85$ (theoretically)

$^{135}_{56}Ba$



the mass parabola for this
lar isobaric family in Fig. ①
For this isobaric family we
ve $^{135}_{56}Ba$ is the most stable isobar
in fig.
The nucleus on either side of the
table isobars are unstable.
These on the lower Z side $(Z<56)$
re β^- (decay) active.
Those on the higher Z-side $(Z>56)$
re β^+ active.

left side

$$^{135}_{52}Te \xrightarrow{\beta^-} {}^{135}_{53}I \xrightarrow{\beta^-} {}_{54}Xe \xrightarrow{\beta^-} {}_{55}Cs \xrightarrow{\beta^-} {}_{56}Ba$$

Right side

$$Ce \xrightarrow{\beta^+} La \xrightarrow{\beta^+} Ba$$

CHAPTER: 3
Excercise (KRANE)

Q-10) For each of the following nuclei use the semiempirical mass formula to compute the total binding energy and Coulomb Energy:

(a) ^{21}Ne

→ **Volume Energy Term**

$$E_v = a_v A$$

for ^{21}Ne $A = 20.991$ and $a_v = 15.5$ MeV

so $E_v = (15.5)(20.993842)(MeV)$ ①

so

$$\boxed{E_v = 325.40\ \text{MeV}}$$

for $A=21$ $E_v = 325.5$ MeV

~~so M = 931.502 MeV~~
~~$E_v = 325.5$~~

→ Surface Energy Term:

$$E_s = -a_s A^{2/3}$$

Here $E_s = -(16.8)(20.993843)^{2/3}$ MeV

$E_s = -127.89$ MeV

→ Coulomb Energy Term:

$$E_c = -a_c \frac{Z^2}{A^{1/3}}$$

$$E_c = -(0.72)\frac{(10)^2}{(20.993843)^{1/3}} \text{ MeV}$$

$E_c = -26.09$ MeV

→ Asymetric Energy Term:

$$E_{asym} = -a_{asym}\frac{(A-2Z)^2}{A}$$

$$E_{asym} = -(23)\frac{(21-20)^2}{(20.993843)^{1/3}}$$

$E_{asym} = -1.095$ MeV

Here $Z = 10$ & $N = 11$ so $E_p = 0$

so,

Total binding energy is,

$E_B = E_v + E_s + E_c + E_{asym}$

$= [325.40 - 127.89 - 26.09 - 1.095]$ MeV

$\boxed{E_B = 170.295 \text{ MeV}}$

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(b) ^{57}Fe

$E_V = a_V A \Rightarrow E_V = (15.7)(57)$ MeV

so $\boxed{E_V = 894.9 \text{ MeV}}$

$E_S = -a_S A^{2/3} \Rightarrow E_S = -(16.8)(57)^{2/3}$ MeV

so $\boxed{E_S = -248.75 \text{ MeV}}$

$E_C = -a_C \dfrac{Z^2}{A^{1/3}} \Rightarrow E_C = \dfrac{-(0.72)(26)^2}{(57)^{1/3}}$

$\boxed{E_C = -126.487 \text{ MeV}}$

$E_{ASYM} = -a_{ASYM} \dfrac{(A-2Z)^2}{A}$

$= -\dfrac{(23)(57-52)^2}{57}$

$\boxed{E_{ASYM} = -10.08 \text{ MeV}}$

Here $Z = 26$ & $N = 31$ so $E_P = 0$.

so,

$E_B = E_V + E_S + E_C + E_{ASYM}$

$= 894.9 - 248.75 - 126.487 - 10.08$ MeV

$\boxed{E_B = 509.583 \text{ MeV}}$

(c) ^{209}Bi

$E_V = a_V A \Rightarrow E_V = (15.7)(209)$ MeV

so $\boxed{E_V = 3,281.3 \text{ MeV}}$

$E_S = -a_S A^{2/3} \Rightarrow E_S = -(16.8)(209)^{2/3}$ MeV

$\boxed{E_S = -591.45 \text{ MeV}}$

$E_c = -a_c \frac{Z^2}{A^{1/3}} \rightarrow E_c = -\frac{(0.72)(83)^2}{(209)^{1/3}}$ MeV

$\boxed{E_c = -835.959 \text{ MeV}}$

$E_{ASYM} = -a_{ASYM} \frac{(A-2Z)^2}{A} \Rightarrow E_{ASYM} = -\frac{(23)(209-166)^2}{209}$

$\boxed{E_{ASYM} = -208.478 \text{ MeV}}$

Here $Z = 83$ & $N = 126$ so $E_P = 0$

so,

$E_B = E_V + E_S + E_C + E_{ASYM}$

$= 3281.3 - 591.45 - 835.959 - 208.478$

$\boxed{E_B = 1650.413 \text{ MeV}}$

(d) ^{256}Fm

$E_V = a_V A \Rightarrow E_V = (15.7)(256)$ MeV

$\boxed{E_V = 4019.2 \text{ MeV}}$

$E_S = -a_S A^{2/3} \Rightarrow E_S = -(16.8)(256)^{2/3}$ MeV

so

$\boxed{E_S = -677.08 \text{ MeV}}$

$E_C = -a_c \frac{Z^2}{A^{1/3}} \Rightarrow E_C = -\frac{(0.72)(100)^2}{(256)^{1/3}}$ MeV

$\boxed{E_C = -1134.14 \text{ MeV}}$

$E_{ASYM} = -a_{ASYM} \frac{(A-2Z)^2}{A}$

$= -\frac{(23)(256-200)^2}{256}$ MeV

$\boxed{E_{ASYM} = -281.75 \text{ MeV}}$

Here $Z = 100$ & $N = 156$ so for

e-e nuclei $E_P = a_P A^{-3/4}$

So

$E_p = (34)(256)^{-3/4}$ MeV

so $\boxed{E_p = 0.53125 \text{ MeV}}$

so $E_B = E_V + E_S + E_C + E_{Asym} + E_p$

$= 4019.2 - 677.08 - 1134.14 - 281.75 + 0.53125$ MeV

$\boxed{E_B = 1926.76 \text{ MeV}}$

Q.11) Compute mass defects of

(a) ^{32}S

$\Delta m = Zm(^1H) + Nm_n - M(A,Z)$

so,

$\Delta m = [(16)(1.00782) + (16)(1.00866) - 31.972071]$ u

$\boxed{\Delta m = 0.291609 \text{ u}}$

(b) ^{20}F

$\Delta m = [9(1.00782) + 11(1.00866) - 19.999981]$

so,

$\boxed{\Delta m = 0.165659 \text{ u}}$

(c) ^{238}U

$\Delta m = [92(1.00782) + 146(1.00866) - 238.050785]$

so,

$\boxed{\Delta m = 1.933015 \text{ u}}$

Q.13) Evaluate the neutron separation energies of,

^7Li

$S_n = [m(^{A-1}_{Z}X_{N-1}) - m(^{A}_{Z}X_N) + m_n]c^2$

$S_n = [m(^6Li_3) - m(^7Li_4) + m_n] \times 931.5$ MeV

$S_n = [6.015121 - 7.016003 + 1.008667] \times 931.5$ MeV

$\boxed{S_n = 7.245207 \text{ MeV}}$

^{91}Zr

$S_n = [m(^{90}Zr_{50}) - m(^{91}Zr_{51}) + m_n] \times 931.5$ MeV/u

$S_n = [89.904703 - 90.905644 + 1.00866] \times 931.5$ MeV

$\boxed{S_n = 7.1902 \text{ MeV}}$

^{236}U

$S_n = [m(^{235}U_{143}) - m(^{236}U_{144}) + m_n] \times 931.5$ MeV/u

$S_n = [235.043924 - 236.045563 + 1.00866]$
$\times 931.5$ MeV

$\boxed{S_n = 6.5400615 \text{ MeV}}$

Evaluate the proton separation energies of,

^{20}Ne

$S_p = [m(^{A-1}_{Z-1}X_N) - m(^A_ZX_N) + m(^1H)]c^2$

$S_p = [m(^{19}_{9}F_{10}) - m(^{20}_{10}Ne_{10}) + 1.00782] \times 931.5$ MeV/u

$S_p = [18.998403 - 19.992436 + 1.00782] \times 931.5$ MeV

$\boxed{S_p = 12.84 \text{ MeV}}$

^{55}Mn

$S_p = [m(^{54}_{24}Cr_{30}) - m(^{55}_{25}Mn_{30}) + m(^1H)]c^2$

$S_p = [53.938882 - 54.938047 + 1.00782] \times 931.5$ MeV

$\boxed{S_p = 8.062 \text{ MeV}}$

^{197}Au

$S_p = [m(^{196}_{78}Pt_{118}) - m(^{197}_{79}Au_{118}) + 1.00782] \times 931.5$ MeV

$S_p = [195.964926 - 196.966543 + 1.00782] \times 931.5$

$\boxed{S_p = 5.778 \text{ MeV}}$

(Me)asurement of Nuclear Size (Radius):

Protons + Neutrons

Scattering experiments are used to measure the size of a nucleus.

(Two) types of (ex)periments
→ Nuclear Charge Distribution
→ Nuclear Matter Distribution

Nuclear Charge Distribution:

In such (ex)periments, the Coulomb interaction of charged particle with the nucleus is measured. (T)hey give us distribution of nuclear charg(e). (E).g high energy e^--scattering experiments, muonic X-rays etc.

→ Nuclear Matter Distribution.

In some experiments e.g Rutherford scattering, α-decay experiments and pionic X-ray experiments, the strong nuclear interaction of nuclear particles is measured which gives us information about nuclear matter distribution, which can be then used to estimate the size of nuclei.

S.B Patel (Ch#4, P#114)

Rutherford Estimation of Nuclear Size :

First estimates of nuclear size (radii) were provided by Rutherford scattering experiment

From this experiment — nuclear radius estimated using classical idea

→ Qualitative Discussion : (low- energy experiment α-particles were carrying low energy)

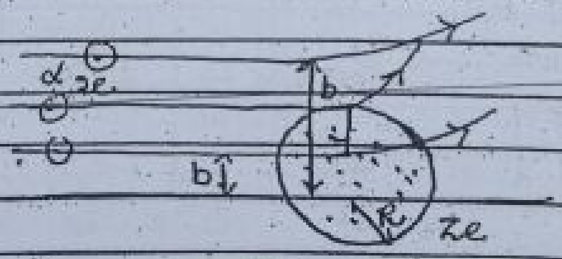
→ Let the gold nucleus in the scattering experiment has radius 'R'.

→ An α-particle's trajectory can be specified by its impact parameter 'b'.

Let us consider 3 cases ...

(i) $b > R$ (ii) $b < R$ (iii) $b \simeq R$

(i) $b > R$:



$F = k\frac{q_1 q_2}{r^2}$

Due to less coulomb repulsion, small deflection of α-particles.

(ii) $b < R$

α-particle goes through the nuclear charge distribution.

→ Nuclear charge above and below the trajectory works in opposite

opposite
irection (i-e they repel in opposite
lirection)

=> The deflection produced in
-particles is small.

(ii) $b \approx R$

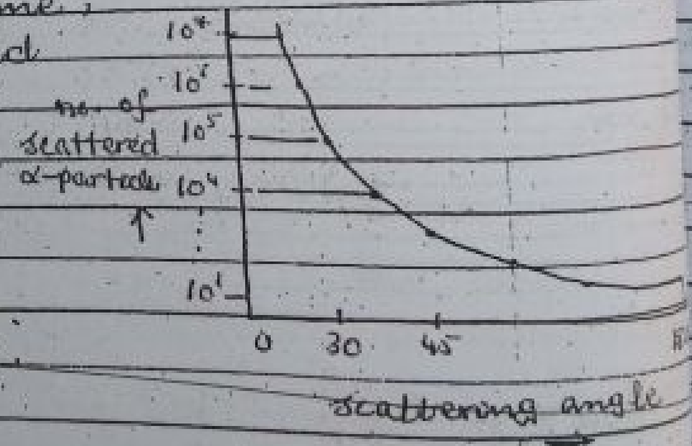
When α-particle just grazes
he nucleus, the coulomb repulsion
becomes maximum.

For $b \sim R$; the large angle
deflections are observed

Note

In fig. see the scattering results from Rutherford scattering experiment

Most of the time, deflections observed by Rutherford in the experiment were of the order of 1 radian (small-angle deflection).



→ **Quantitative Discussion:**

- Consider trajectory corresponding to a small angle deflection.

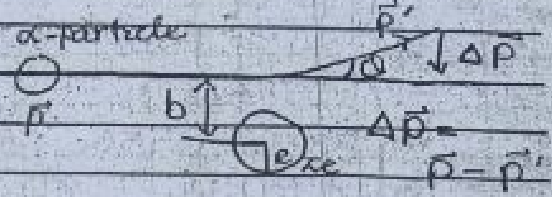
When α-particle approaches the nucleus, the coulomb repulsion is,

$$F = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^2} = \frac{1}{4\pi\epsilon_0}\frac{(2e)(Ze)}{b^2} \quad \to (1)$$

with $z=79$ for gold nucleus

→ This repulsive force can be approximately considered to the operating prependicular to the incident direction over a distance b

→ If α-particle has velocity 'v' then the time during which this force operates is



$\vec{\Delta p} = \vec{p} - \vec{p}'$

$$\boxed{\Delta t = \frac{b}{v}} \rightarrow (2)$$

→ This force produces a momentum 'Δp' in the direction prependicular to the incident direction

Then by Newton's Law

$$F = \frac{\Delta p}{\Delta t}$$

$$\Delta p = F \Delta t \rightarrow (3)$$

Using (1) & (2) in (3)

$$\Delta p = \frac{1}{4\pi\epsilon_0} \frac{(2e)(79e)}{b^2} \cdot \frac{b}{v}$$

$$\Delta p = \frac{1}{4\pi\epsilon_0} \frac{158e^2}{bv} \rightarrow (4)$$

From fig. $\tan\theta = \frac{\Delta p}{p}$

From experimental results 'θ' is small

$$\Rightarrow \tan\theta \sim \theta \simeq \frac{\Delta p}{p}$$

$$\theta \sim \frac{1}{4\pi\epsilon_0} \frac{(2e)(79e)}{bv} \frac{1}{m_\alpha v}$$

m_α → Nuclear mass of α-particle

$$\theta = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{bmv^2}$$

or $\Rightarrow$ $$\boxed{b = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{mv^2} \frac{1}{\theta}}$$

here $v = 10^7$ m/sec (velocity of the α-particle coming from α-source)

He used

$\theta \approx 1$ rad & $b = R$

$$R \sim \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{mv^2}$$

Using all values he estimated

$$R \sim 10^{-14}\ m$$

Rutherford repeated his experiment, for light and heavier nuclei (target) and discovered that

$$R \propto A^{1/3}$$

$$R = R_0 A^{1/3}$$

where

$$R_0 \approx 1.2 \times 10^{-15}\ m$$

Nuclear Angular Momentum: (Nuclear Spin)

Nucleus — Protons + Neutrons (Nucleons)

For Nucleons,

Nucleons posses
- intrinsic spin angular momentum $s = 1/2$ (in units of $\hbar$).
- quantized orbital angular momentum "L" about the centre of mass of nucleus.

$$\theta = \frac{1}{4\pi\epsilon_0}\,\frac{2Ze^2}{bmv^2}$$

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- intrinsic spin angular momentum $s = 1/2$ (in units of $\hbar$).
- quantized orbital angular momentum "L" about the centre of mass of nucleus.

→ The total angular momentum of a nucleon
$\vec{J} = \vec{L} + \vec{S}$. $|L-S| \le J \le (L+S)$

J_1, J_2, J_n

In Quantum mechanical sense
one can label every nucleon with the corresponding Q-nos l, s, j

For Nucleus

The total angular mom. of a nucleus (having 'A' nucleons)
= vector sum of angular momenta of all nucleons

$$\vec{I} = \vec{J}_1 + \vec{J}_2 + \cdots \vec{J}_A$$

$$\vec{I} = \sum_{i=1}^{A} \vec{J}_i = \sum_{i=1}^{A} (\vec{L}_i + \vec{S}_i)$$

where i the nucleon.
& $\vec{J}_i$: total angular momentum of a single nucleon.

→ The total angular mom. of the nucleus → nuclear spin ($\vec{I}$).

For a nucleus having angular mom. '$\vec{I}$', let 'i' is the corresponding Q-no.

$$I^2 = i(i+1)\hbar^2 \; ; \; i = 0, 1, 2, \ldots$$

$$I_z = m_i \hbar \quad \text{where } m_i = -i, \ldots, +i$$

$$[I^2, I_x] = [I^2, I_y] = [I^2, I_z] = 0$$

→ Often nucleus behaves as if it were a single entity with an intrinsic angular mom. of 'I'.

Note:

Remember that in atomic physics "in small magnetic fields"

$j \to 2j+1$ (j-j coupling)

$(l+s)$

"in strong magnetic field" (l-s coupling)

$s \rightarrow 2s+1$

$l \rightarrow 2l+1$

nuclear physics

in ordinary magnetic fields — nuclear zeman effect"

tate 'i' split up into 2i+1 individual substates.

'in very strong magnetic field each individual 'j' split into 2j+1 ubstates (not possible to acheive such strong fields yet)

→ We observe behaviour of 'I' as if the nucleus were only a single spinning particle

So Spin (total angular momentum) I & corresponding Q.no i are used to describe nuclear states.

→ Allowed values of I

Restriction on allowed values of I → comes from considering possible z-components of the total angular mom. of individual nucleons.

→ As for nucleons

$s = 1/2$ (half integral)

& $l = 0, 1, 2, \ldots$ (integral)

⇒ $j = (l+s)$; $(1/2, 3/2, 5/2, \ldots)$ (half odd integral)

$\vec{I} = \vec{J_1} + \vec{J_2} + \ldots$

$I_z = J_{1z} + J_{2z} + \ldots$

⇒ only possible z-components of total angular momentum of individual nucleon carry half odd integral values.

→ Now consider e.g. a nucleus of even 'A'
→ there will be an even no. of ½-integral components.
⇒ z- component of total I

$$I_z = \sum_{i=1}^{A} (J_{z_i}) \Rightarrow \text{integral values}$$

⇒ total 'I' of even nuclei will also be an integral value.

→ Now consider a nucleus having odd no. of nucleons i.e 'A' then there will be odd no. of ½-integral components
⇒ z-component of total I $I_z = \sum_{i=1}^{A} (J_{z_i}) \Rightarrow$ half odd integral

⇒ total 'I' of odd nuclei will also be an half odd integral.
⇒ The value of 'I' depends on the mass no. 'A' of a nucleus.
→ The value of 'I' is integer – even-A nuclei

→ The value of 'I' is half odd integral – odd-A nuclei

→ Nuclear Spin can also give some information about the nuclear structure.
e.g Spin = 0 nuclei ; even Z – even N
(like nucleons form pairs with equal and oppositely alligned momenta to make spin = 0 pairs)
→ For odd A nucleus;
(o-e , e-o) Nuclear Spin = j of the odd proton or odd neutron (which are unpaired)

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o-o nuclei (even A): Nuclear spin = sum of the J's of the unpaired 'p' & 'n'

r example ${}^{12}_{6}C\ (I=0)$; ${}^{16}_{8}O\ (I=0)$; ${}^{13}_{6}C_{7}\ (I=1/2)$

${}^{1}_{1}H_{0}\ (I=1/2)$

${}^{2}_{1}H_{1}\ (I=1)$

Nuclear Parity: CH # 1 + CH # 3 Krane

rity operation is an operation which uses reflection of all the space coordinates through the origin

$$\vec{r} \xrightarrow{P} -\vec{r}$$

- In cartesian coordinates

$x \to -x\ ,\ y \to -y\ ,\ z \to -z$

- In spherical coordinates (r, θ, ϕ)

$r \to r\ ,\ \theta \to \pi - \theta\ ,\ \phi \to \pi + \phi$

- Right handed coordinate system $\xrightarrow[\text{operation}]{\text{Parity}}$ left-handed coordinate system

et $\Psi(x, y, z)$ is wave fn. of a system

$$\hat{\pi}\,\Psi(x,y,z) = \Psi(-x,-y,-z) \quad \text{and}$$

~~applying~~ again $\pi^2\,\Psi(+x,+y,+z) = \hat{\pi}\,\Psi(-x,-y,-z)$

$$= \Psi(x, y, z)$$

$$\Rightarrow \pi = \pm 1$$

$$\Psi(-x,-y,-z) = \Psi(x,y,z)$$

$\Rightarrow$ wave fn has even parity

& if $\Psi(-x,-y,-z) = -\Psi(x,y,z)$

$\Rightarrow$ wave fn has odd parity.

For many particle system

Parity of the system = Product of parities of all the particles of the system

$\Pi = \Pi_1 \Pi_2 \Pi_3 \ldots \Pi_n$ where Π_i ($i=1,2,\ldots n$ are the parities of individual particle in the system).

Π of the system – even
(for any no of even parity particles)
&
Π of the system – even/odd
(for even no/odd no of odd parity particles)

➡ Nucleus – Many particle system (consists of protons & neutrons).
If we know the W.F of every nucleon, then we can determine the nuclear parity by multiplying together the parities of each of the A-nucleons.

Nuclear Parity – even or odd.

Note:
To assign a definite W.F of every nucleon with known parity – not possible!
Like Spin,
Parity (Π) – an overall property of the whole nucleus.

→ There are several techniques which can be used to determine nuclear parity (e.g nuclear reactions & nuclear decays).
For systems where potential is central i.e. (depends only of magnitude of $\vec{r}$ and independent of θ, ϕ)

$$\Pi = (-1)^l$$

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where l is orbital Q. no.
if l- even $\Rightarrow$ Parity $\rightarrow +1$ (even)
l-odd $\Rightarrow$ Parity $\rightarrow -1$ (odd)

For even-A (e-e)
$l = 0$; $\pi - +1$ (even)

For odd-A nuclei (e-o, o-e)

$\pi = (-1)^l$; l- orbital angular mom. of unpaired nucleon.

For even-A nuclei (o-o)
where 'l' has
$\pi = (-1)^l$ contribution from 'l' of unpaired (p) & unpaired neutron

Convention

For nuclei, Parity is denoted by '+' or '-' sign as superscript to the nuclear spin I^{π}
e.g. 2^- ; $1/2^+$, 0^+.

Quantum Statistics (CH # 01, CH # 03) KRANE

→ The concept of statistics related to the behaviour of large no. of particles.

→ In classical physics
Maxwell-Boltzman Statistics

→ In Quantum Physics
Two new forms of statistics
① Fermi-Dirac Statistics
② Bose-Einstein Statistics
(Add Differences)

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For nuclei

Consider a collection of large no. of nuclei

If ① nuclei are of $1/2$-integral spin (i.e. of odd A) e.g. ^1H, ^7Li, ^{23}Na

↳ Follow Fermi-Dirac Statistics.

② nuclei of integral spin (i.e. of even A) ^2H, ^4He, ^{12}C etc

↳ Follow Bose-Einstein Statistics.

Nuclear Electromagnetic Moments

An Introduction.

→ Distribution of electric charges produces Electric field.

→ Distribution of electric currents produces Magnetic field.

These fields vary with distance in a characteristic fashion.

Note. Multipole expansion of electric fields (CED course)

$$V(r) = \frac{1}{4\pi\epsilon_0}\left\{\frac{Q}{r} + \frac{\vec{P}\cdot\vec{r}}{r^3} + \cdots\right\}$$

$$V(r) = V(r)_{mono} + V(r)_{dip} + V(r)_{quad} + \cdots$$

As $\quad E = -\nabla V$

$$E(r) = E(r)_{mono} + E(r)_{dipole} + E(r)_{quad} + \cdots$$

$E(r)_{mono} \propto \frac{1}{r^2}$ arises from monopole moment (L=0, zeroth order of multipole moment)

$E(r)_{dipole} \propto \frac{1}{r^3}$ " " dipole (L=1, 1st order of multipole moment)

$E(r)_{quad} \propto \frac{1}{r^4}$ " " quadrupole (L=2, 2nd order of " ")

ere 'L' represents order of a
ultipole moment
Magnetic multipole moments behave
milarly with the exception of
monopole moment → (does not exist)
Dipole magnetic moment (lowest order).

The simplest distribution of charges
& currents give only the lowest
order multipole fields.
eg the spherical charge distribution
gives us only monopole field.

lculating various Electric and Magnetic
Multipole moments:

→ Classically
- Using EM theory
→ Quantum Mechanically
treat the multipole moments in
operator form ($\hat{O}$)
Calculate the expectation value of
the moments $\int \psi^* \hat{O} \psi \, dV$
where $\hat{O}$ is appropriate EM moment operator
One can also compare these expectation
values with the experimental values

→ Nuclear multipole moments
- Nucleus consists of protons
↓ charged particles (electric field)
- Since nucleons in a nucleus
are in motion → magnetic field
(orbital + spin)

EM multipole moments

Naturally nuclei tend to acquire a possible simple and symmetric structure.

⇒ Only lowest order multipole moments are necessary to be calculated to characterize EM properties of nuclei.

→ **Restriction on the multipole moments of Nuclei:**

- comes about from the symmetry of the nuclei.
- Directly related to the parity of the nuclear states.
- Each EM multipole moment has parity (which can be determined by checking the behaviour of respective multipole operator ($\hat{O}$) when $\vec{r} \to -\vec{r}$).

$\vec{p} = m\vec{v} = m(d\vec{r}/dt)$

$\pi\vec{p} = -m\, d\vec{r}/dt$ (odd)

$\vec{L} = \vec{r} \times \vec{p}$

$\pi\vec{L} = -\vec{r} \times -\vec{p} = \vec{r} \times \vec{p} = \vec{L}$ (even)

- The parity of electric multipole moments $= (-1)^L$ & the parity of magnet multipole moments $= (-1)^{L+1}$
  where L = order of the multipole moment.

Note: (From Q.M) If $\hat{O}$ has odd parity then integral vanishes (expectation value = 0).

→ **Vanishing & Non-Vanishing Nuclear EM moments:**

Electric Multipole moment	Parity $(-1)^L$	Expectation Value
$L=0$ monopole	Even	non-zero
$L=1$ dipole	Odd	zero
$L=2$ quadrupole	Even	non-zero

Magnetic Multipole moment	Parity $(-1)^{l+1}$	Expectation Value
$l=0$ monopole	odd	zero
$l=1$ dipole	even	non-zero
$l=2$ quadrapole	odd	zero
⋮		

$$E(r) = E_{mon} + 0 + E_{quad} + 0 + \cdots$$
$$B(r) = 0 + B_{dipole} + 0 + B_{octopole} + \cdots$$

For nuclei important moments are:

- Electric monopole moment → spherical charge distribution for nuclei $= +Ze$
- Magnetic dipole moment
- Electric quadrapole moment

Magnetic Dipole Moment:

Magnetic dipole moment of a nucleus arises due to orbital and spin motion of nucleons (Protons, neutrons). Total nuclear magnetic dipole moment is

$$\mu = \mu_l + \mu_s$$

μ_l: contribution due to orbital motion of nucleons. μ_s: contribution due to spin motion of nucleons.

$Z_p\ N_n$ μ

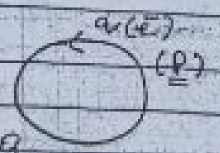
$$= \sum_{i=1}^{Z} \mu_{lp} + \underbrace{\sum_{i=1}^{N} \mu_{ln}}_{=0} + \sum_{i=1}^{Z} \mu_{sp} + \sum_{i=1}^{N} \mu_{sn}$$

contribution due to orbital motion of protons; contribution due to spin motion of protons; contribution due to spin motion of neutrons

→ Classical Concept:

A circular loop carrying current 'i' and enclosing area 'A' has magnetic moment of magnitude

$$|M| = iA \rightarrow (1)$$



If current is caused by a particle of charge 'q' moving with speed 'v' in a circle of radius 'r' then

$$i = \frac{q}{t} = \frac{qv}{2\pi r}$$

& $A = \pi r^2$

Using in eq. (1)

$$|M| = \frac{qv}{2\pi r} \times \pi r^2$$

$$|M| = \frac{qvr}{2} \rightarrow (2)$$

Let $q = e$ (which is charge on e^-) & multiply and divide by 'm_e'

$$|M| = \frac{evr}{2} \cdot \frac{m_e}{m_e} = \frac{e}{2m_e}(m_e vr)$$

where $\vec{L} = \vec{r} \times \vec{p}$

$$= \vec{r} \times m\vec{v}$$

$$|\vec{L}| = mvr$$

Using in last eq.

$$\boxed{|M| = \frac{e}{2m_e}|\vec{L}|}$$ → mag. of classical angular mom. which gives to 'M' for a charged particle.

→ Quantum Concept:

the observable magnetic moment corresponds to the direction of greatest component of L (L has its maximum projection along z-axis)

$$\mu = \frac{e}{2m_e} L_z \qquad \because L_z = m_l \hbar$$

$$\mu = \frac{e}{2m_e}(m_l \hbar)$$

where $(m_l)_{max} = +l$

where $m_l = -l \ldots\ldots +l$

$$\boxed{\mu = \frac{e\hbar}{2m_e} l} \rightarrow (3)$$

angular mom. Q no.

$$\boxed{\mu = \mu_B l} \rightarrow (4) \quad \text{where } \mu_B = \frac{e\hbar}{2m_e}$$

$$\mu_B = \frac{e\hbar}{2m_e} \qquad \hbar = 1.0546 \times 10^{-34}\ J.s$$

$$\mu_B = 9.27 \times 10^{-24}\ J/tesla$$

is called Bohr magneton.

For proton, we have to simply replace 'm_e' by 'm_p' in eqr. (3)

$$\mu = \frac{e\hbar}{2m_p} l$$

where $\mu_N = \frac{e\hbar}{2m_p}$

so $\boxed{\mu_{lp} = \mu_N l} \rightarrow (6)$

$\{\mu_N << \mu_B$

This is the contribution into magnetic dipole moment due to orbital motion of a single proton.

For neutron

$$\boxed{\mu_{ln} = 0} \rightarrow (7)$$

The generalization of (6) & (7) is $\boxed{\mu_l = g_l \mu_N l}$

where $g_\ell = 1$ for proton
$g_\ell = 0$ for neutron (i.e. no contribution to 'μ' from orbital motion of neutrons).

g_ℓ : g-factor associated with orbital angular momentum.

$\Rightarrow$ net contribution from orbital motion of protons

$$\sum_{i=1}^{Z} (\mu_{\ell p})$$

$\rightarrow \mu_{sp}, \mu_{sn} = ?$ (i.e. contribution to 'μ' due to spin motion)

for neutrons & protons $S = 1/2$
& due to spin motion of these nucleons, we have

$$\boxed{\mu_s = g_s \mu_N S}$$ — spin quantum no. $(= 1/2)$

g_s: g factor associated with spin motion of nucleons

where

$$\begin{cases} g_s = 5.586 & \text{for proton} \\ g_s = -3.826 & \text{for neutron} \end{cases}$$

where these values were set after comparison with the observed value of 'μ_s' for protons & neutrons.

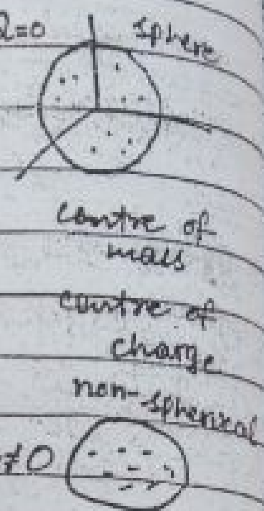
Now the total nuclear magnetic dipole moment is

$$\mu = \sum_{i=1}^{Z} (\mu_{\ell p})_i + \sum_{i=1}^{A} (\mu_s)_i$$

$$\boxed{\mu = \sum_{i=1}^{Z} (g_\ell \mu_N \ell) + \sum_{i=1}^{A} (g_s \mu_N S)}$$

Nuclear Electric Quadropole moment :

The next non-vanishing term in nuclear EM multipole moments.



Electric Quadropole moment (Q)

↓

arises due to non-spherical charge distribution in the nucleus

i.e. we can say

A measure of the deviation of the nuclear charge distribution from sphericity → electric quadropole moment

→ For a spherical shape of nucleus → Centre of mass of the nucleus coincides with centre of charge of the nucleus

→ For a non-spherical shape of nucleus → Centre of mass & Centre of charge of a nucleus do not coincide

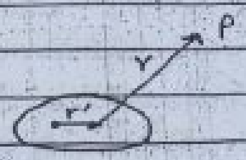
Consider a non-spherical nucleus (charge distribution)

Assume that

Centre of mass is at the origin of the nucleus &

Centre of charge is not at the origin

The potential at any point (p) outside the nucleus of a distance 'r' from the origin is



$$V(r) = \frac{1}{4\pi\epsilon_0}\frac{e}{|\vec{r}-\vec{r}'|} \rightarrow (1)$$

where 'e' — is the total charge of charge distribution of nucleus, located at a distance 'r'' from the origin. Now,

$$|\vec{r}+\vec{r}'| = [(\vec{r}-\vec{r}')^2]^{1/2}$$

$$= [r^2 + r'^2 - 2(\vec{r}).(\vec{r}')]^{1/2}$$

$$= [r^2 + r'^2 - 2rr'\cos\theta]^{1/2}$$

θ is the angle b/w $\vec{r}$ & $\vec{r}'$ substituting this in eq. (1) we get,

$$V = \frac{1}{4\pi\epsilon_0}\frac{e}{[r^2+r'^2-2rr'\cos\theta]^{1/2}}$$

$$= \frac{e}{4\pi\epsilon_0 r}\left(1+\frac{r'^2}{r^2}-\frac{2r'}{r}\cos\theta\right)^{-1/2}$$

Since r' is much smaller than r, we can carry out binomial expansion in above eq. and obtain

$$V = \frac{e}{4\pi\epsilon_0 r}\left[1+\left(-\frac{1}{2}\right)\left(\frac{r'^2}{r^2}-\frac{2r'}{r}\cos\theta\right)+\frac{3}{8}\left(\frac{r'^2}{r^2}-\frac{2r'}{r}\cos\theta\right)^2+\dots\right]$$

$$= \frac{e}{4\pi\epsilon_0 r}\left[1-\frac{r'^2}{2r^2}+\frac{r'}{r}\cos\theta+\frac{3}{8}\left(\overset{\text{Neglecting the power of } r'}{\cancel{\frac{r'^4}{r^4}}}+\frac{4r'^2}{r^2}\cos^2\theta - \overset{\text{Neglecting}}{\cancel{\frac{4r'^3}{r^3}\cos\theta}}\right)+\dots\right]$$

$$V = \frac{e}{4\pi\epsilon_0 r}\left[1-\frac{r'^2}{2r^2}+\frac{r'}{r}\cos\theta+\frac{3}{2}\frac{r'^2}{r^2}\cos^2\theta+\dots\right]$$

$$V = \frac{e}{4\pi\epsilon_0 r}+\frac{er'\cos\theta}{4\pi\epsilon_0 r^2}+\frac{er'^2}{4\pi\epsilon_0 r^2}\left(\frac{3}{2}\cos^2\theta-\frac{1}{2}\right)+\dots$$

The first term on R.H.S is contribution potential from single charge; the second term to dipole, its dipole moment is $er'\cos\theta$; the third term represents the contribution due to electric quadrupole whose quadrupole moment

$$Q = e(r')^2\left(\frac{3}{2}\cos^2\theta - \frac{1}{2}\right)$$

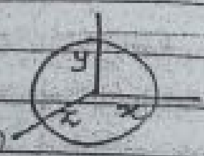
$$= e\left(\frac{3}{2}r'^2\cos^2\theta - \frac{r'^2}{2}\right)$$

Q for sphere
Q for non-sphere shape

$$\boxed{Q = e\left(\frac{3}{2}z^2 - \frac{r'^2}{2}\right)} \quad \because z = r'\cos\theta$$

$\hookrightarrow$ (1)

For a spherical shape nucleus $x = y = z$



Using eq $x^2 = y^2 = z^2 = \frac{r'^2}{3}$ in (1)

$r'^2 = x^2 + y^2 + z^2$

$x^2 = r'$

$y^2 =$

$z^2 =$

$$Q = e\left(\frac{3}{2}\left(\frac{r'^2}{3}\right) - \frac{r'^2}{2}\right)$$

$$\boxed{Q = 0}$$

If $(Q > 0)$ +ve

nucleus is $\Rightarrow$ prolate spheroid

Two out of 3 principle axes are equal and the third one, which is non-equal is very large as compared to the other two

$r'^2 = x^2 + y^2 + z^2$

$r'^2 \approx z^2$

$x = y \neq z$

$\Downarrow$

much larger in value than x & y.

now eq $Q \cong e\left(\frac{3}{2}(r'^2) - \frac{r'^2}{2}\right)$

$$\boxed{Q \cong er'^2} \text{ +ve}$$

If $(Q<0)$ -ve

nucleus is oblate spheroid.
Two out of 3 principle axes are equal and the third one which is non-equal is very small as compared to the other two now

e.g $Q \simeq e\left(-\frac{r'^2}{2}\right)$

$r'^2 = x^2+y^2+z^2$

$r'^2 \propto z^2$ $\quad x = y \neq z$

much smaller in value than x and y.

$\boxed{Q \simeq \frac{-er'^2}{2}}$ -ve

Section - II

Radioactive Decays

Radioactivity:
The phenomenon of spontaneous decay of a nucleus (heavy) accompanied by the emission of α, β or γ-rays.

Note. In α or β-decay processes
- an unstable nucleus emitts an α or β-particle, and tries to become a more stable nucleus.

While in γ-decay process
- an excited state decays towards ground state.
- nuclear species do not change.

α-emission / α-decay: (CH # 8 Krane).
- A process in which the parent nucleus spontaneously disintegrates into

daughter nucleus and an α-particle

The α-decay process is

${}_{Z}X_{N} \to {}_{Z-2}Y_{N-2} + {}_{2}He$ (α-particle) + Q

total energy released in the process → α-disintegration energy

example

$${}^{226}_{88}Ra_{138} \to {}^{222}_{86}Rn_{136} + {}^{4}_{2}He$$

Q- Why α-decay occurs?

It normally occurs in heavy nuclei ($A>160$)

Recall → $E_v \propto A$

appears because of the attractive nuclear force b/w nucleons

short-range force

Eg → $E_c \propto Z^2$

appears because of coulomb repulsive force b/w protons. long-range force

Consider Ra
$A = 226$
$Z = 88$
$Z^2 = 7744$

→ For heavy nuclei

Short range attractive force b/w nucleons is hardly able to counter-balance to neutral repulsion of protons and thus they are unstable and try to become stable e.g via α-emission).

Why α-emission? (Why not individual proton or ${}^{3}_{2}He$ nucleus)?

Ans. The high B.E of α-particle (28 MeV) favours their "spontaneous emission".

Spontaneous Process:- A process in which some K.E suddenly appears in the

$$Q_\alpha = T_Y + T_\alpha \rightarrow ②$$

we apply law of conservation of momentum

initial mom. = final mom.

$$\vec{P}_i = \vec{P}_f$$

$$\vec{P}_X (=0) = \vec{P}_Y + \vec{P}_\alpha$$

$$|\vec{P}_Y| = |\vec{P}_\alpha|$$

Y and α emitts in opposite direction carrying same value of mom.

$$P_Y^2 = P_\alpha^2 \rightarrow ③$$

$T = P^2/2m$, $P^2 = 2mT$

$$2 m_Y T_Y = 2 m_\alpha T_\alpha$$

$$\boxed{T_Y = \frac{m_\alpha T_\alpha}{m_Y}} \rightarrow ④$$

Use eqn. ④ in ②

$$Q_\alpha = \frac{m_\alpha T_\alpha}{m_Y} + T_\alpha$$

$$\boxed{T_\alpha = \frac{Q_\alpha}{\left(1 + \frac{m_\alpha}{m_Y}\right)}} \rightarrow ⑤$$

can be used to determine 'T_α'

Consider $\frac{m_\alpha}{m_Y} \simeq \frac{4}{(A-4)}$ use in ⑤

$$T_\alpha = Q_\alpha \Big/ 1 + \frac{4}{A-4}$$

$$= \frac{Q_\alpha (A-4)}{A}$$

$$T_\alpha = Q_\alpha \left(\frac{A-4}{A}\right)$$

$$T_\alpha = Q_\alpha \left(1 - \frac{4}{A}\right)$$

where $\frac{4}{A} \ll 1$

$$Q_\alpha = T_Y + T_\alpha \rightarrow ②$$

Next we apply law of conservation of linear momentum

initial mom. = final mom

$$\vec{P}_i = \vec{P}_f$$

$$\vec{P}_i (=0) = \vec{P}_Y + \vec{P}_\alpha$$

$$|\vec{P}_Y| = |\vec{P}_\alpha|$$

X. $\uparrow Y$ $\downarrow \alpha$

i.e. Y and α emitts in opposite direction carrying same value of mom

$$P_Y^2 = P_\alpha^2 \rightarrow ③$$

Since as $T = P^2/2m$, $P^2 = 2mT$

$$\cancel{2} m_Y T_Y = \cancel{2} m_\alpha T_\alpha$$

$$\boxed{T_Y = \frac{m_\alpha T_\alpha}{m_Y}} \rightarrow ④$$

Use eq. ④ in ②

$$Q_\alpha = \frac{m_\alpha T_\alpha}{m_Y} + T_\alpha$$

$$\boxed{T_\alpha = \frac{Q_\alpha}{\left(1 + \frac{m_\alpha}{m_Y}\right)}} \rightarrow ⑤$$

⑤ can be used to determine 'T_α'

Consider $\frac{m_\alpha}{m_Y} \approx \frac{4}{(A-4)}$ use in ⑤

$$T_\alpha = Q_\alpha \Big/ \left(1 + \frac{4}{A-4}\right)$$

$$= \frac{Q_\alpha (A-4)}{A}$$

$$T_\alpha = Q_\alpha \left(\frac{A-4}{A}\right)$$

$$T_\alpha = Q_\alpha \left(1 - \frac{4}{A}\right)$$

where $\frac{4}{A} \ll 1$

i.e. α-particle carries more energy as compared to daughter nucleus.

e.g

$$\boxed{\begin{array}{l} Q_\alpha = 5.085 \text{ MeV} \\ T_\alpha = 5.000 \text{ MeV} \\ T_Y = 0.085 \text{ MeV} \end{array}}$$

α-particle carries about 98% of the Q_α-value.

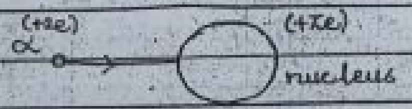
Typical Q_α-value 4-10 MeV

→ Two body decay problem.

THEORY of α-emission / α-decay

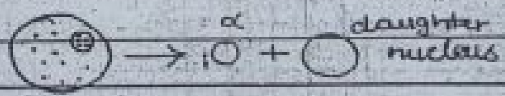
Note: The interaction b/w α-particle and nucleus at various distances.

→ In α-scattering experiment



- As α-particle comes closer to the nucleus — coulomb repulsion increases.
- When α-particle and nucleus just touching each other — repulsion max.
- If incoming α-particle have large K.E → α will overcome coulomb repulsion and enters the nucleus
- Inside the nucleus — strong nuclear force.
- α-particle attracted by

→ In radioactive α-decay

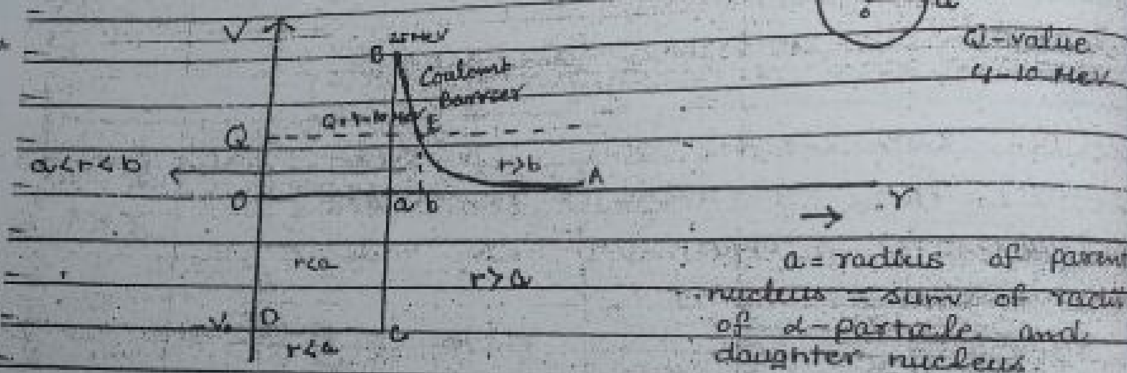


- In radioactive nuclei, 2P and 2n form an α-particle
- As long as an α-particle is inside the nucleus → acted upon by strong nuclear force (attractive).
- As α-particle arrives near the nuclear surface and comes out → acted upon by coulomb repulsive force (max at

Nucleons in the nucleus. Nature of interaction changes from coulomb repulsion to attractive strong nuclear force.	nuclear surface) Eq decreases as the α-particle moves away. • Nature of interaction : Attractive → Repulsive

Potential energy curve (for α-decay)

In general, the interaction b/w α-particle and nucleus at various distances can be represented by potential energy curve.



a = radius of parent nucleus = sum of radii of α-particle and daughter nucleus.

In fig.

Horizontal dashed line → Q (total available energy) α-disintegration energy.

→ Outside the nucleus $(r > a)$.

acts like a barrier. ← { Coulomb potential Energy → +ve.

$V_c(r) = \frac{1}{4\pi\epsilon_0} \frac{2Z'e^2}{r}$ which is represented by line AEB.

→ { Inside the nucleus $(r < a)$

there is short range attractive potential (-ve) constant value $= V_0$

Acts like a potential well { represented by line BCD

→ At nuclear surface $(r=a)$

transition from Coulomb repulsive potential to the attractive potential

Point B - represents the largest value of coulomb potential 'V_c').

Note

Usually for radioactive nuclei the max height of coulomb barrier ~ 25 MeV and for such nuclei Q = 4 - 10 MeV $R = R_0 A^{1/3}$

Classical Description for α-emission.

3 Regions of interest:

Region 1. $r < a$

Potential well of depth $-V_0$, in this energy

$$E = T + V$$
$$T = E - V$$
$$T = Q + V_0 \rightarrow +ve \text{ value.}$$

i.e α-particle can move in this region with K.E = T

⇒ Classically allowed region.

Region 2 $a < r < b$

→ Potential barrier of varying height which is determined by

$$V_c(r) = \frac{1}{4\pi\epsilon_0} \frac{2z'e^2}{r}$$

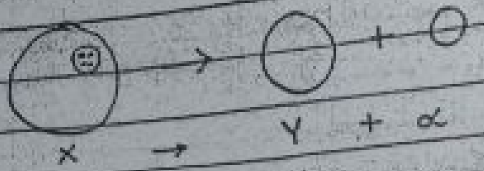
In this region $E = T + V$

$$T = E - V$$
$$T = Q - V_c(r)$$
$$\boxed{T = -ve} \quad V_c \gg Q$$

assically forbidden region.
[classically α-particle cannot enter this region]

assical theory fails to explain the emission/α-decay process, so we have switch to QM to explain the -decay process

amow Theory of α-decay/emission



-mechanical theory of α-emission was iven by Gamow in 1928.

is Assumptions:

• An α-particle may exist as an α-particle within the heavy nucleus.
Such an α-particle is in const. motion nd is held in the nucleus by potential barrier.
There is a small but definite probability that α-particle may tunnel through the barrier, each time it collides with the barrier

→ Based on these assumptions, Gamow derived a relation for probability (P) that α-particle may tunnel through the barrier

• **Barrier Penetration Probability (P):**
→ Obtained from Q.M calculations similar to 1-D regular barrier

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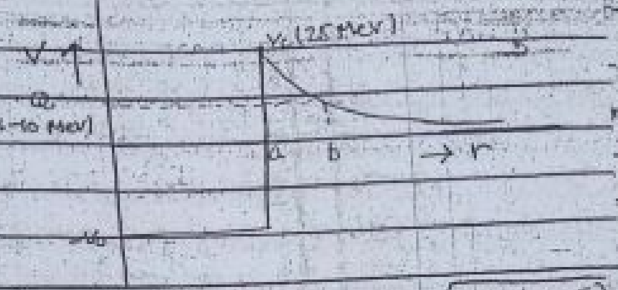
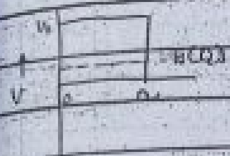
Note For the case of rectangular potential barrier
Q.M results



Transition probability



$$T \simeq \frac{16(V_0 - E)}{V_0^2} e^{-2K_2 a} \quad \text{where } K_2 = \sqrt{\frac{2m(V_0-E)}{\hbar^2}}$$

a : width of barrier
V_0 : height of barrier

$$(P)\ T \simeq e^{-2K_2 a} \qquad \left\{ \frac{16E(V_0-E)}{V_0^2} \sim 1 \right.$$

$$P \cong e^{-2K_2 a} \rightarrow ①$$

the transition probability for a barrier of fixed height 'V_0'

For our case, (α-emission process), we have Coulomb barrier of variable height

where $V_c(r) = \dfrac{2Z'e^2}{4\pi\epsilon_0 r}$

i.e height varies as $1/r$

→ We can think of the coulomb barrier made up of a sequence of infinitesimal rectangular barrier of height

$$V(r) = \frac{2Z'e^2}{4\pi\epsilon_0 r} \quad \text{\& width } dr$$

→ Now Consider any one infinitesimal rectangular potential barrier, For this the probability to penetrate → dP

$$dP \simeq e^{-2K_2 dr} \rightarrow ②$$

$$\text{where } K_2 = \sqrt{\frac{2m(V_0-E)}{\hbar^2}} = \sqrt{\frac{2m_\alpha(V_c - Q)}{\hbar^2}} \rightarrow ③$$

...ular momentum & Parity in α-decay

$$X \rightarrow Y + \alpha$$

conservation of total angular mom.

$$I_P = I_D + I_\alpha$$

$$\hookrightarrow ? \quad L_\alpha + S_\alpha \, (2P+2n)$$

$$\downarrow = 0$$

$$I_\alpha = l_\alpha$$

$$|I_P - I_D| \leq l_\alpha \leq (I_P + I_D)$$

$$l \neq 0 \Rightarrow V_l = \frac{l(l+1)\hbar^2}{2m_\alpha r^2} \qquad V_c + V_l$$

The probability to penetrate complete ...rrier is,

$$P = e^{-2\int_a^b k_2 \cdot dr}$$

$$P = e^{-2\int_a^b \sqrt{\frac{2m_\alpha (V_c(r) - Q)}{\hbar^2}} dr}$$

$$P = e^{-2G} \rightarrow (4)$$

where the Gamow factor 'G' is

$$G = \int_a^b \sqrt{\frac{2m_\alpha (V_c(r) - Q)}{\hbar^2}} \, dr \rightarrow (5)$$

where $V_c(r) = \frac{2z'e^2}{4\pi\epsilon_0 r}$ & $Q = \frac{2z'e^2}{4\pi\epsilon_0 b}$

Using in (5)

$$G = \int_a^b \sqrt{\frac{2m_\alpha}{\hbar^2}\left\{\frac{2z'e^2}{4\pi\epsilon_0 r} - \frac{2z'e^2}{4\pi\epsilon_0 b}\right\}} \, dr$$

$$G = \sqrt{\frac{2m_\alpha}{\hbar^2}} \sqrt{\frac{2z'e^2}{4\pi\epsilon_0}} \int_a^b \sqrt{\frac{1}{r} - \frac{1}{b}} \, dr$$

$$G = \sqrt{\frac{2m_\alpha}{\hbar^2} \frac{2z'e^2}{4\pi\epsilon_0}} \, I \rightarrow (6)$$

Take $I = \int_a^b \sqrt{\frac{1}{r} - \frac{1}{b}} \, dr \rightarrow (7)$

Let us take $\frac{1}{r} = \frac{1}{b}\sec^2 x$

$b = r\sec^2 x$ ...

For limits: at $r = a$

$$a = b\cos^2 x \Rightarrow x = \cos^{-1}\sqrt{\frac{a}{b}}$$

at $r = b$ $\quad b = b\cos^2 x \Rightarrow \cos^2 x = 1$

$$x = \cos^{-1}(1) \Rightarrow x = 0$$

Using above results in eq. (7)

$$I = \int_{\cos^{-1}\sqrt{a/b}}^{0} \sqrt{\left(\frac{1}{b}\sec^2 x - \frac{1}{b}\right)}\,(-2b\sin x\cos x\,dx)$$

$$I = \frac{-2b}{\sqrt{b}} \int_{\cos^{-1}\sqrt{a/b}}^{0} \sqrt{\sec^2 x - 1}\,\sin x\cos x\,dx$$

$$I = \frac{2b}{\sqrt{b}} \int_{0}^{\cos^{-1}\sqrt{a/b}} \sqrt{\tan^2 x}\cdot\sin x\cos x\,dx$$

$$I = \frac{2b}{\sqrt{b}} \int_{0}^{\cos^{-1}\sqrt{a/b}} \tan x\cdot\sin x\cdot\cos x\,dx$$

$$I = 2\sqrt{b} \int_{0}^{\cos^{-1}\sqrt{a/b}} \frac{\sin^2 x}{\cos x}\cdot\cos x\,dx$$

$$I = 2\sqrt{b} \int_{0}^{\cos^{-1}\sqrt{a/b}} \sin^2 x\,dx$$

$$I = 2\sqrt{b} \int_{0}^{\cos^{-1}\sqrt{a/b}} \left(\frac{1-\cos 2x}{2}\right) dx$$

$$I = \sqrt{b}\left[\int_{0}^{\cos^{-1}\sqrt{a/b}} dx - \int_{0}^{\cos^{-1}\sqrt{a/b}} \cos 2x\,dx\right]$$

$$I = \sqrt{b}\left[x\Big|_{0}^{\cos^{-1}\sqrt{a/b}} - \frac{\sin 2x}{2}\Big|_{0}^{\cos^{-1}\sqrt{a/b}}\right]$$

$$I = \sqrt{b}\left[\cos^{-1}\sqrt{\frac{a}{b}} - \frac{\cancel{2}\sin x\cos x}{\cancel{2}}\Big|_{0}^{\cos^{-1}\sqrt{a/b}}\right]$$

$$I = \sqrt{b}\left[\cos^{-1}\sqrt{\frac{a}{b}} - \sin\left(\cos^{-1}\sqrt{\frac{a}{b}}\right)\cos\left(\cos^{-1}\sqrt{\frac{a}{b}}\right)\right]$$

$$I = \sqrt{b}\left[\cos^{-1}\sqrt{\frac{a}{b}} - \sin\left(\cos^{-1}\sqrt{\frac{a}{b}}\right)\sqrt{\frac{a}{b}}\right]$$

Use this in eq. (6)

$$G = \sqrt{\frac{2m_e}{\hbar^2}\,\frac{2Z'e^2}{4\pi\epsilon_0}}\,\sqrt{b}\left[\cos^{-1}\sqrt{\frac{a}{b}} - \sqrt{\frac{a}{b}}\,\sin\left(\cos^{-1}\sqrt{\frac{a}{b}}\right)\right]$$

$\therefore$ From $Q = \frac{2z'e^2}{4\pi\epsilon_0 b} \Rightarrow Qb = \frac{2z'e^2}{4\pi\epsilon_0}$

$\therefore G = \sqrt{\frac{2m_\alpha Qb}{\hbar^2}}\sqrt{b}\left[\cos^{-1}\sqrt{\frac{a}{b}} - \sqrt{\frac{a}{b}}\sin\left(\cos^{-1}\sqrt{\frac{a}{b}}\right)\right]$

$G = b\sqrt{\frac{2m_\alpha}{\hbar^2}}\sqrt{Q}\left[\cos^{-1}\sqrt{\frac{a}{b}} - \sqrt{\frac{a}{b}}\sin\left(\cos^{-1}\sqrt{\frac{a}{b}}\right)\right] \rightarrow (8)$

As $\frac{Q}{B} = \frac{\frac{2z'e^2}{4\pi\epsilon_0 b}}{\frac{2z'e^2}{4\pi\epsilon_0 a}} = \frac{\frac{1}{b}}{\frac{1}{a}} \Rightarrow \boxed{\frac{Q}{B} = \frac{a}{b}}$

Use in (8)

$G = b\sqrt{\frac{2m_\alpha}{\hbar^2}}\sqrt{Q}\left[\cos^{-1}\sqrt{\frac{Q}{B}} - \sqrt{\frac{Q}{B}}\sin\left(\cos^{-1}\sqrt{\frac{Q}{B}}\right)\right] \rightarrow (9)$

let $y = Q/B$

$G = b\sqrt{\frac{2m_\alpha}{\hbar^2}}\sqrt{Q}\left[\cos^{-1}\sqrt{y} - \sqrt{y}\sin(\cos^{-1}\sqrt{y})\right] \rightarrow (10)$

Let $\sqrt{x} = \cos^{-1}\sqrt{y} \Rightarrow \sqrt{y} = \cos\sqrt{x} \Rightarrow y = \cos^2\sqrt{x}$

$\because \sin^2\sqrt{x} + \cos^2\sqrt{x} = 1$

$\sin^2\sqrt{x} + y = 1$

$\sin^2\sqrt{x} = 1 - y \Rightarrow \sin\sqrt{x} = \sqrt{1-y}$

so $\sin(\cos^{-1}\sqrt{y}) = \sqrt{1-y}$

Using in eq. (10)

$G = b\sqrt{\frac{2m_\alpha}{\hbar^2}}\sqrt{Q}\left[\cos^{-1}\sqrt{y} - \sqrt{y}\sqrt{1-y}\right] \rightarrow (11)$

As $y = \frac{Q}{B} = \frac{a}{b}$ as $b \gg 1$

so $\frac{a}{b} \ll 1$ or $\frac{Q}{B} \ll 1$ or $y \ll 1$

then for this case

$\cos^{-1}\sqrt{y} \simeq \frac{\pi}{2} - \sqrt{y}$

$\because y^2 \lll 1$ can be ignored

so eq. (11) becomes

$G = b\sqrt{\frac{2m_\alpha}{\hbar^2}}\sqrt{Q}\left[\frac{\pi}{2} - \sqrt{y} - \sqrt{y - y^2}\right]$

$$G = b\sqrt{\frac{2m_\alpha}{\hbar^2}}\sqrt{Q}\left[\frac{\pi}{2} - \sqrt{y} - \sqrt{y}\right] \quad \because 2y^2 \ll 1$$

$$G = b\sqrt{\frac{2m_\alpha}{\hbar^2}}\sqrt{Q}\left[\frac{\pi}{2} - 2\sqrt{y}\right]$$

$$G = b\sqrt{\frac{2m_\alpha}{\hbar^2}}\sqrt{Q}\left[\frac{\pi}{2} - 2\sqrt{\frac{Q}{B}}\right] \rightarrow (12)$$

As $P = \exp(-2G)$

$$P = \exp\left[-2b\sqrt{\frac{2m_\alpha}{\hbar^2}}\sqrt{Q}\left\{\frac{\pi}{2} - 2\sqrt{\frac{Q}{B}}\right\}\right] \rightarrow (13)$$

(13) shows that tunneling probability has strong dependance on Q ($Q\uparrow = P\uparrow$).

→ **Decay Constant (λ)**

Disintegration constant of an α-emission (λ) is given by

$$\lambda = fP \rightarrow (14)$$

where P = Probability of transition through barrier.

& f = collision frequency

→ Collision frequency (f): The no. of attempts α-particle makes on average to penetrate the barrier before it escapes.

→ Suppose at any moment an α-particle exists in a nucleus and that it moves back and forth

then $\because S = vt$

$a = vt \qquad \because S = a$

$$1/t = v/a \Rightarrow f = v/a \rightarrow (15)$$

where v: relative velocity of α-particle

a: radius of nucleus (parent)

→ 'v' can be calculated from K.E of the α-particle.

for $r<a$ As $T = Q + V_0$ for $r<a$

$$\frac{1}{2} m_\alpha v^2 = Q + V_0$$

$$\Rightarrow v^2 = \frac{2(Q+V_0)}{m_\alpha}$$

$$v = \sqrt{\frac{2(Q+V_0)}{m_\alpha}} \quad \text{or} \quad m_\alpha v^2 = 2(Q+V_0)$$

for $Q = 6\,MeV$, $V_0 = 35\,MeV$ $f \approx 6\times10^{21}/sec$

$\Rightarrow$ so decay constant

$$\lambda = \frac{v}{a} e^{-2G}$$

Half life ($t_{1/2}$)

As $t_{1/2} = \frac{0.693}{\lambda}$

$$t_{1/2} = \frac{0.693}{\frac{v}{a} e^{-2G}} = \frac{0.693}{v} a\, e^{2G}$$

$$t_{1/2} = 0.693\, a \sqrt{\frac{1}{v^2}}\, e^{2G}$$

$$= 0.693 \frac{a}{c} \sqrt{\frac{c^2}{v^2}}\, e^{2G}$$

$$t_{1/2} = 0.693 \frac{a}{c} \sqrt{\frac{m_\alpha c^2}{m_\alpha v^2}}\, e^{2G}$$

where $\sqrt{m_\alpha v^2} = \sqrt{2(Q+V_0)}$ in region I

$$t_{1/2} = 0.693 \frac{a}{c} \sqrt{\frac{m_\alpha c^2}{2(V_0+Q)}}\, e^{2G} \rightarrow (16)$$

→ **Discussion:**

(From Notes)

→ β-de
unstab
nucleu
Types
β⁻-dec
is em
→ Fun
→ Nu
β⁺-dec
is
→ Fun
→ Nu
Electr
an
captu
(i) n
(ii) p
(iii) p

→ **β-decay**: Krane Ch#9.

A mechanism by which an unstable nucleus changes to a stable nucleus, by keeping a balance in N/Z ratio.

Types of β-decay:

(i) β^--decay
(ii) β^+-decay
(iii) Electron Capture.

β^--decay

In this process, β^--particle (electron) is emitted by the nucleus.

→ Fundamental β^--decay

$$n \text{ (free neutron)} \rightarrow p + e^-$$

→ Nuclear β^--decay

$$^{A}_{Z}X_{N} \rightarrow {}^{A}_{Z+1}Y_{N-1} + e^-$$

β^+-decay

In this process, β^+-particle (positron) is emitted by the nucleus.

→ Fundamental β^+-decay

$$p \rightarrow n + e^+$$

→ Nuclear β^+-decay

$$^{A}_{Z}X_{N} \rightarrow {}^{A}_{Z-1}Y_{N+1} + e^+$$

Electron Capture:

The process in which an e^- in the lower most shell is captured by the nucleus.

$$p + e^- \rightarrow n \text{ (orbital } e^- \text{ capture)}$$

(i) $n \rightarrow p + e^-$
(ii) $p \rightarrow n + e^+$
(iii) $p + e^- \rightarrow n$

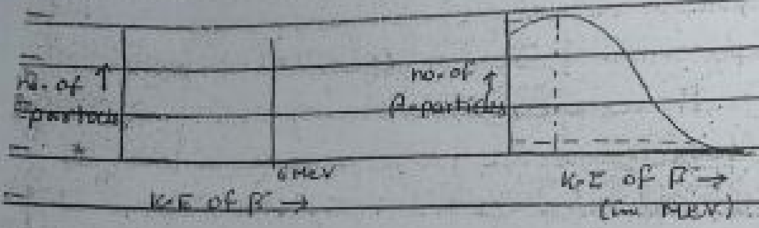
} not yet complete

Note: β^+-decay and e^--capture processes occur only for those protons bound in the nucleus.

Problems in Understanding β-decay :

(i) Puzzle-1 (Conservation of energy)

- Energy release in β-decay
- Unlike α-particles, β-particles emitted having different values of K.E (for a particular β-decay) range of energies (from 0 to an upper limit).



This was an unexpected result (expectation was that all β-particles emit with a fixed K.E)

⇒ According to experimental observations, mostly e^-s are emitted with lower K.E than expected → Where went this missing energy.

As $Q = T_{\bar{\nu}} + T_{e^-}$

(ii) Puzzle-2 (Conservation of angular mom.)

e.g in $n \rightarrow p + e^-$

$p \rightarrow n + e^+$

$p + e^- \rightarrow n$

- In this process, one fermion ($s = 1/2$) on one side of the eq. and two fermions on the other side of the eq. therefore

$L+S$ on one side of the eq → angular

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Properties

(i) should
(ii) Highly
(iii) spin -
(iv) Two
(v) ν̄ an
emitt
So now
complete
β-decay pro
→ neutrino

momentum is ½-odd integral. while on the other side of the eqn.
→ angular momentum is integral.
→ Problem with conservation of angular momentum.

Several ideas were proposed to resolve these issues. Among them the successful one was the 'Neutrino Hypothesis' proposed by Pauli.

Neutrino Hypothesis:

Pauli proposed that β-decay is a 3 body problem process (3-particles in the final state)

$$n \rightarrow p + e^- + c$$

this third particle 'c' was later named as neutrino by Fermi;

neutrino it carries missing energy.

$$Q = T_Y + T_{e^-} + T_\nu$$

Properties of Neutrino

(i) Should be neutral (from conservation of charge)
(ii) Highly Penetrating - not stopped within calorimeter → its energy not recorded.
(iii) spin - ½ Particle (from conservation of angular momentum).
(iv) Two different kinds of neutrino $\nu, \bar{\nu}$
(v) $\bar{\nu}$ emitted in β^- decay & ν emitted in β^+ decay.

so now our complete β-decay processes

$$n \rightarrow p + e^- + \bar{\nu}$$
$$p \rightarrow n + e^+ + \nu$$
$$p + e^- \rightarrow n + \nu$$

→ neutrino's rest mass is almost zero (ignored)

(Es)timation of mass of neutrino

Consider the decay of a free neutron

$$n \rightarrow p + e^- + \bar{\nu} \rightarrow (1)$$

Q-value for this process, $m_{\bar{\nu}}$ unknown

$$Q = (m_n - m_p - m_{e^-} - m_{\bar{\nu}})c^2 \rightarrow (2)$$

also for this process

$$Q = T_p + T_{e^-} + T_{\bar{\nu}} \rightarrow (3)$$

As $m_p \gg m_{e^-}$

⇒ Proton shares very little of the total available energy (Proton recoil energy) since it is very small, so it can be ignored

$$Q \simeq T_{e^-} + T_{\bar{\nu}} \rightarrow (4)$$

⇒ e^- and $\bar{\nu}$ share most of the available energy

Note electrons emitted with minimum K.E correspond to max K.E neutrinos & electrons emitted with max. K.E correspond to min. K.E neutrinos

if $T_{\bar{\nu}} \rightarrow 0$

$$T_{e^-} \rightarrow Q = (T_{e^-})_{max} \simeq 0.782 \text{ MeV}$$

From experimental graph, the max. possible K.E of e^-'s was found to be 0.782 ± 0.013 MeV

$$(T_{e^-})_{max} \simeq Q \simeq 0.782 \pm 0.013 \text{ MeV} \rightarrow$$

Use this Q-value in eq. (2)

$$Q = (m_n - m_p - m_{e^-} - m_{\bar{\nu}})c^2$$

Using rest masses of neutron, proton e^- and the Q-value given above the mass of $\bar{\nu}$ was estimated to be equal to zero.

Theory of β-decay:

Differences in α and β decay

α-decay	β-decay
(i) In α-decay, a bound structure (made of 2P +2N) is emitted.	(i) In β-decay, single particles e^- and ν are emitted.
(ii) Coulomb barrier in α-decay (because of coulomb repulsion b/w α and daughter nucleus).	(ii) In β^--decay ($n \rightarrow P + e^- + \bar{\nu}$) no such coulomb barrier (no repulsion b/w e^- and P). In β^+-decay coulomb barrier present (small height) → but very easily penetrable. For β^+-decay $P \simeq 1$
(iii) Discrete energy spectrum for a certain α-decay process E_α → fixed value.	(iii) Continuous energy spectrum. For a certain β-decay process emitted β-particles have a range of energies.
(iv) Velocities of emitted α-particle can be treated classically	(iv) e^- and ν must be treated relativistically {ν travels with speed comparable to 'c'}.
(v) In theory of α-emission pre-formation of α-particle was supposed	(v) e^- and ν do not exist before decay so must account for formation of these particles

we have to use a completely different
[ap]proach for calculation of 'λ' for
[β]-decay.

[Fe]rmi theory of β-decay

- by Fermi in 1934.
- based on Pauli's neutrino
hypothesis

$$\begin{cases} n \to p + e^- + \bar{\nu} & (\beta^- \text{-decay}) \\ p \to n + e^+ + \nu & (\beta^+ \text{-decay}) \end{cases}$$

The transition probability ($p \to n$ or $n \to p$)
[b/]w two states is given by Fermi's
[g]olden rule

$$\underbrace{\lambda}_{\text{transition probability}} = \frac{2\pi}{\hbar} |V_{fi}|^2 \, f(E_f) \to ①$$

where E_f : Energy of final states
$f(E_f)$: Density of final states,
which is defined as

$$f(E_f) = \frac{dn}{dE_f} \to ②$$

[t]he no ('dn') of final states in the
energy range dE_f

V_{fi} : matrix element

$$V_{fi} = \int \psi_f \hat{V} \psi_i \, dV \to ③$$

where

$\hat{V}$: operator for the interaction
potential which causes the transition
from initial state 'ψ_i' to final
state ψ_f.

ψ_i : initial state W.F (describes parent
nucleus).

ψ_f: Final state W.F {describes the product of W.F for daughter nucleus times W.F for $e^-(e^+)$ and W.F for $\bar{\nu}(\nu)$.}

→ To determine 'λ' for a certain β-decay process we need to specify / determine all the involved factors (e.g. $f(E_f)$, ψ_i, V, ψ_f ... etc.)

→ <u>How to find $f(E_f)$?</u>

As $$f(E_f) = \frac{dn}{dE_f}$$

To determine density of final states "We need to know the no. of final states accessible to decay products."

→ In QM, we deal with phase space (6N-dim) <u>Remember</u> {A space in which all possible states of system are represented, and each state of the system corresponds to a unique point in the phase space (each point corresponds to a particular position and momentum).}

If there are N particles, then phase space of N-particles has 6N dimensions

{ − 3N dimension for configurational space.
{ − 3N dimension for momentum space.

→ Consider the phase space of a single particle then the unit volume in phase space is given by

$$\text{Unit vol.} = \Delta x\, \Delta y\, \Delta z\, \Delta P_x\, \Delta P_y\, \Delta P_z$$

$$\text{Unit vol.} = (\Delta x\, \Delta P_x)(\Delta y\, \Delta P_y)(\Delta z\, \Delta P_z)$$

by Uncertainty Principle

$$\Delta x\, \Delta P_x \sim \hbar = \frac{h}{2\pi} \sim h$$

Unit vol in phase space $= h^3$
vol. occupied by a single particle
state in phase space.)

In case of β-decay

$E_f = E_N + E_{e^-} + E_{\bar{\nu}}$

E_N: ignorable (very small)

$E_f \simeq E_{e^-} + E_{\bar{\nu}} \Rightarrow$ Total available
total energy is mostly shared b/w e^- (e^+) and $\bar{\nu}(\nu)$.

e^- and $\bar{\nu}$ emit with different energies

Since $E^2 = p^2c^2 + m^2c^4$

e^- and $\bar{\nu}$ they have corresponding different values of momenta.
↳ (diff states).

no. of final states which have simultaneously an e^- & a $\bar{\nu}$ with proper momenta is

$$\boxed{dn = dn_e \, dn_{\bar{\nu}}} \rightarrow (4)$$

To determine dn_e and $dn_{\bar{\nu}}$, we determine the corresponding volume occupied by these states.

As volume occupied by one state $= h^3$

total no. of e^--states $= dn_e$

& similarly total volume occupied by dn_e states $= h^3 dn_e$ (5a)

total volume occupied by $dn_{\bar{\nu}}$ states $= h^3 dn_{\bar{\nu}}$ (5b)

→ determining dn_e and $dn_{\bar{\nu}}$:

In a $\beta^-(\beta^+)$ decay → Suppose that $e^-(e^+)$ emitted with mom. 'p', $\bar{\nu}(\nu)$ emitted with mom. 'q'.

Since total vol. in phase space = (Vol. in coordinate space) × (Vol. in mom. space)
= V (" " ?)

→ Imagine a coordinate system whose axes are labelled by P_x, P_y and P_z

$$\vec{P} = P_x\hat{e_x} + P_y\hat{e_y} + P_z\hat{e_z}$$

$$P^2 = P_x^2 + P_y^2 + P_z^2 \rightarrow (6)$$

(1a)

This is eq. of a sphere.

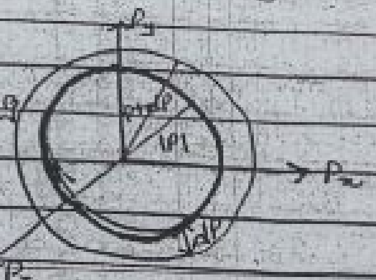
⇒ The locus of points representing a specific value of 'P' represents a sphere of radius $|P|$

where $|P| = \sqrt{P_x^2 + P_y^2 + P_z^2}$

If mom changes from P to $P+dP$ then radius of sphere also changes from P to $P+dP$.

→ More specifically

the locus of points representing momenta in the range 'P' to $P+dP$ in a spherical shell of radius 'P' and thickness 'dP'



As vol. of sphere = $\frac{4}{3}\pi P^3$

$$V_P = \frac{4}{3}\pi P^3$$

$$dV_P = \frac{4}{\cancel{3}}\pi \cdot \cancel{3}P^2\, dP$$

$$dV_P = 4\pi P^2\, dP \rightarrow (7)$$

Using (7) in last eq.

the total phase space volume occupied by states = $V(4\pi P^2 dP)$ → (8a)

of e⁻'s emitting with momenta

in the range ...

~~slarly~~ total phase space volume ~~speed by~~ $\bar{\nu}$ states emitting $= V(4\pi a^2 da)$ → (8b)
~~w momenta~~ in the range
a and $a+da$

Comparing (5a) and (8a)

$$\Rightarrow V(4\pi P^2 dP) = h^3 dn_e$$

$$dn_e = \frac{V(4\pi P^2 dP)}{h^3} \rightarrow (9a)$$

e^- states

Similarly comparing (5b) & (8b)

$$dn_{\bar{\nu}} = \frac{V(4\pi a^2 da)}{h^3} \rightarrow (9b)$$

Now using (9a) and (9b) in eq.

$$f(E_F) = \frac{dn}{dE_F} = \frac{dn_e\, dn_{\bar{\nu}}}{dE_F}$$

$$f(E_F) = \frac{(4\pi)^2 V^2\, p^2 dP\, a^2 da}{h^6\, dE_F} \rightarrow (10)$$

$|V_{fi}| = ?$

As $V_{fi} = \int \psi_f^* \hat{V} \psi_i \, dV$
↳ interaction pot.

where $\hat{V} = g\,\hat{O}_x$

~~here~~ g: a constant which determines the strength of interaction.

$\hat{O}_x$: mathematical operator to represent the interaction potential.

∵ To check type of operator apply parity on the operator.

Note Types of Operators (x)

① If $x = V$ (vector)

$V \xrightarrow{P} -V$ (e.g linear mom.)

② $x = A$ (Axial vector)

$A \xrightarrow{P} A$ (e.g angular mom.)

weak interaction case → potential (Axial-vector)

③ $X = S$ (scalar)

$S \xrightarrow{\hat{P}} S$ (e.g. temperature).

④ $X = P$ (Pseudoscalar)

$P \xrightarrow{\hat{P}} -P$ (e.g volume)

⑤ $X = T$ (Tensor)

(Fermi tried all possible forms (5 possible) of this operator.)

Studies revealed that, correct form of $\hat{O}_x$ should be V-A form

so $V_{fi} = g \int \Psi_f^* \hat{O}_x \Psi_i \, dV$

where $\Psi_f = \Psi_Y \phi_e \phi_{\bar{\nu}}$

$\Psi_f^* = \Psi_Y^* \phi_e^* \phi_{\bar{\nu}}^*$

so

$$V_{fi} = g \left[\int (\Psi_Y^* \phi_e^* \phi_{\bar{\nu}}^*) \right] \hat{O}_x \Psi_x \, dV \rightarrow (11)$$

→ The form of wave functions (ϕ_e, $\phi_{\bar{\nu}}$?)

(ignoring coulomb interaction) → (just to keep calculations simple)

If the coulomb interaction of emitted e^- with the product nucleus is ignored, then e^- & $\bar{\nu}$ W.Fs have the usual free particle form, normalized with the volume (V)

$$\phi_{e^-}(r) = \frac{1}{\sqrt{V}} e^{i\vec{p}\cdot\vec{r}/\hbar}$$

$$\phi_{\bar{\nu}}(r) = \frac{1}{\sqrt{V}} e^{i\vec{q}\cdot\vec{r}/\hbar} \quad \text{where}$$

$\vec{p}$ and $\vec{q}$ are momenta of emitted e^- and $\bar{\nu}$.

$$\frac{\vec{p}\cdot\vec{r}}{\hbar} \;\&\; \frac{\vec{q}\cdot\vec{r}}{\hbar} \ll 1$$

Taking only the first term

$$e^{i\vec{p}\cdot\vec{r}/\hbar} = 1 + i\frac{\vec{p}\cdot\vec{r}}{\hbar} + \dots \simeq 1$$

$$e^{i\vec{q}\cdot\vec{r}/\hbar} = 1 + i\frac{\vec{q}\cdot\vec{r}}{\hbar} + \dots \simeq 1$$

so $\phi_e(\vec{r}) = \frac{1}{\sqrt{V}} \rightarrow (12a)$ This approximation is called as allowed approximation.

$\phi_{\bar{\nu}}(\vec{r}) = \frac{1}{\sqrt{V}} \rightarrow (12b)$

Use (12a) and (12b) in (11)

$$V_{fi} = g\int \Psi_Y^* \left(\frac{1}{\sqrt{V}}\right)\left(\frac{1}{\sqrt{V}}\right) O_x \Psi_x \, dV$$

$$= \frac{g}{V}\underbrace{\int \Psi_Y^* O_x \Psi_x \, dV}_{M_{fi} \rightarrow \text{nuclear matrix element}}$$

$$\boxed{V_{fi} = \frac{gM_{fi}}{V}} \rightarrow (13)$$

Partial Decay Rate:

The partial decay rate for e^-'s & $\bar{\nu}$'s with the proper momenta is

$$d\lambda = \frac{2\pi}{\hbar}|V_{fi}|^2 d\rho(E_f)$$

Using results from (11) and (13)

$$d\lambda = \frac{2\pi}{\hbar}\frac{g^2|M_{fi}|^2}{\cancel{V^2}}\frac{(4\pi)^2\cancel{V^2}p^2dp\,q^2dq}{h^6\,dE_f} \rightarrow (14)$$

As final energy is given by

$$E_f = E_{e^-} + E_{\bar{\nu}}$$

Using relativistic energy expression

$$E^2 = p^2c^2 + m^2c^4$$

for e^-;

$$E_{e^-} = (p^2c^2 + m_e^2c^4)^{1/2}$$

for $\bar{\nu}$: $E_{\bar{\nu}}^2 = q^2c^2 + m_{\bar{\nu}}^2 c^4 \rightarrow 0$ (rest mass of $\bar{\nu} = 0$)

$\Rightarrow E_{\bar{\nu}} = qc$

so $E_f = (P^2c^2 + m_e^2c^4)^{1/2} + qc$

$\frac{dE_f}{dq} = c \Rightarrow \frac{dq}{dE_f} = \frac{1}{c}$ Using in (14)

$$\boxed{d\lambda = \frac{2\pi}{\hbar}\, \frac{g^2|M_{fi}|^2}{1}\, \frac{(4\pi)^2 P^2 dP\, q^2}{h^6 c}} \rightarrow (15)$$

All the factors, except momenta of e^- and $\bar{\nu}$, can be combined into a constant 'A'

so $\boxed{d\lambda = AP^2q^2dP} \rightarrow (16)$

→ Write this expression only in terms of mom. of e^-

$$d\lambda = \frac{A}{c^2} P^2 \left[Q - \sqrt{P^2c^2 + m_e^2c^4} + m_ec^2\right]^2 dP$$

This is the decay rate which is equal to the number of e^-s emitted with mom b/w P & $P+dP$.

$$d\lambda = N(P)dP = \frac{A}{c^2} P^2 \left[Q - \sqrt{P^2c^2 + m_e^2c^4} + m_ec^2\right]^2 dP \rightarrow (17)$$

As $Q = T_{e^-} + T_{\bar{\nu}}$

$Q = \sqrt{P^2c^3 + m^2c^4} - m_ec^2 + qc$

$q = \frac{Q - \sqrt{P^2c^2 + m^2c^4} + m_ec^2}{c}$

∴ Inserting in eq. (16).

$E_{e^-} = m_ec^2 + T_{e^-}$

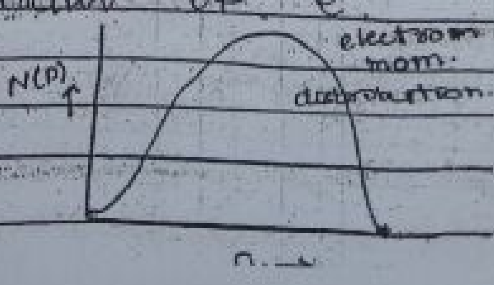
$T_{e^-} = E_{e^-} - m_ec^2$

$T_{e^-} = \sqrt{P^2c^2 + m_e^2c^4} - m_ec^2$

∵ $E_{\bar{\nu}} = m_{\bar{\nu}}c^2 + T_{\bar{\nu}}$

$qc = T_{\bar{\nu}}$

This function can explain the complete momentum spectrum of e^-.



at $P = 0$, $N(P) = 0$

where $P = P_{max}$ $(T_{e^-} = T_{max}) = Q$

again $N(P) = 0$

Electron Energy distribution

Consider eq

$$d\lambda = Aq^2p^2\,dP$$

Write this eq in terms of T_{e^-}

Hint: $E = (p^2c^2+m^2c^4)^{1/2}$

$$T_{e^-}+m_ec^2 = (p^2c^2+m_e^2c^4)^{1/2}$$

$p = \ldots$

$dP = \ldots\, dT_{e^-}$

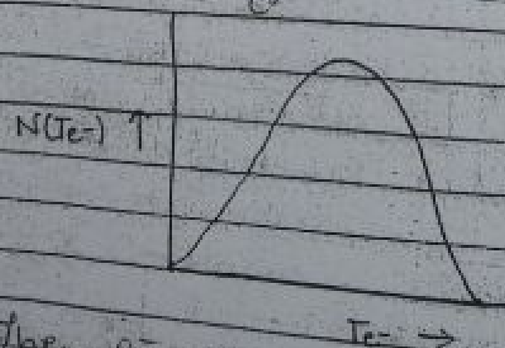
...al expression

$$d\lambda = A\left(\frac{(Q-T_{e^-})^2}{c^2}\right)\left(\frac{(T_{e^-}+m_ec^2)}{c^2}\right)\left(\frac{T_{e^-}^2+2T_{e^-}m_ec^2}{c^2}\right)^{1/2} dT_{e^-}$$

This eq is purely a function of T_{e^-} and thus gives the no of e^-s emitted with energy b/w T_{e^-} and $T_{e^-}+dT_{e^-}$

$$N(T_{e^-})\,dT_{e^-} = A\left(\frac{(Q-T_{e^-})^2}{c^2}\right)\left(\frac{(T_{e^-}+m_ec^2)}{c^2}\right)\left(\frac{T_{e^-}^2+2T_{e^-}m_ec^2}{c}\right)^{1/2} dT_{e^-}$$

$$N(T_{e^-}) = \frac{A}{c^5}(T_{e^-}+m_ec^2)(Q-T_{e^-})^2(T_{e^-}^2+2T_{e^-}m_ec^2)^{1/2}$$



when $T_{e^-}=0$, $N(T_{e^-})=0$ or visible in the graph.

$T_{e^-} = T_{e^-}(max) = Q$

$N(T_{e^-}) = 0$

The e^- energy distribution (e_q e^- mom distri) as predicted by Fermi theory of β-decay

From these graphs, we can see e^-s

emit with a certain range of K.E (mom.
This is the result as we also
observe experimentally.

Draw Experimental mom. and energy
distributions of e^-/e^+ (for Cu-decay)

$$\begin{cases} Cu \rightarrow Y + e^- + \bar{\nu} \\ Cu \rightarrow Y + e^+ + \nu \end{cases}$$

Considering eq. $d\lambda = A\, q^2 p^2\, dP \rightarrow$ (A)

Write the eq. in terms of T_{e^-}

As $E^2 = p^2c^2 + m^2c^4$

for e^- $\quad E_{e^-}^2 = p^2c^2 + m_{e^-}^2c^4$

Also $(m_{e^-}c^2 + T_{e^-})^2 = p^2c^2 + m_{e^-}^2c^4$

so $\quad P^2 = \dfrac{(m_{e^-}c^2 + T_{e^-})^2 - m_{e^-}^2c^4}{c^2} \rightarrow$ (i)

$$\rightarrow \quad P = \frac{\sqrt{(m_{e^-}c^2 + T_{e^-})^2 - m_{e^-}^2c^4}}{c}$$

$$dP = \frac{1}{\cancel{2}}\,\frac{[(m_{e^-}c^2 + T_{e^-})^2 - m_{e^-}^2c^4]^{-1/2}}{c}\,(m_{e^-}c^2 + T_{e^-})\cdot\cancel{2}\cdot dT_{e^-}$$

so $\quad dP = \dfrac{(T_{e^-} + m_{e^-}c^2)}{c\,\sqrt{(m_{e^-}c^2 + T_{e^-})^2 - m_{e^-}^2c^4}}\,dT_{e^-} \rightarrow$ (ii)

Also Considering for $\bar{\nu}$

$$E_{\bar{\nu}}^2 = q^2c^2 + \cancel{m_{\bar{\nu}}^2}c^4$$

Also $\quad (\cancel{m_{\bar{\nu}}}c^2 + T_{\bar{\nu}})^2 = q^2c^2$

$$T_{\bar{\nu}}^2 = q^2c^2 \rightarrow \text{(iii)}$$

& $\quad Q = T_{e^-} + T_{\bar{\nu}}$

$\Rightarrow \quad T_{\bar{\nu}} = (Q - T_{e^-}) \Rightarrow T_{\bar{\nu}}^2 = (Q - T_{e^-})^2$

so $\quad q^2c^2 = (Q - T_{e^-})^2 \rightarrow$ (iv) by (iii).

Multiplying & Dividing eq. (A) by c^2
and substituting eq. (i), (ii) & (iv), we get

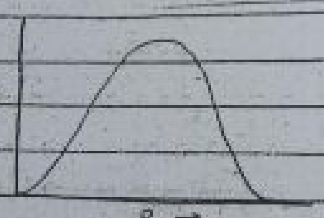
$$d\lambda = \frac{4A\,p^2c^2}{c^2}\left[\frac{(T_{e^-}+m_{e^-}c^2)^2-m_{e^-}^2c^4}{c^2}\right]\left[\frac{(T_{e^-}+m_{e^-}c^2)}{c}\right]\frac{dT_{e^-}}{\sqrt{(T_{e^-}+m_{e^-}c^2)^2-m_{e^-}^2c^4}}$$

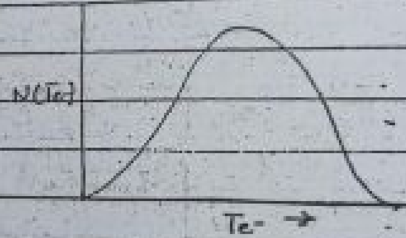
$$d\lambda = \frac{A}{c^5}(Q-T_{e^-})^2\left[(T_{e^-}+m_{e^-}c^2)^2-m_{e^-}^2c^4\right]^{1/2}(T_{e^-}+m_{e^-}c^2)\cdot dT_{e^-}$$

$$d\lambda = \frac{A}{c^5}(Q-T_{e^-})^2\left[T_{e^-}^2+\cancel{m_{e^-}^2c^4}+2m_{e^-}c^2T_{e^-}-\cancel{m_{e^-}^2c^4}\right]^{1/2}(T_{e^-}+m_{e^-}c^2)\cdot dT_{e^-}$$

$$d\lambda = \frac{A}{c^5}(T_{e^-}+m_{e^-}c^2)(Q-T_{e^-})^2(T_{e^-}^2+2T_{e^-}m_{e^-}c^2)^{1/2}\,dT_{e^-}$$

which is the required result.





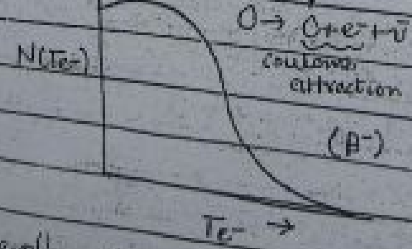
i.e. $e^-(e^+)$ emit having a range of momenta/energy.

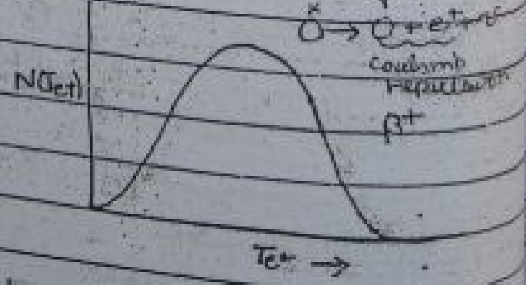
(Theoretical curves/results obtained by using Fermi Theory) (applicable for both β^-/β^+ decay)

→ Coulomb interaction neglected.

⇒ Allowed approximation used.

Experimental Results:





β^+ ⇒ General shape for theoretical and experimental curves agree.

In experimental results, please note that
→ For β^- decay : curve peak closer to the origin.
→ For β^+ decay : curve peak away from the origin

Semiclassical explanation

Difference originate because of Coulomb interaction b/w daughter nucleus and the emitted β^-/β^+ - particles

For β^- emission : Coulomb attraction of β^- by nucleus

⇓

results in a decrease in T_e/P_e of β^-.

⇓

Curve pushed towards origin.

For β^+ emission : Coulomb repulsion of β^+ by nucleus

⇓

gives an increase in T_e and P_e.

⇓

Curve pushed away from origin.

→ Improving theory by taking into account - Coulomb interaction :

If Coulomb interaction is to be considered - there will be change in the plane wave eq. brought about by the Coulomb potential

- The effect of the nuclear Coulomb field modifies the mom./energy spectrum by introducing an additional factor;

- Fermi function $F(z', P)$ or $F(z', T_e)$

z' : atomic no. of daughter nucleus.

$$N(P) = \frac{A}{c^2} P^2 [Q - \sqrt{P^2c^2 + m_e^2c^4} + m_ec^2]^2 \cdot F(z', P)$$

$$N(T_e) = \frac{A}{c^5} (T_e + m_ec^2)(Q - T_e)^2 (T_e^2 + 2T_e m_ec^2)^{1/2} F(z', T_e)$$

- Improved Fermi theory

So far, we used allowed approximation,
where $e^{i\vec{P}\cdot\vec{r}/\hbar} \cong 1 \Rightarrow \phi_e(\vec{r}) = 1/\sqrt{V}$
$e^{i\vec{q}\cdot\vec{r}/\hbar} \cong 1 \Rightarrow \phi_\nu(\vec{r}) = 1/\sqrt{V}$

So we had

$$V_{fi} = \frac{g}{V} \int \psi_x^* O_x \psi_x \, dV$$

$M_{fi} \rightarrow$ was constant before
$\rightarrow$ sometimes diverge.

- where M_{fi} : nuclear matrix element
independent of P and q

$\Rightarrow$ constant (does not influence the shape of the mom/energy spectrum)

- Mostly this approximation works but there are few cases in which M_{fi} vanishes in the allowed approximation
$\Rightarrow$ does not give any spectrum.

$\therefore$ For such cases, we must take the next term of the plane wave expansion equations;

$$e^{i\vec{P}\cdot\vec{r}/\hbar} = 1 + i\vec{P}\cdot\vec{r}/\hbar + \cdots$$

$$e^{i\vec{q}\cdot\vec{r}/\hbar} = 1 + i\vec{q}\cdot\vec{r}/\hbar + \cdots$$

- So in such cases M_{fi} has mom dependence (i.e. not a constant) so it will effect the mom/energy distribution.

$$|M_{fi}|^2 \rightarrow |M_{fi}|^2 S(P, q)$$

in allowed approximation

longer half lives

→ Such cases are called "forbidden decays" (not absolutely forbidden, but less likely to occur than allowed decays)

Note:

→ If only 1st term ($i\vec{p}\cdot\vec{r}/\hbar$) is included to avoid vanishing of M_{fi} → first forbidden decays.

→ If also 2nd term is to be considered → 2nd forbidden decays

Fermi-Kurie Plot

→ In allowed approximation:

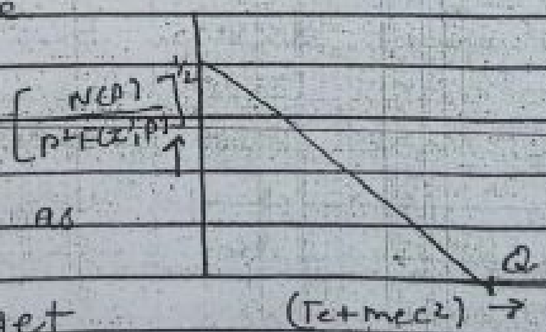
$$\begin{cases} S(P,Q) = 1 \\ |M_{fi}|^2 - \text{constant} \end{cases}$$

So $$N(P) \propto P^2 \left[Q - \sqrt{P^2c^2 + m_e^2c^4} + m_e c^2\right]^2 F(Z',P)$$

$$\left[Q - \sqrt{P^2c^2 + m_e^2c^4} + m_e c^2\right]^2 \propto \frac{N(P)}{P^2 F(Z',P)}$$

Plotting $\left[\dfrac{N(P)}{P^2 F(Z',P)}\right]^{1/2}$ against $(T_e + m_e c^2)$

→ Gives a straight line which intercepts the x-axis at the decay energy Q. This plot is known as Fermi-Kurie plot (A convenient way to get decay energy Q.)



→ Total decay rate:

It can be calculated by integrating the partial decay rate over all values of e^- mom

$$d\lambda = A P^2 (Q - T_e)^2 dP$$

$$\lambda = A\int_0^{p_{max}} p^2 (Q - T_{e^-})^2 \, dP$$

It can be used to calculate $t_{1/2}$ by using
* forbidden approximation * completeness of β-decay

HW-b
KRANE

Gamma Decay

→ Introduction
→ Energetics of γ-decay

Introduction:

→ α and β decays of radioactive nuclei usually leave the daughter nuclei in an excited state

$$X \rightarrow Y^* + \alpha$$
$$X \rightarrow Y^* + e^- + \bar{\nu}$$

→ If excitation energy of the daughter nucleus is not sufficient for further particle emission, then it loses its excess energy by following processes:
(i) Gamma decays
(ii) Internal Conversion
(iii) Internal Pair Conversion

→ Gamma decay takes place more often than the other two processes

(i) **Gamma decay**

When excited states decay to the ground state via emission of one or more γ-rays

$$Y^* \rightarrow Y + \gamma \quad \text{(2 body decay)}$$

E_Y, E_γ - fixed
(discreete energy spectrum)

→ **Properties of γ-rays:**

(i) Gamma rays are EM waves just like X-rays or visible light.

(ii) They travel with speed of light.

(iii) They do not cause any appreciable ionization and are not deflected by electric or magnetic fields.

(iv) Energies of γ-rays range b/w 0.1 MeV – 10 MeV

(v) The corresponding wave lengths $\lambda = hc/E$ → (0.01 nm – 0.0001 nm)

(vi) High penetrating power, much larger than α or β-particle.

• **Energetics of γ-decay:**

→ Let us consider the decay of nucleus of mass 'm' (which is initially at rest), from an initial excited state 'E_i' to final state 'E_f' $(Y^* \to Y + \gamma)$

→ Then according to law of conservation of energy

$$E_i = E_f$$

$$E_{Y^*} = E_Y + T_Y + E_\gamma$$

$$E_i = E_f + T_Y + E_\gamma \quad \rightarrow \text{let } E_Y = E_f \quad E_{Y^*} = E_i$$

where, T_Y = recoil K.E of daughter nucleus.

By law of conservation of mom.

$$0 = \vec{P}_Y + \vec{P}_\gamma$$

i.e. Y and γ should emit with equal and opposite momenta

$$|\vec{P}_Y| = |\vec{P}_\gamma| \rightarrow ②$$

$$T_Y = \frac{|P_Y|^2}{2M_Y} \text{ (non-relativistic as Y is heavy)}$$

$T_Y = \frac{1}{2} M_Y v^2 = \frac{1}{2} M_Y^2 v^2$ ...

Take eqn ①

$$E_i - E_f = E_\gamma + T_R$$

$$\Delta E = E_\gamma + \frac{|P_\gamma^2|}{2M_\gamma} \rightarrow ③$$

Since γ have speed of light using relativistic expression of energy

$$E_\gamma^2 = P_\gamma^2 c^2 + m_\gamma^2 c^4$$

$$E_\gamma = P_\gamma c \Rightarrow P_\gamma = \frac{E_\gamma}{c}$$

Use this in ③

$$\Delta E = E_\gamma + \frac{E_\gamma^2}{2M_\gamma c^2}$$

$$\frac{E_\gamma^2}{2M_\gamma c^2} + E_\gamma - \Delta E = 0$$

$$E_\gamma^2 + 2M_\gamma c^2 E_\gamma - 2M_\gamma c^2 \Delta E = 0$$

Quadratic eq. in E_γ

$$E_\gamma = \frac{-2M_\gamma c^2 \pm \sqrt{4M_\gamma^2 c^4 + 8 M_\gamma c^2 \Delta E}}{2}$$

$$E_\gamma = \frac{-2M_\gamma c^2 + 2M_\gamma c^2 \sqrt{1 + \frac{2\Delta E}{M_\gamma c^2}}}{2}$$

$$E_\gamma = -M_\gamma c^2 + M_\gamma c^2 \sqrt{1 + \frac{2\Delta E}{M_\gamma c^2}} \rightarrow ④$$

where $\Delta E = E_i - E_f = E_{ex}$

$(m_\gamma c^2 + E_{ex} - m_\gamma c^2)$

Difference in initial and final energies small → typically of the order of MeV

Eg $M_\gamma c^2$ = rest mass energy of γ ($\simeq A \times 10^3$ MeV)

$\Rightarrow \Delta E << M_\gamma c^2$ A = mass no.

So we can use binomial expansion

$\{(1+x)^{1/2} = 1 + \frac{1}{2}x + \frac{1}{2}(\frac{1}{2}-1)x^2/2! + \ldots\}$

Take $\left(1 + \frac{2\Delta E}{M_\gamma c^2}\right)^{1/2} = 1 + \frac{\Delta E}{M_\gamma c^2}$ ← neglecting higher terms

Use this results in (4)

$$E_\gamma \simeq -Mc^2 + Mc^2\left(1 + \frac{\Delta E}{Mc^2}\right)$$

$$E_\gamma \simeq -Mc^2 + Mc^2\left(1 + \frac{\Delta E}{Mc^2}\right)$$

we keep only positive term so

$$E_\gamma \simeq -\cancel{Mc^2} + \cancel{Mc^2} + \Delta E$$

$$E_\gamma \simeq \Delta E \rightarrow (5)$$

($\because$ for -ve term E_γ = -ve meaningless)

• **Internal Conversion:**

→ The process in which γ^* nucleus directly transfers its energy to one of the orbital electrons by which the orbital e^- is ejected from the atom, is known as "internal conversion". The ejected e^-'s known as conversion e^-'s

Explanation

Add from book Gupta (Copy).

• **Internal Pair Conversion:**

In addition to γ-emission and internal conversion, there is another competing process known as internal pair conversion (for the de-excitation of the nucleus)

→ If excitation energy of the nucleus (ΔE) is greater than $2m_ec^2$ (1.022 MeV), then the nucleus may de-excite by the creation of $e^- e^+$ pair. such a process is known as internal pair conversion. (Add explanation)

Theory of gamma decay
- Classical theory
- Q.M theory

Recall:
In γ-decay, a nucleus makes a transition from an excited state to a state of lower energy, by emitting γ-rays (EM radiations)

Mechanism of emission of EM radiation?
As we know that a nucleus carries a finite electric charge (protons) which is distributed according to the density distribution in the nucleus.

→ These charges set up a
- Charge density distribution &
- Current density distribution (due to motion of charges)

An oscillation of the charges and currents leads to the emission of gamma rays (EM-radiations)

$O^* \to O + \gamma$ → Oscillating electric & magnetic fields

→ For analyzing these EM radiations emitted by nuclei one needs Q.M theory of radiation.

→ To understand quantum calculations of EM radiations, we first review the classical description.

→ **Classical theory of EM radiation:**

Recall: ① Static distribution of electric charges & currents → Produce static electric & magnetic fields, can be analyzed in term of the multipole moments (dipole moment, quadrupole moment & so on)

These multipole moments give characteristic fields
↳ (dipole field, quadrupole field and so on)
& ② (Oscillating) Time-varying → Produce → Oscillating
distribution of charges & current
radiation electric and magnetic fields
↓
can be analyzed (like static fields) in
terms of their multipole character.
↓
they give rise to dipole radiation fields,
quadrupole radiation field and so on.

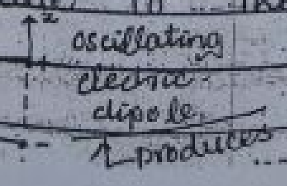
Note:
Lowest multipole order → dipole radiation
in this case field.

Next steps:
- We consider dipole radiation field
- Discuss its important characteristics from
classical theory
- then we will extend our results to higher
order multipole radiation fields.
- then we will make transition to Q-M.

→ **Dipole field**

{ Recall for static electric dipole : $d = qz$
{ Recall for magnetic electric dipole : $\mu = iA$

For Oscillating electric dipole:
If we allow charges
to oscillate along z-axis (e.g by connecting
them to a current source of frequency (ω)
(there should be no contribution to radiation
due to the current source.)



electric dipole moment
for varying / oscillating

Oscillating magnetic dipole :

Attach the current op to some external current source of frequency (ω), then the varying magnetic dipole moment can be calculated using $\mu(t) = I_0 A \cos \omega t$.

Characteristics of the dipole radiation field

3 characteristics are important

Diff multipole radiation fields have diff. angular mom. distribution in a properly chosen direction.

Although electric and ~~multipole~~ magnetic dipole radiation fields gives the same angular distribution, but they have opposite parity.

The average radiated power (energy radiated / unit time) is

$$P = \frac{1}{12\pi\epsilon_0} \frac{\omega^4}{c^3} d^2$$

for electric dipoles and

$$P = \frac{1}{12\pi\epsilon_0} \frac{\omega^4 \cdot \mu^2}{c^5}$$

for magnetic dipoles. Here d and μ represent the amplitudes of the time-varying dipole moments.

Extension of above results to multipole radiation fields :

We first define the index L of the radiation so that 2^L is the multipole order ($L=1$ for dipole, $L=2$ for quadrupole and so on). With E for electric and M for magnetic, we generalize above three properties of dipole radiation.

① The angular distribution of 2^L-pole radiation relative to a properly chosen direction, is governed by the Legendre Polynomial $P_{2L}(\cos\theta)$. The most common cases are dipole, for which $P_2 = \frac{1}{2}(3\cos^2\theta - 1)$ and quadrupole with $P_4 = \frac{1}{8}(35\cos^4\theta - 30\cos^2\theta + 3)$.

② The parity of the radiation field is

$$\pi(ML) = (-1)^{L+1}$$
$$\pi(EL) = (-1)^L$$

Notice that electric and magnetic multipoles of the same order always have opposite parity

③ The radiated power is, using $\sigma = E$ or M to represent electric or magnetic radiation

$$P(\sigma L) = \frac{2(L+1)c}{\epsilon_0 L[(2L+1)!!]^2}\left(\frac{\omega}{c}\right)^{2L+2}[m(\sigma L)]^2$$

where $m(\sigma L)$ is the amplitude of the (time varying) electric or magnetic multipole moment. The generalized multipole moment $m(\sigma L)$ differs for $L=1$ from d and μ.

Transition to Q.M:

We start with the classical result

$$P(\sigma L) = \frac{2(L+1)c}{\epsilon_0 L[(2L+1)!!]^2}\left(\frac{\omega}{c}\right)^{2L+2}[m(\sigma L)]^2 \rightarrow ①$$

→ To carry the results to Q.M Replace the amplitude of multipole moments by appropriate multipole operators, that change the nucleus from its initial ate ψ_i to the final state 'ψ_f'

$m(SL) \rightarrow m_{fi}(SL) = \int \psi_f^* \hat{m}(SL)\, \psi_i\, dv \rightarrow (2)$

classical (EM changes nucleus from initial ψ_i

final state ψ_f while simultaneously

rating photon of proper energy and parity

..(1) can be considered as the energy

diated per unit time in the form

photons

here energy of each photon $= \hbar\omega$

Probability per unit time for photon

mission is (i.e decay constant)

$$\lambda(SL) = \frac{P(SL)}{\hbar\omega} = \frac{2(L+1)}{\epsilon_0 \hbar L[(2L+1)!!]^2}\left(\frac{\omega}{c}\right)^{2L+1}$$

$$[m_{fi}(SL)]^2 \rightarrow (3)$$

r exact calculation of $\lambda(SL)$ — one needs
o evaluate matrix element $m_{fi}(SL)$

s evaluation requires the knowledge of
ψ_i and ψ_f (difficult)

- Estimates of $\lambda(SL)$ (Weisskoff estimates)

We can estimate the γ-ray emission probabilities, if we assume that the transition ($X^* \rightarrow X + \gamma$) is due to a single photon. We do separately for electric and magnetic transitions.

$$\lambda(EL) = \frac{2(L+1)}{\epsilon_0 \hbar L[(2L+1)!!]^2}\left(\frac{\omega}{c}\right)^{2L+1}[m_{fi}(SE)]^2$$

* the multipole operator $m(EL)$ includes the terms of the form

$$\hat{m}(EL) = \underbrace{e\, r^L}_{\text{about radial distribution}}\ \underbrace{Y_{LM}(\theta, \phi)}_{\text{about angular distribution}} \rightarrow (5)$$

As an approximation (estimate) we take the radial parts of ψ_i & ψ_f to be constant up to the nuclear radius 'R' and zero for $r > R$. then in

$$m_{fi}(EL) = \int_0^R \psi_f^* \, \hat{m}_i(EL) \, \psi_i \, dV$$

$$m_{fi}(EL) = \int_0^R \psi_f^* \, e r^L Y_{LM}(\theta,\phi) \, \psi_i \, r^2 dr \sin\theta \, d\theta \, d\phi$$

$$= \int (\phi_f \chi_f)^* e r^L Y_{LM}(\theta,\phi) (\phi_i \chi_i) \, r^2 dr \sin\theta \, d\theta \, d\phi$$

where ϕ_f, ϕ_i = radial parts of the W.F

χ_f, χ_i = angular parts of the W.F

~~For constant ϕ_i and ϕ_f upto 'R' the~~

$$\cancel{m_{fi}(EL) \propto e \int_0^R r^2 r^L dr \int_0^{2\pi}\int_0^{\pi} \chi_f^* Y_{LM}(\theta,\phi) \chi_i \sin\theta \, d\theta \, d\phi}$$

~~Multiplying and Dividing~~ by the factor $\int_0^R r^2 dr$ ~~for normalization~~

$$m_{fi}(EL) = \frac{e \int_0^R r^2 r^L dr \int_0^{2\pi}\int_0^{\pi} \chi_f^* Y_{LM}(\theta,\phi) \chi_i \sin\theta \, d\theta \, d\phi}{\int_0^R r^2 dr} \quad (2)$$

The term $\int_0^{2\pi}\int_0^{\pi} \chi_f^* Y_{LM}(\theta,\phi) \chi_i \sin\theta \, d\theta \, d\phi = 1$

So $$m_{fi}(EL) = \frac{e \int_0^R r^2 r^L dr}{\int_0^R r^2 dr}$$

$$= \frac{e \left[R^L \cdot R^3 / L+3 \right]}{R^3/3}$$

$$\boxed{m_{fi}(EL) = \frac{3e R^L}{L+3}}$$

In eq. (1) the term in the denominator is

...r

...luded for normalization and the ... factor comes from the volume element. In... the matrix element ... cling this factor in the angular integrals by ... replacing the ...

$$\lambda(EL) = \frac{2(L+1)}{\epsilon_0 \hbar L[(2L+1)!!]^2}\left(\frac{\omega}{c}\right)^{2L+1}\left(\frac{3}{L+3}\right)^2 e^2 R^{2L}$$

As $E = \hbar\omega \Rightarrow \omega = \frac{E}{\hbar}$

$$\therefore \lambda(EL) = \frac{2(L+1)}{\epsilon_0 \hbar L[(2L+1)!!]^2}\left(\frac{E}{\hbar c}\right)^{2L+1}\left(\frac{3}{L+3}\right)^2 e^2 R^{2L} \cdot \frac{4\pi}{4\pi}$$

$$\lambda(EL) = \frac{8\pi(L+1)c}{L[(2L+1)!!]^2}\underbrace{\left(\frac{e^2}{4\pi\epsilon_0\hbar c}\right)}_{\approx 1/137}\left(\frac{E}{\hbar c}\right)^{2L+1}\left(\frac{3}{L+3}\right)^2 R^{2L}$$

where $R = R_0 A^{1/3}$

For $L=1$

$$\lambda(E1) = \frac{8\pi(1+1)c}{1[(2+1)!!]^2}\left(\frac{1}{137}\right)\left(\frac{E}{\hbar c}\right)^3\left(\frac{3}{4}\right)^2 R^2$$

$$\simeq \left(\frac{1}{137}\right)\left(\frac{16\pi \cancel{c}}{\cancel{9}}\right)\frac{E^3}{\hbar^3 c^{\cancel{3}2}}\left(\frac{\cancel{9}}{\cancel{16}}\right) R_0^2 A^{2/3}$$

$$\boxed{\lambda(E1) \simeq \frac{1}{137}\left(\frac{\pi}{\hbar^3 c^2}\right) R_0^2 E^3 A^{2/3}}$$

$$\lambda(E1) = 1.0 \times 10^{14} A^{2/3} E^3$$

after inserting values of all constants.

Similarly,

for $L = 2, 3, 4, \ldots$

$$\left\{\begin{aligned} \lambda(E2) &\simeq 7.3 \times 10^7 A^{4/3} E^5 \\ \lambda(E3) &\simeq 34 A^2 E^7 \\ \lambda(E4) &\simeq 1.1 \times 10^{-5} A^{8/3} E^9 \end{aligned}\right\}$$

where E is in MeV.

For magnetic transitions

$$\lambda(ML) = \frac{2(L+1)}{\epsilon_0 \hbar L [(2L+1)!!]^2} \left(\frac{\omega}{c}\right)^{2L+1} [m_{fi}(\delta M)]^2$$

Here for magnetic transitions

$$\hat{m}(ML) = e r^{L-1} Y_{LM}(\theta, \phi) \left\{ \left(\mu_p - \frac{1}{L+1}\right)\left(\frac{\hbar}{m_p c}\right) \right\}$$

related to magnetic moment of proton

magnetic moment of proton

mass of Proton

Preceeding in the same way, as we did for electric transitions calculate

$$m_{fi}(ML) = \int \psi_f^* \, \hat{m}(ML) \, \psi_i \, dV$$

$$\lambda(ML) \cong \frac{8\pi(L+1)}{L[(2L+1)!!]^2} \left(\mu_p - \frac{1}{L+1}\right)^2 \left(\frac{\hbar}{m_p c}\right)^2 \left(\frac{e^2}{4\pi\epsilon_0 \hbar c}\right) \left(\frac{E}{\hbar c}\right)^{2L+1} \left(\frac{3}{L+2}\right)^2 c R^{2L-2}$$

where the factor

$$\left(\mu_p - \frac{1}{L+1}\right)^2 \simeq 10$$

for $L = 1, 2, 3, 4$,

$\lambda(M1) \cong 5.6 \times 10^{13} E^3$

$\lambda(M2) \cong 3.5 \times 10^{7} A^{2/3} E^5$

$\lambda(M3) \cong 16 A^{4/3} E^7$

$\lambda(M4) \cong 4.5 \times 10^{-6} A^2 E^9$

These estimates for the transition rates are known as Weisskoff estimates.

→ Conclusions:

Based on Weisskoff estimates, we can draw following main conclusions about transition probabilities

① lower multipoles are dominant.

– Increasing the multipole order by one

it reduces the transition probability by a factor of roughly 10^{-5}

) For a given multipole order (e.g $L=1$ or $L=2$), the electric radiation is more likely than magnetic radiation (by ~ 2 orders of magnitude)

Multipolarity of γ-rays: (V. Imp)
(Angular Momentum and Parity Selection Rules)

Note: As we know that
A classically EM field produced by oscillating charges and current transmit energy. this result in Quantum limit can be considered as energy radiated per unit time in the form of photons.

→ A classical EM field also transmit angular momentum
{In Quantum limit we consider it as each photon carries a definite angular momentum

→ Recall, the multipole operator of order 'L' $\hat{m}(\sigma L)$ also includes the factor $Y_{LM}(\theta, \varphi)$ which is associated with an angular mom. L
⇒ a multipole of order 'L' transfers an angular mom. of $L\hbar$ per photon.

→ Consider a γ-decay process
$$X^* \rightarrow X + \gamma$$
where the initial excited state nucleus carries an angular mom 'I_i' and parity 'π_i' and transforms via γ-emission into ground state nucleus which carries angular mom 'I_f' and parity 'π_f'

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we will consider different cases depending on whether angular momenta and parities of the nuclei change or not.

Cases

(i) when $I_i \neq I_f$

(a) $\Pi_i = \Pi_f$ ($\Pi_i = \Pi_f = (\Delta\Pi) = no$)

(b) $\Pi_i \neq \Pi_f$ ($\Pi_i - \Pi_f = (\Delta\Pi) = Yes$)

(ii) When $I_i = I_f$

(iii) When $I_i = 0$ or $I_f = 0$

(iv) When $I_i = I_f = 0$

Assume **Case (i)**

When $I_i \neq I_f$

then by conservation of angular mom

$$\vec{I}_i = \vec{I}_f + \vec{L}$$

$$\vec{L} = \vec{I}_i - \vec{I}_f$$

$$\text{or} \quad \vec{L} = \vec{I}_i + (-\vec{I}_f)$$

The possible values of $\vec{L}$ are restricted to

$$|I_i - I_f| \leq L \leq (I_i + I_f)$$

smallest possible value of L — Largest possible value of L.

Example

if $I_i = 3/2$ & $I_f = 5/2$

then $|3/2 - 5/2| \leq L \leq (3/2 + 5/2)$

$$1 \leq L \leq 4$$

$$L = 1, 2, 3, 4$$

in this case, the radiation field would consist of a mixture of dipole ($L=1$), quadrupole ($L=2$), octopole ($L=3$) and hexadecapole ($L=4$) radiation fields.

→ The parity selection rules will tell us about the nature/type of the radiation.

(electric or magnetic)

If there is no change in parity

i.e $\Pi_i = \Pi_f$ $(\Delta\Pi = \text{no})$

→ by conservation of parity

$\Pi_i = \Pi_f \, \Pi_\gamma$

radiation field (γ) must have an even parity

If there is change in parity

i.e $\Pi_i \neq \Pi_f$ $(\Delta\Pi = \text{Yes})$

→ by conservation of parity

$\Pi_i = \Pi_f \, \Pi_\gamma$

radiation field must have an odd parity.

we by eq.

$$\Pi(EL) = (-1)^L$$

& $$\Pi(ML) = (-1)^{L+1}$$

Electric field or electric transitions are even parity if L = even

while

Magnetic field or magnetic transitions are even parity if L = odd

therefore

a $\Delta\Pi = \text{no}$ transition would consist of even electric multipoles and odd magnetic multipoles

&

a $\Delta\Pi = \text{Yes}$ transition on the other hand would consist of odd electric and even magnetic multipoles

In example

$$X^* \rightarrow X + \gamma$$

$I_i = \frac{3}{2}$ & $I_f = \frac{5}{2}$

$L: 1, 2, 3, 4$

If $\Delta\pi$ = no

then
$$\begin{cases} L=1 & \text{radiation must be of magnetic character} \\ L=2 & \text{" " electric "} \\ L=3 & \text{" " magnetic "} \\ L=4 & \text{" " electric "} \end{cases}$$

Thus for this example the radiation field (EM field) must be

M1, E2, M3, E4 radiation

Thus we have the following angular mom. & parity selection rules.

$$|I_i - I_f| \leq L \leq (I_i + I_f)$$

$\Delta\pi$ = no, even electric, odd magnetic

$\Delta\pi$ = yes, even magnetic, odd electric.

Assume Case (ii)

When $I_i = I_f$

then by $|I_i - I_f| \leq L \leq I_i + I_f$

$$0 \leq L \leq (I_i + I_f)$$

Here occurs exception to the angular momentum selection rule, because there are no monopole ($L=0$) transitions.

For such cases ($I_i = I_f$), then the lowest possible γ-ray multipole order is always dipole ($L=1$)

In example

$I_i = 3/2$ & $I_f = 3/2$

$$0 \leq L \leq 3$$

$$L: 0, 1, 2, 3$$

If $\Delta\pi$ = no M1, E2, M3

If $\Delta\pi$ = Yes E1, M2, E3

Case (iii) If $I_i = 0$ or $I_f = 0$
then by $|I_i - I_f| < L < (I_i + I_f)$
$I_f < L < I_f$

Only a pure multipole radiation is emitted (i.e no mixing of other multipoles).

Case (iv) If $I_i = 0$ and $I_f = 0$
then $|I_i - I_f| < L < (I_i + I_f)$
$0 < L < 0$

$L = 0$ which is not permitted for radiative transitions.
These states decay through internal conversion or internal pair conversion.

CHAPTER # 11 (KRANE) NUCLEAR REACTIONS CHAPTER # 10, 11 (Ghoshal)

{ An introduction
{ Types of nuclear reactions

→ **An Introduction**

$$\underset{\text{projectile}}{a} + \underset{\text{target}}{X} \rightarrow \underset{\text{residual nucleus}}{Y} + \underset{\text{emitted particle}}{b}$$

Definition

The process of bombarding a stable nucleus (called target) by fast moving charged/neutral particles (called projectiles) and the subsequent interaction b/w the two, which alters the composition, energy etc. of the target nucleus is known as "nuclear reaction." This is also known as transformation of a nucleus.

Note: In this process, no. of protons or neutrons or both inside the target nucleus may change

→ If the proton no. 'Z' is changed
↳ transformation from one element to other
→ If the neutron no. 'N' is changed
↳ transformation from one isotope of an element into another of the same element

→ In general a nuclear reaction can be represented by an equation in the following form:

$a + X \rightarrow Y + b$ or simply as $X(a,b)Y$

In many cases, the projectile 'a' and the emitted are light nuclei/particles, such as proton (p), neutron (n), deutron (d), α-particles or γ-rays.

$A \leq 4$ – light particles

→ These days, also heavy ion collisions: $A > 4$, $A < 40$ are also performed for certain studies.

Note:

If more than one particles are emitted in a nuclear reaction

$a + X \rightarrow Y + b_1 + b_2 + b_3$

$X(a, b_1, b_2, b_3)Y$

Depending upon the energies of incident projectiles, we have following 3 categories of nuclear reactions

Low energy reactions $E < 10$ MeV
Medium " " $10\,\text{MeV} < E < 1\,\text{GeV}$
High " " $E > 1\,\text{GeV}$

Such high energies can be achieved by using particle accelerators.

...es of Nuclear Reactions: There are several ways of classifying nuclear reactions. Two commonly used ways are:
Reactions based on reaction mechanism.
Reactions based on the mass of projectile

Reactions based on reaction mechanism:

① Elastic scattering ② Inelastic scattering
③ Compound nuclear reactions
④ Radiative Capture ⑤ Photodisintegration
⑥ Direct Reactions
↓
Transfer Reactions
(a) Stripping reactions
(b) Picking reactions
7) Knock out reactions
8) Heavy-ion reactions
9) Many-body reactions
10) Resonance Reactions
11) Fission / Fusion Reactions
12) Radioactive decays

① **Elastic Scattering:**

- In elastic scattering, the ejected particle 'b' is the same as the projectile 'a'.
- There is no loss or gain of energy of the incident projectile.
- The residual nucleus 'Y' is same as

the target nucleus 'X' and is left in the same state as X (i.e. ground state)

$$a + X \rightarrow X + a \qquad X(a,a)X$$

The direction of the incident particle generally changes.

Example

Rutherford scattering Experiment

$${}^{4}_{2}He + {}^{197}_{79}Au \rightarrow {}^{197}_{79}Au + {}^{4}_{2}He$$

$${}^{1}_{0}n + {}^{208}_{92}U \rightarrow {}^{208}_{92}U + {}^{1}_{0}n$$

② In-elastic Scattering:

– The ejected particle 'b' is the same as the incident one 'a' but it has different energy and different angular mom.

– In inelastic scattering the incident particles loses some of its K.E to the target nucleus The raises the internal energy of the target nucleus and leaves it in an excited state.

$$a + X \rightarrow X^* + a \qquad X(a,a)X^*$$

This excited nucleus decays by γ-emission

$$X^* \rightarrow X + \gamma$$

e.g. $${}^{1}_{1}H + {}^{27}_{13}Al \rightarrow {}^{27}_{13}Al^* + {}^{1}_{1}H$$
$$\hookrightarrow {}^{27}_{13}Al + \gamma$$

③ Compound Nuclear Reactions:

– The incident nucleus is absorbed by the target nucleus and a compound system is formed.

This compound system lives for sometime

10^{-14} to 10^{-15} sec and then a particle group of particles is ejected from.

st. $a + X \rightarrow [C] \rightarrow Y + b$
[compound nucleus]

e.g. ${}^{4}_{2}He + {}^{27}_{13}Al \rightarrow [{}^{31}_{15}P] \rightarrow {}^{30}_{15}P + {}^{1}_{0}n$

${}^{4}_{2}He + {}^{60}_{28}Ni \rightarrow [{}^{64}_{30}Zn] \rightarrow {}^{62}_{29}Cu + {}^{1}_{1}H + {}^{1}_{0}n$

(4) **Radiative Capture:**

In such reactions, the incident projectile is (absorbed) captured by the target nucleus to form an excited compound nucleus, which subsequently goes down to the ground state by emission of γ-rays.

$a + X \rightarrow [Y]^* \rightarrow Y + \gamma$

e.g. ${}^{4}_{2}He + {}^{12}_{6}C \rightarrow [{}^{16}_{8}O]^* \rightarrow {}^{16}_{8}O + \gamma$

(5) **Photodisintegration** (nuclear photoeffect)

High energy γ-rays photons are absorbed by the target nucleus and then nucleus ejects a particle or a group of particles.

$\gamma + X \rightarrow Y + b \quad X(\gamma, b)Y$

e.g. $\gamma + {}^{2}_{1}H \rightarrow {}^{1}_{1}H + {}^{1}_{0}n$

(6) **Direct Reactions**

- Only very few nucleons take part in the reaction, with the remaining nucleons of the target serving as passive spectators.

- The typical reaction time of direct reactions

is from $\sim 10^{-21}$ to 10^{-22} sec.
An important class of direct reactions
is: Transfer reactions:

In this class of nuclear reactions, one or two nucleons are transferred b/w the projectile and the target. There are two categories.

1) Stripping reactions
2) Pick up reactions

→ Stripping Reactions:

If nucleons are transferred from projectile to the target

e.g $^{2}_{1}H + ^{16}_{8}O \rightarrow ^{17}_{9}F + ^{1}_{0}n$

(1 P of the projectile transferred to the target) $(a + X \rightarrow Y + b)$

$^{4}_{2}He + ^{14}_{7}N \rightarrow ^{17}_{8}O + ^{1}_{1}H$

(1 P and 2n are transferred from projectile to the target)

→ Picking Reactions:

In picking reactions target transfers some of its nucleons to the projectile (or some nucleons are picked up by projectile).

e.g $^{1}_{1}H + ^{7}_{3}Li \rightarrow ^{6}_{3}Li + ^{2}_{1}H$

$a + X \rightarrow \quad + b$

(1n is picked up by projectile)

③ Knockout Reactions:

Sometimes 'a' and 'b' is the reaction are the same particles, but

uses get another nucleon to be
jected separately (3 particles in final
state).

$a + X \rightarrow Y + a + c$ $\quad X(a, a, c)Y$

e.g $\quad p + {}^{16}_{8}O \rightarrow {}^{15}_{7}N + p + p$

$\quad p + {}^{16}_{8}O \rightarrow {}^{15}_{8}O + p + n$

8) **Heavy-Ion Reactions:**

The reactions in which
incident particle is a fast moving
heavy ion (heavier than α) is known
as heavy ion reactions. Such reactions
usually take place at fairly high
energies of the projectile.

e.g $\quad {}^{10}_{5}B + {}^{16}_{8}O \rightarrow {}^{22}_{11}Na + {}^{4}_{2}He$

9) **Many-Body Reactions:**

When the K.E of
an incident particle (projectile) is high,
two or more particles can come out.

e.g $\quad {}^{1}_{1}H + {}^{16}_{8}O \rightarrow {}^{15}_{7}N + {}^{1}_{1}H + {}^{1}_{1}H$

10) **Resonance Reactions:**

Between the two
extremes (of direct reactions and compound
nuclear reactions) are resonance reactions
in which particle forms a quasi-bound
state before outgoing particle is (less-stable)
ejected.

11) **Fission Reactions:**

When heavy nuclei capture

incident particle (usually neutron), they break into generally two smaller fragments with the emission of few particles

e.g $\; {}^{1}_{0}n + {}^{235}_{92}U \rightarrow {}^{143}_{56}Ba + {}^{90}_{36}Kr + 3\,{}^{1}_{0}n$

Fusion Reactions:

When two light nuclei combine to form a heavy nucleus, the reaction is known as fusion reaction

e.g $\; {}^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{3}_{2}He + {}^{1}_{0}n$

+ Reactions based on the mass of Projectile:

(a) Reactions induced by particle of mass '1' (A=1) — P, n

- **Proton induced reactions**

$a = P({}^{1}_{1}H)$ $\quad P + X \rightarrow Y + P \quad X(P,P)Y$

$Y + n \quad X(P,n)Y$

$Y + {}^{2}_{1}H \quad (P, {}^{2}_{1}H)$

$Y + {}^{3}_{1}H \quad (P, {}^{3}_{1}H)$

$Y + {}^{4}_{2}He \quad (P, \alpha)$

- **Neutron Induced Reactions**

(n, P), (n, n), (n, α) $(n, {}^{3}_{1}H)$ - - -

- ~~Deut~~ (b) Reactions induced by Particle of mass 2 (A=2) (Deutron)

- **Deutron induced reactions**

(d,p), (d,n), (d,α) - - -

for mass = 3 $\quad {}^3_1H$, ${}^3_2He$
$({}^3_1H, p)$, $({}^3_1H, n)$, - - -

for mass = 4
(α, p), (α, n), (α, α), (α, d), - -

for mass = 0
γ-ray induced reactions. (Photodisintegration)
Add from Ghoshal → All above.

Conservation laws of Nuclear Reactions

Conservation laws tell whether a nuclear reaction will occur or not. There are certain physical quantities which should remain conserved in a nuclear reaction.

1) Conservation of energy
2) " " Linear momentum.
3) Conservation " angular "
4) Conservation of Charge (Proton no)
5) " " nucleons (Mass no).
6) " " Spin
7) " " Parity
8) " " Lepton no

Energetics of Nuclear Reactions

- Q-value of a nuclear reaction
- Threshold energy

→ Consider a general nuclear reaction

$$a + X \rightarrow Y + b \rightarrow (1)$$

(reactants) (Products)

→ By Conservation of energy

$$E_i = E_f$$

$$E_a + E_X = E_Y + E_b$$

$$m_a c^2 + T_a + m_X c^2 + T_X = m_Y c^2 + T_Y + m_b c^2 + T_b$$

where T's are K.E's and m's are rest masses

$$[(m_a + m_X) - (m_Y + m_b)]c^2 = (T_Y + T_b) - (T_a + T_X)$$

$$(m_i - m_f)c^2 = T_f - T_i \rightarrow (2a)$$

where Q-value $Q = (m_i - m_f)c^2$ diff. in the rest-mass energies of reactants & products

$Q = T_f - T_i$ (excess K.E of final products).

↳ (2b)

$$Q = (T_Y + T_b) - (T_a + T_X)$$

→ Q-value may be +ve or -ve.

Case 1. $Q > 0$

$\Rightarrow m_i > m_f$ or $T_f > T_i$

Such reactions — Exoergic or Exothermic reactions.

↳ excess mass appears as K.E of products.

Case 2. $Q < 0$

$\Rightarrow m_i < m_f$ or $T_i > T_f$

→ Reactions — Endoergic or Endothermic reactions.

↳ initial K.E (of reactants) is converted into nuclear mass (of products).

Note:

Endoergic reactions will not occur unless a certain min K.E is carried into the system

the Projectile — min. amount of energy
needed to initiate a process.
(threshold energy)

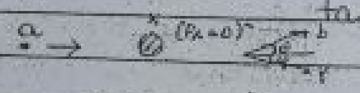
exoergic reactions initial K.E of projectile
only needed to approach the target
to overcome the Coulomb barrier (if
projectile is +vely charged) but reaction in
itself occurs spontaneously.

The changes in mass and energy are
related by $\Delta E = \Delta mc^2$

Eq-s (2a) and (2b) are valid in any
frame of reference in which we want
to work.

Lab frame of reference:

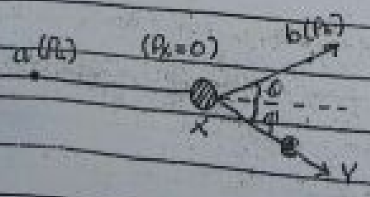
target is at rest



Centre of mass frame of reference:

(velocity of com is zero).

• Consider lab frame of reference
Target is at rest $(P_X = 0)$.



→ We define reaction plane by the direction of the incident projectile and one of the outgoing particles 'b'

• the 2nd outgoing particle 'Y' must lie in the same plane for conservation of momentum.
Conserving linear mom. along and perpendicular to the incident direction.

$$\begin{cases} P_a = P_b\cos\theta + P_Y\cos\phi \rightarrow (3a) \\ 0 = P_b\sin\theta - P_Y\sin\phi \rightarrow (3b) \\ Q = T_Y + T_b - T_a \rightarrow (3c) \end{cases}$$

* these eq.s Q - known by $(m_i - m_f)c^2$
T_a - parameter which we control.

(3a), (3b), (3c) three eq/s, 4 - unknows.

Q, ϕ, T_b and T_Y.

* We eliminate ϕ and T_Y from these eq.s to find a relationship b/w T_b and Q.

the eq. $Q = T_Y + T_b - T_a \Rightarrow \underline{\underline{T_Y = Q + T_a - T_b}}$

$$P_Y\cos\phi = P_a - P_b\cos\theta$$
$$P_Y\sin\phi = P_b\sin\theta$$

Squaring and Adding last two eq.s

$$P_Y^2\cos^2\phi + P_Y^2\sin^2\phi = P_a^2 + P_b^2\cos^2\theta - 2P_aP_b\cos\theta + P_b^2\sin^2\theta$$

$$P_Y^2 = P_a^2 + P_b^2 - 2P_aP_b\cos\theta$$

As $T = P^2/2m \Rightarrow P^2 = 2mT$

$$2m_YT_Y = 2m_aT_a + 2m_bT_b - 2(4m_aT_am_bT_b)^{1/2}\cos\theta$$

$$2T_Y = \frac{2m_a}{m_Y}T_a + \frac{2m_b}{m_Y}T_b - \frac{2}{m_Y}(4m_aT_am_bT_b)^{1/2}\cos\theta$$

or $$T_Y = \frac{m_a}{m_Y}T_a + \frac{m_b}{m_Y}T_b - \frac{2}{m_Y}(m_aT_am_bT_b)^{1/2}\cos\theta$$

$$Q + T_a - T_b = \frac{m_a}{m_Y}T_a + \frac{m_b}{m_Y}T_b - \frac{2}{m_Y}(m_aT_am_bT_b)^{1/2}\cos\theta$$

$$Q = \left(\frac{m_b}{m_Y} + 1\right)T_b - \left(1 - \frac{m_a}{m_Y}\right)T_a - \frac{2}{m_Y}(m_am_bT_aT_b)^{1/2}\cos\theta \rightarrow (4)$$

It is called $\underline{\text{Q-value}}$ equation quadratic in $\sqrt{T_a}$ in $\sqrt{T_b}$.

Let us take it to be quadratic in $\sqrt{T_b}$ as for a given value of T_a.

$$\underbrace{\left(1 + \frac{m_b}{m_Y}\right)}_{a}T_b - \underbrace{\frac{2}{m_Y}(m_am_bT_aT_b)^{1/2}\cos\theta}_{b} - \underline{\left\{\left(1 - \frac{m_a}{m_Y}\right)T_a + Q\right\}} = 0$$

e $a = \left(1 + \frac{m_b}{m_Y}\right)$ $b = -\frac{2}{m_Y}(m_a m_b T_a)^{1/2}\cos\theta$

and $c = -\left[Q + T_a\left(1 - \frac{m_a}{m_Y}\right)\right]$

Using quadratic formula

$$\sqrt{T_b} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{2/m_Y\,(m_a m_b T_a)^{1/2}\cos\theta \pm \sqrt{4/m_Y^2\,(m_a m_b T_a)\cos^2\theta + 4/m_Y^2\,(m_Y + m_b)\{Q m_Y + T_a(m_Y - m_a)\}}}{2/m_Y\,(m_Y + m_b)}$$

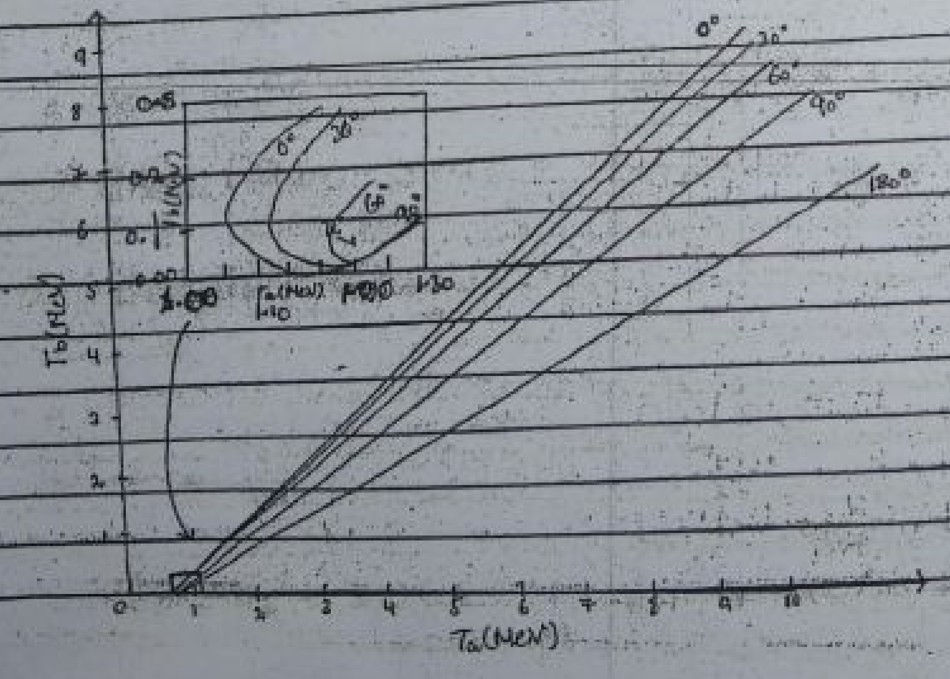
$$= \frac{2/m_Y\,(m_a m_b T_a)^{1/2}\cos\theta \pm 2/m_Y\sqrt{m_a m_b T_a\cos^2\theta + (m_Y + m_b)\{Q m_Y + T_a(m_Y - m_a)\}}}{2/m_Y}$$

?, we get

$$= \frac{(m_a m_b T_a)^{1/2}\cos\theta \pm \left[m_a m_b T_a\cos^2\theta + (m_Y + m_b)\{Q m_Y + (m_Y - m_a)T_a\}\right]^{1/2}}{m_Y + m_b} \quad \hookrightarrow (5)$$

eq. (5) gives relationship b/w T_b and θ (independent of ϕ and T_Y). (Fig 11-9 KRANE).

Graph b/w T_b Vs. T_a.



→ Plot of T_b Vs T_a ($^3H(P,n)^3He$) $P + {}^3H \rightarrow {}^3He + n$

Q<0 Endothermic reaction $m_f > m_i$

- For a given 'T_a' there is one to one correspondance b/w 'T_b' and 'θ' {i.e. keeping the incident energy fixed, choosing a value of 'θ' to observe the outgoing particles, automatically select their energy.}
- Exception in a very small energy region (1.019 - 1.147 MeV) → We observe double-valued behaviour.

→ **Features of Graph:**

① Threshold energy

② Double - Valued Behaviour

③ θ_m (max. angle) at which double-valued behaviour occurs.

These features of graph can be shown explicitly from eq. ⑤

① **Threshold Energy**

— For reactions with Q<0 there is an absolute min value of 'T_a', below which reaction is not possible → threshold energy.

- Threshold condition always occurs for $\theta = 0°$ ($\theta = 0$) (no energy wasted by giving b and Y, a mom. transverse to the beam direction).

→ $T_{th} = ?$

Use $\theta = 0°$ in eq. ⑤

$$T_b^{1/2} = \frac{(m_a m_b T_a)^{1/2} \pm [m_a m_b T_a + (m_Y + m_b)\{Q m_Y + (m_Y - m_a) T_a\}]^{1/2}}{m_Y + m_b}$$

Note: (T_b can never be -ve or imaginary).

For this example we have $Q = -Q$

As a 1st-step e.g. let us take $T_a = 0$

At this value $T_b^{1/2} = 0 \pm \sqrt{0 + (m_Y + m_b)(-m_Y Q)}$

$\Rightarrow T_b^{1/2} \rightarrow$ imaginary (impossible).

$\Rightarrow$ We have to increase T_a:

For $T_a > 0$

① $m_a m_b T_a \rightarrow +ve$

② $(m_x + m_b) m_x Q \rightarrow -ve$

③ $(m_x + m_b)(m_y - m_a) T_a \rightarrow +ve$

That value of T_a at which sum of 1st and 3rd term = 2nd term.

In that case imaginary root can be avoided.

And this value of $T_a \Rightarrow T_{th}$.

$\Rightarrow$ at $T_a = T_{th}$

$\sqrt{m_a m_b T_{th} + (m_x + m_b)\{m_x Q + (m_x - m_a) T_{th}\}} = 0$

$m_a m_b T_{th} + (m_x + m_b) m_x Q + (m_x + m_b)(m_y - m_a) T_{th}]^{1/2} = 0$

$T_{th} [m_a m_b + m_x^2 - m_a m_x + m_b m_x - m_a m_b]$

$= -Q m_x (m_x + m_b)$

$T_{th}\, m_x [m_x - m_a + m_b] = -Q\, m_x (m_x + m_b)$

$$\boxed{T_{th} = \frac{-Q(m_x + m_b)}{m_x - m_a + m_b}} \rightarrow ⑥$$

2) **Double Valued Behaviour:**

$\hookrightarrow$ occurs at values of incident energies b/w T_{th} and T_a' (upper limit)

- Only occurs for reactions with $Q < 0$.

$\rightarrow$ Range of double-valued behaviour:

$T_a' - T_{th} = ?$

① and

$\Rightarrow$ Such reactions neither have and nor a double value

$\Rightarrow$ curves are single-valued

$m_i > m_f \Rightarrow Q > 0$

Problems

CH #11 (Krane)

$a + X \rightarrow Y +$

$Q = (m_0 - m$

Example: $m \Rightarrow$

$a + X \rightarrow Y^*$

$Q = [(m_a + m_X) - ($

$Q = [m_a + m_X -$

$Q = Q_0 -$

S.Q Q-value for a nucleus is left in

Cross section of a

What is probability of a nuclear Reaction to occur

just single parti[cle] particles "a" . So[me] other goes with probability

Note:

Dominant and nu[clear]

Nuclear D

$\Rightarrow T_b^{1/2} \rightarrow$ imaginary (impossible).

$\Rightarrow$ We have to increase T_a?

For $T_a > 0$

① $m_a m_b T_a \rightarrow +ve$

② $(m_x + m_b) m_x Q \rightarrow -ve$

③ $(m_x + m_b)(m_y - m_a) T_a \rightarrow +ve$

That value of T_a at which sum of 1st and 3rd term = 2nd term in that case imaginary root can be avoided and this value of $T_a \rightarrow T_{th}$.

$\Rightarrow$ at $T_a = T_{th}$

$\sqrt{m_a m_b T_{th} + (m_x + m_b)\{m_x Q + (m_y - m_a) T_{th}\}} = 0$

$[m_a m_b T_{th} + (m_x + m_b) m_x Q + (m_x + m_b)(m_y - m_a) T_{th}]^{1/2} = 0$

$T_{th} [\cancel{m_a m_b} + m_x^2 - m_a m_x + m_b m_y - \cancel{m_a m_b}]$

$= -Q m_x (m_y + m_b)$

$T_{th}\, m_x [m_x - m_a + m_b] = -Q\, m_x (m_y + m_b)$

$$\boxed{T_{th} = \frac{-Q (m_y + m_b)}{m_x - m_a + m_b}} \rightarrow ⑥$$

→ $T_a^{1/2}$ → imaginary (impossible)

⇒ We have to increase "T_a"

for $T_a > 0$

① $m_a m_b T_a \to +ve$

② $(m_Y + m_b) m_Y Q \to -ve$

③ $(m_Y + m_b)(m_Y - m_a) T_a \to +ve$

That value of T_a at which sum of 1st and 2nd term = 2nd term in that case imaginary root can be avoided and this value of $T_a \to T_{th}$

→ at $T_a = T_{th}$

$$\sqrt{m_a m_b T_{th} + (m_Y + m_b)\{m_Y Q + (m_Y - m_a) T_{th}\}} = 0$$

$$m_a m_b T_{th} + (m_Y + m_b) m_Y Q + (m_Y + m_b)(m_Y - m_a) T_{th}]^{1/2} = 0$$

$$T_{th} [m_a m_b + m_Y^2 - m_a m_Y + m_b m_Y - m_a m_b] = -Q m_Y (m_Y + m_b)$$

$$T_{th}\, m_Y [m_Y - m_a + m_b] = -Q\, m_Y (m_Y + m_b)$$

$$\boxed{T_{th} = \frac{-Q (m_Y + m_b)}{m_Y - m_a + m_b}} \to ⑥$$

2) Double Valued Behaviour:

↳ occurs at values of incident energies b/w T_{th} and T_a' (upper limit)

- Only occurs for reactions with $Q < 0$.

⇒ Range of double-valued behaviour:

$$T_a' - T_{th} = ?$$

Where T_{th} as given in eq. ⑥ and T_a' is unknown

To find T_a' the condition is

$(m_Y + m_b) m_Y Q + (m_Y \ldots$

→ Such reactions neith...
and nor a double...
→ Curves are single-...
$m_i > m_e$ ...

Problems CH # 11 (Kra...

a + X ...

Q = (...

a + X ...

Q = [(ma + m...

Q = [ma + m...

Q = ...

S.Q Q-value for ... nucleus is left...

Cross section of ...

What is probability of a nuclear Reactio... to occur

just single par... particles "a". S... other goes with... probability.

Note:

① Dominant ... and nuc...

Nuclear De...

A → ...

$\Rightarrow T_b^{1/2} \rightarrow$ imaginary (impossible)

$\Rightarrow$ We have to increase "T_a"

For $T_a > 0$

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That value of T_a at which sum of 1st and 3rd term = 2nd term in that case Imaginary root can be avoided and this value of $T_a \rightarrow T_{th}$.

$\Rightarrow$ at $T_a = T_{th}$

$\sqrt{m_a m_b T_{th} + (m_x + m_b)\{m_x Q + (m_x - m_a) T_{th}\}} = 0$

$[m_a m_b T_{th} + (m_x + m_b) m_x Q + (m_x + m_b)(m_x - m_a) T_{th}]^{1/2} = 0$

$T_{th}[m_a m_b + m_x^2 - m_a m_x + m_b m_x - m_a m_b] = -Q m_x (m_x + m_b)$

$T_{th}\, m_x [m_x - m_a + m_b] = -Q\, m_x (m_x + m_b)$

$$\boxed{T_{th} = \frac{-Q(m_x + m_b)}{m_x - m_a + m_b}} \rightarrow ⑥$$

2) Double Valued Behaviour:

↳ occurs at values of incident energies b/w T_{th} and T_a' (upper limit)

- Only occurs for reactions with $Q < 0$.

$\rightarrow$ Range of double-valued behaviour:

$$T_a' - T_{th} = ?$$

where T_{th} as given in eq-⑥ and T_a' is unknown

To find T_a' the condition is

$(m_x + m_b) m_x Q + (m_x + m_b)(m_x - m_a) T_a' = 0$

$$T_a' = \frac{-Q m_x (m_x + m_b)}{(m_x + m_b)(m_x - m_a)}$$

such reactions neither have a threshold and nor a double valued behaviour. curves are single-valued for all θ & T_a.

$m_i > m_f \Rightarrow Q > 0$ Fig. 11.3

Problems CH #11 (Krane) 6, 7, 9, 10.

$$a + X \rightarrow Y + b$$

$$Q = (m_i - m_f)c^2$$

$$a + X \rightarrow Y^* + b$$

$$Q = [(m_a + m_x) - (m_Y + E_{ex}/c^2 + m_b)]c^2$$

$$Q = [m_a + m_x - (m_Y + m_b)]c^2 - E_{ex}$$

$$Q = Q_0 - E_{ex}$$

is Q-value for a reaction where recoiled nucleus is left in an excited state.

Cross section of a Nuclear Reaction

What is probability of a nuclear Reaction to occur

$a + X \rightarrow Y + b$

σ = prob

In lab there is not just single particle but ~~but~~ beam of particles "a". Some interact with "x" and other goes with interacting so average / probability.

Note:

Dominant differences b/w nuclear decay and nuclear reaction.

Nuclear Decay $A \rightarrow B + b$	Nuclear Reaction $a + X \rightarrow Y + b$
(i) It is a spontaneous process.	(i) It is not a spontaneous process (external aid is needed to start a

Maximum angle at which double-valued behaviour occurs:

Consider

$$\sqrt{T_b} = \frac{(m_a m_b T_a)^{1/2}\cos\theta \pm \{m_a m_b T_a \cos^2\theta + (m_y + m_b)(m_y Q_1 + (m_y - m_a) T_a)\}^{1/2}}{(m_y + m_b)}$$

will remain real, +ve and double valued only when the term in the square root is smaller than 1st term.

The value of 'θ_m' is determined for a given 'T_a' in the permitted range by vanishing of the argument in the square root i.e.

$$m_a m_b T_a \cos^2\theta_m + (m_y + m_b)[m_y Q_1 + (m_y - m_a) T_a] = 0$$

$$\cos\theta_m = -\left\{\frac{m_y + m_b [m_y Q_1 + (m_y - m_a) T_a]}{m_a m_b T_a}\right\}^{1/2}$$

$$\theta_m = \cos^{-1}\left[-\left\{\frac{(m_y + m_b)[m_y Q_1 + (m_y - m_a) T_a]}{m_a m_b T_a}\right\}^{1/2}\right]$$

double valued, $T_a' = T_{th}$

where $I = I_a N$

I_a: current of incident particles (no. of incident particles per unit time)

N: no. of target nuclei per unit area exposed to the incident beam.

⇒ Units of cross-sections are unit of area (m^2)

→ Nuclear cross-sections are normally very small of the order of $10^{-27} - 10^{-28}\ m^2$.

↳ convenient to use a unit called barn.

where $1b(barn) = 10^{-28}\ m^2$ smaller units are milli barn, microbarn etc.

Q. What are different reaction cross sections? How they are measured? What are their possible applications?

Ans: In nuclear reactions, we come across different types of reaction cross sections.

7.

on a given target, projectile and incident energy combination

$$\begin{cases} {}^1_1H + {}^7_3Li \to {}^7_4Be + {}^1_0n \\ {}^1_1H + {}^7_3Li \to {}^4_2He + {}^4_2\alpha \end{cases}$$

Several different nuclear reactions (elastic scattering, inelastic scattering, radiative capture ... etc) are possible simultaneously

$$a + X \to X + a$$
$$a + X \to X^* + a$$
$$a + X \to Y_1 + b_1$$
$$a + X \to Y_2 + b_2 \quad \text{--- etc.}$$

→ For each such reaction, there is a definite reaction cross-section denoted by $\sigma_1, \sigma_2, \sigma_3, \ldots$

↳ These cross sections are known as "partial" or "reaction" cross section.

Let I_a : Current of the incident particles (no. of particles / unit time)

N : target nuclei per unit area.

R_b : rate of specific outgoing particle 'b' (no. of reactions per unit time for outgoing particle 'b').

then reaction cross-section for $(a + X \to Y + b)$ is

$$\left(\sigma_b = \frac{R_b}{I_a N}\right)$$

proportional to reaction probability.

units — barn

② **Differential reaction Cross-Section :**

When incoming particles (from the incident beam) interact with the target nuclei, reaction or scattering takes place.

→ The outgoing particles will not in general be emitted uniformly in all directions.

(anisotropic distribution) i.e no. of outgoing particles at different angles will be different

→ We measure the no. of outgoing particles per second into a solid angle $d\Omega$, making an angle 'θ' with the incident direction, and thus only a fraction of cross section '$d\sigma$' is deduced.

where
θ is a polar angle w.r.t z-axis and
ϕ is an azimuthal angle w.r.t x-axis

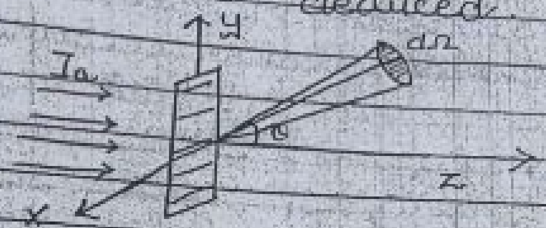

Fig: Reaction geometry showing incident beam, target and outgoing beam going into solid angle $d\Omega$ at θ, ϕ.

Units of differential cross section: $\frac{\text{barn}}{\text{steradian}}$

~~→ We measure the no. of outgoing particles per second into a solid angle $d\Omega$, making an angle 'θ' with the incident direction, and thus only a fraction of cross section '$d\sigma$' is deduced.~~

→ This cross-section depends upon the angle (θ) (and also possibly on ϕ) is called differential cross section.
Normally denoted by $\sigma(\theta,\phi) = \frac{d\sigma}{d\Omega}$.

→ **Expression** ? Let $\sigma(\theta,\phi)$: represents angular distribution function of utgoing particles

$$d\ell_b = r(\theta,\phi)\frac{d\Omega}{4\pi}$$

then $d\sigma = \frac{d\ell_b}{I_a N} = \frac{r(\theta,\phi)\,d\Omega}{4\pi I_a N}$

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{r(\theta,\phi)}{4\pi I_a N}}$$

$\frac{d\sigma}{d\Omega}$: gives us important information on the angular distribution of reaction products

The reaction cross section (σ_b) can be obtained by integrating $\frac{d\sigma_b}{d\Omega}$ over all angles.

$$\boxed{\sigma_b = \int \frac{d\sigma_b}{d\Omega}\, d\Omega}$$

Differential cross section in energy $\left(\frac{d\sigma}{dE}\right)$

If we don't observe direction of outgoing particle e.g b and we are just interested to know, what is probability for b to emit with a certain energy then we use differential cross section in energy ($d\sigma/dE$).

3) Double-Differential Cross Section: ($d^2\sigma/dE\,d\Omega$)

In some nuclear physics applications, we are not simply concerned with the probability to find particle 'b' emitted at a certain angle but also want to find it with a certain

energy ...
energy ...
→ The ...
cross s...
probabi...
angular ...
the en...
double ...

4) Total ...

target ...
add t...
all p...
matter ...
energy ...

σ_T ...

where b...
particles ...
reaction ...
→ σ_T t...
incident ...
reaction ...
and th...
of inc...
→ σ_T ca...
measuri...
incoming ...
certain ...
Table 11.1 ...

energy corresponding to a particular energy of the residual nucleus 'Y'.
→ The corresponding definition of cross section which gives the probability to observe b in the angular range of $d\Omega$ and in the energy range dE_b is called double differential cross section $d^2\sigma/d\Omega dE_b$

Total Cross Section (σ_t):

For a given energy target and projectile combination "we add the reaction cross sections for all possible outgoing particles b no matter what their direction or energy be.

$$\sigma_T = \sigma_{b_1} + \sigma_{b_2} + \sigma_{b_3} + ---$$

where b_1, b_2, b_3 are different outgoing particles corresponding to different reactions.

→ σ_T tells us the probability for an incident particle a to have any reaction at all with the target, and thus be removed from the beam of incident particles.

→ σ_T can be deduced directly by measuring the loss in intensity of incoming beam in passing through a certain thickness of the target material.

Table 11.1 (Krane)

oss ctions	Symbol	Technique	Possible Application
tal	σ_T	Attenuation of beam	Shielding
action	σ	Integrate over all angles and all energies of b (all excited states of Y)	Production of radioisotope Y in a nuclear reaction
fferential Angular)	$\frac{d\sigma}{d\Omega}$	Observe b at (θ,ϕ) but integrate over all energies	Formation of beam of b particles in a certain direction (or recoil of Y in a certain direction)
fferential (Energy)	$\frac{d\sigma}{dE}$	Don't observe b but observe excitation of Y by subsequent γ-emission	Study of decay of excited states of Y.

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ouble ifferential	$\frac{d^2\sigma}{dE\, d\Omega}$	Observe b at (θ, φ) at a specific energy.	Information on excited states of Y by angular distribution of b

$$Q = [m(^9Be) + m(P) - m(d) - m(^8Be)]c^2$$

where $\frac{559.5\times10^3 \text{ eV u}}{931.502\times10^6 \text{ eV}} = 9.012182 + 1.007825 - 2.014102 - m(^8Be)$

$m(^8Be) = 8.005905 - 6.006428\times10^{-4}$ u

$$\boxed{m(^8Be) = 8.0053044 \text{ u}}$$

Q.7 (a) Calculate Q value of the reaction

$$p + {}^4He \rightarrow {}^2H + {}^3He$$

$m(P) = 1.007825$ u $\quad m(^2H) = 2.014102$ u

$m(^4He) = 4.002603$ u $\quad m(^3He) = 3.016029$ u

As we know that

$$Q = (m_i - m_f)c^2 = [m(P) + m(^4He) - m(^2H) - m(^3He)]c^2$$

$$Q = [1.007825 + 4.002603 - 2.014102 - 3.016029] \times 931.502 \text{ MeV}$$

$$\boxed{Q = -18.35 \text{ MeV}} \quad \text{or} \quad \boxed{Q = -18350 \text{ keV}}$$

(b) What is the threshold energy for protons incident on He? For α's incident on hydrogen?

$$a + X \rightarrow Y + b$$

As we know that

$$T_{th} = \frac{-Q(m_Y + m_b)}{m_Y - m_a + m_b}$$

Threshold energy for protons:

Here

$$T_{th} = \frac{18.350 \text{ MeV}(2.014102 + 3.016029)}{(2.014102 + 3.016029 - 1.007825)}$$

$$T_{th} = \frac{18.35(5.030131)}{(5.030131 - 1.007825)} \text{ MeV}$$

$$T_{th} = 22.94775779$$

or $\boxed{T_{th} = 22.95 \text{ MeV}}$

Threshold energy for α:

ere. $T_{th} = \frac{18.35(1.007825 + 9.016029)}{(1.007825 + 4.002603 + 3.016029)}$

$T_{th} = \frac{18.35(4.023854)}{0.091299}$

$\boxed{T_{th} = 46.099 \text{ MeV}}$

9(a) Calculate Q-value for
$^6Li + p \rightarrow {}^3He + {}^4He$
) $^{59}Co + p \rightarrow n + {}^{59}Ni$
) $^{40}Ca + \alpha \rightarrow n + {}^{43}Ti$

lution: (a)

$Q = [m(^6Li) + m(p) - m(^3He) - m(^4He)]c^2$

$= [6.015121 + 1.007825 - 3.016029 - 4.002603] \times 931.5 \text{ MeV}$

$\boxed{Q = 4.018 \text{ MeV}}$

) $Q = [m(^{59}Co) + m(p) - m(n) - m(^{59}Ni)]c^2$

$Q = [58.933198 + 1.007825 - 1.00866 - 58.934349] \times 931.5 \text{ MeV}$

$\boxed{Q = -1.84996 \text{ MeV}}$

) $Q = [m(^{40}Ca) + m(\alpha) - m(n) - m(^{43}Ti)]c^2$

$Q = [39.962591 + 4.002603 - 1.00866 - 42.968523] \times 931.502 \text{ MeV}$

$\boxed{Q = -11.16 \text{ MeV}}$

.10) Find Q-value and Threshold K.E;
(a) $^7Li + p \rightarrow {}^7Be + n$

$$T_{th} = \frac{1.639(7.016928 + 1.00866)}{(7.016928 + 1.00866 - 1.007825)}$$

$$T_{th} = \frac{1.639(8.025588)}{7.017763}$$

$$\boxed{T_{th} = 1.874 \text{ MeV}}$$

(b) $^{12}C + p \rightarrow n + ^{12}N$

$Q = [m(^{12}C) + m(P) - m(n) - m(^{12}N)]c^2$

$Q = [12 + 1.007825 - 1.00866 - 12.018613] \times 931.502 \text{ MeV}$

$\boxed{Q = -18.11 \text{ MeV}}$

Here $a = P$, $Y = ^{12}N$, $b = n$

$$T_{th} = \frac{18.11(12.018613 + 1.00866)}{(12.018613 + 1.00866 - 1.007825)}$$

$$\boxed{T_{th} = 19.628 \text{ MeV}}$$

(c) $^{35}Cl + \alpha \rightarrow n + ^{38}K$

$Q = [m(^{35}Cl) + m(\alpha) - m(n) - m(^{38}K)]c^2$

$Q = [34.968853 + 4.002603 - 1.00866 - 37.969080] \times 931.502 \text{ MeV}$

$\boxed{Q = -5.828 \text{ MeV}}$

Here $a = \alpha$, $Y = ^{38}K$ and $b = n$

$$T_{th} = \frac{5.828(1.00866 + 37.969080)}{(1.00866 + 37.969080 - 4.002603)}$$

$$\boxed{T_{th} = 6.49 \text{ MeV}}$$

5: Shielding (a wall or housing of concrete or lead built around a nuclear detector/reactor to prevent the escape of radiation).

(Variation) of (Cross Section) with Thickness of Target:

↳ unit – barn ($10^{-28}\ m^2$) → area

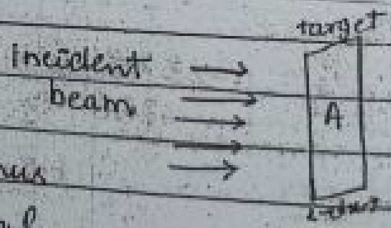
Note:
The reaction cross section can be visualized as an effective area around the target nucleus, such that if the incident particle 'a' crosses that area, a nuclear reaction takes place otherwise not.

The effective area is not exactly the same as the geometrical area (πr^2) ; r – radius of the nucleus.

It can be larger, smaller or equal to the geometrical area.

→ Let us consider an incident beam of particles of flux 'I' (no. of particles crossing a unit area of target per unit time) incident on a thin sheet of target material having thickness 'dx' and face area 'A' (Volume = Adx).

Note (Assume that thickness 'dx' is so small that none of the nuclei of target overlap each other) (thus each nucleus has an equal probability to cause a nuclear reaction with the incident beam)



→ Now let for a certain reaction to occur 'σ' be the reaction cross-section

Here one can consider that 'σ' is the effective area of each of the target nucleus. This area is such that when the incident particles crosses this area the reaction definitely takes place.
→ Let 'n' be the no. of target nuclei per unit volume in this thin sheet (of target material) then
total no. of nuclei in the target sheet $= nA\,dx$
total effective area available for the reaction $= \sigma n A\,dx$
∴ fractional effective area $= f = \frac{\sigma n A\,dx}{A} = \sigma n\,dx$

If the fundamental effective area is large then there will be more probability for a reaction to occur ⇒ more incident particles take part in the reaction ⇒ corresponding decrease in the flux 'I' is large and vice versa.
→ It can be mathematically summarized as
fractional change in flux = fractional effective area.

$$\frac{dI}{I} = -n\sigma\,dx$$

-ve sign is introduced because as dx increases, the flux I decreases.
Assume that $I = I_0$, at $x = 0$ (projectile has not reached the target).
Integrating eq-(1) we get

$$\int^{I} \frac{dI}{I} = -n\sigma \int_0^x dx$$

Proposed by ...

Fundamental assumption of theory

Bohr assumed that a nuclear reaction takes place in two steps

1) Incident particle is absorbed by the target nucleus to form a compound nucleus.

$$a + X \rightarrow C^*$$

2) The compound nucleus then disintegrates by ejecting a particle (P, n, α ---) or a γ-ray, leaving the final or product nucleus.

$$C^* \rightarrow Y + P$$

$$\rightarrow Y + n$$

$a + X \rightarrow C^*$

$d + W \rightarrow C^*$

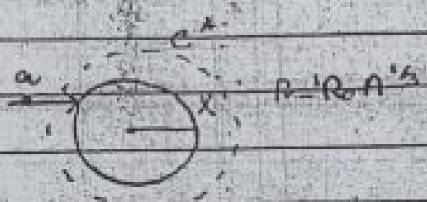
depends on the properties of C^* itself (such as its energy* & angular momentum etc).

step → Formation of Compound Nucleus

– The incident particle interacts through strong nuclear force with the nucleons present in the target nucleus & gives away / shares its energy to all the nucleus and thereby loses its identity and becomes a part of the nucleus.

This new nucleus which is formed in highly excited state and is called compound nucleus.

$a + X \rightarrow C^*$



Excitation energy of compound nucleus:

→ on being captured the incident particle / projectile 'a' makes available a certain amount of excitation 'E_e' to the compound nucleus which is equal to sum of two terms

① Fraction of the energy of the captured particle, E_a' (some of the energy of the incident particle is used in supplying K.E to the compound nucleus).

② It's binding energy in the compound nucleus 'E_B' i.e.

$E = E_a' + E_B \rightarrow ①$ $\quad E_a' < E_a$:

Calculation of E_a':

- Before collision, target 'X' is at rest 'a' of energy 'E_a' (moving with velocity 'v_a') strikes 'X'
after absorption of 'a' the compound nucleus C^* recoils with velocity 'V'.
then for the process

$$a + X \rightarrow C^*$$

By conservation of mom.

$$m_a v_a = MV$$

$$\Rightarrow \boxed{V = \frac{m_a}{M} v_a} \rightarrow (2)$$

The portion of energy of the incident particle that is used in the excitation of the compound nucleus is given by

$$E_a' = E_a - E_{cm}$$

$$= E_a - \frac{1}{2} MV^2$$

$$E_a' = \frac{1}{2} m_a v_a^2 - \frac{1}{2} M \left(\frac{m_a}{M} v_a \right)^2$$

putting $M \approx m_x + m_a$

$$E_a' = \frac{1}{2} m_a v_a^2 - \frac{1}{2} \frac{(m_x + m_a) m_a^2}{(m_x + m_a)^2} v_a^2$$

$$E_a' = \frac{1}{2} m_a v_a^2 \left[1 - \frac{m_a}{m_x + m_a} \right]$$

$$E_a' = \frac{1}{2} m_a v_a^2 \left[\frac{m_x}{m_x + m_a} \right]$$

$$\boxed{E_a' = E_a \left(\frac{m_x}{m_x + m_a} \right)} \rightarrow (3)$$

where $m_x + m_a > m_x$

$$\Rightarrow E_a' < E_a$$

Calculation of E_B:

for $a + x \rightarrow C^*$ (sum of mass of components is greater than the mass of the product).

Since $m_i > m_f$ then the corresponding available $B \cdot E = (m_i - m_f)c^2 \rightarrow (4)$

So $E_{ex} = E_a' + E_B$

$$E_{ex} = E_a\left(\frac{m_x}{m_x + m_a}\right) + [(m_a + m_x) - M]c^2 \rightarrow (5)$$

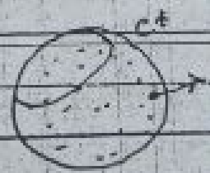
Given ${}^{1}_{0}n + {}^{27}_{13}Al \rightarrow {}^{28}_{13}Al^*$

for $E_a = 1\,MeV$ Calculate E_{ex} of ${}^{28}_{13}Al^*$.

→ **Decay of Compound Nucleus**

– After the formation of compound nucleus the available excitation energy is randomly distributed or shared among the nucleons (i.e. at a given instant, it may be shared among several nucleons and at the later time may be concentrated in a single nucleon or combination of nucleons).

– If excitation energy is large enough, one nucleon or a combination of nucleons may escape.



→ Compound nucleus disintegrates into the product nucleus and the outgoing particle.

→ The energy needed to separate a nucleon or a group of nucleons from the compound nucleus is called separation or dissociation energy (usually ~ 8 MeV).

Why Bohr assumed that mode of disintegration of compound nucleus is independent of the way it was formed?

Lifetime of compound nucleus is relatively longer than natural nuclear one.

Planation

Life time of Compound Nucleus and Natural Nuclear time.

tural Nuclear time

The time that would be quired for a particle to vel across the nucleus

$S = Vt$

$t = \frac{S}{V} = \frac{2R}{V}$

where by $t = \frac{S}{V}$ = diameter of the nucleus / speed of the incident particles

diameter of nucleus $\approx 10^{-14}$ m (10^{-12} cm)

speed of the incident particle

g for an incident projectile $\simeq 10^{9}$ cm/sec of 1 MeV)

$$t \simeq \frac{10^{-12}\text{ cm}}{10^{9}\text{ cm/sec}} \sim 10^{-21}\text{ sec}$$

i.e the time required for 1 MeV particle to cross the nucleus is of e order of 10^{-21} sec.

relatively longer compared with the time required for a particle to travel across the nucleus.

This time is found to be 10^{-14} sec $- 10^{-15}$ sec which is $10^{6} - 10^{7}$ times longer than the natural nuclear time.

→ During its relatively longer lifetime, the C.N. forgets the way it was formed and the disintegration is independant of the mode of its formation

$$\left\{\begin{array}{l} a + X \rightarrow C^{*} \\ d + W \rightarrow C^{*} \end{array}\right.$$

→ A given C.N may decay in a variety of different ways and the relative probability for decay into a specific set of final products is independant of the means of its formation.

$$C^{*} \rightarrow b + Y$$
$$C^{*} \rightarrow e + Z$$

Example.

→ Consider a specific example

① The compound nucleus (Zn^{*}) can be formed through several reaction processes, including

$$p + {}^{63}_{29}Cu \rightarrow {}^{64}_{30}Zn^{*}$$ {With

$${}^{4}_{2}\alpha + {}^{60}_{28}Ni \rightarrow {}^{64}_{30}Zn^{*}$$ appropriately

chosen energies of p and α to reach the same excitation energy of Zn^{*}}.

② This C.N (Zn^{*}) can decay in a variety of different ways.

$${}^{64}_{30}Zn^{*} \rightarrow {}^{63}_{30}Zn + {}^{1}_{0}n$$
$$\quad '' \rightarrow {}^{63}_{29}Cu + {}^{1}_{0}n + {}^{1}_{1}p$$
$$\quad '' \rightarrow {}^{62}_{29}Cu + {}^{1}_{0}n + {}^{1}_{1}p$$
$$\quad '' \rightarrow {}^{62}_{30}Zn + 2\,{}^{1}_{0}n$$

compound nucleus theory

this model (i.e. compound nucleus theory of nuclear reaction) is correct, then the relative cross section/probability for a certain decay channel/mode

e.g. $p + {}^{64}Cu \rightarrow Zn^* \rightarrow {}^{63}Zn + n$ would be same

$\alpha + Ni \rightarrow Zn^* \rightarrow Zn + n$

would be same at the incident energies that give the same excitation energy to Zn^*.

As we can see from the fig, the agreement b/w 3-pairs of cross-sections is fairly good.

↳ decay of Zn^* into any specific set of final products is almost independent of the way, how it was originally formed.

Characteristics of Compound Nucleus Model:

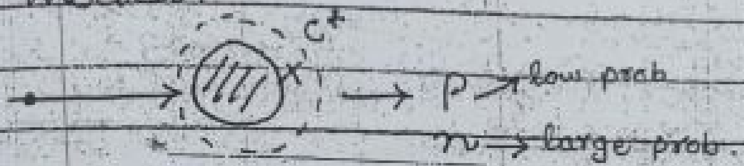
① It works best for ...

NUCLEAR PHYSICS

DIRECT REACTIONS

At low energies, (d,p) reactions showed large cross-sections than (d,n) reactions.

↳ This result could not be explained by C-N model.



→ Oppenheimer and Philipp suggested that, a new type of reaction taking place, which could not be explained within the context of C-N model.

→ Their explanation